



<Product name and version number>

Author		 中科时代 基于PC技术的工智机新时代 West Tower of Baidu International Building, Yuehai Sub-district, Nanshan District, Shenzhen Official website: www.sinsegye.com.cn E-mail: Sales@sinsegye.com.cn
Date	2025.4.17	
Version	1.2.1	
Email		

		Hotline: 400-013-2158
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Update instructions:

This document has recorded the changes to Xvisual.

<Online Date> / <Version Number>

Function	Type of change	Description	Related documents
	Refer to the following change types: <Addition> <Modification> <Optimization> <Scrap> <Deletion> <Repair> <Known Issue>	Describe the specific changes.	List related documents and provide hyperlinks.



Preface

I. Document Description

The instructions have been developed for technical experts in control and automation, who are familiar with the relevant national standards and who have received professional training.

Be sure to peruse all relevant documents and the following instructions when installing and commissioning components.

Qualified personnel should always operate with the latest available documentation.

The responsible person must ensure that the application or use of the product described is in full compliance with all safety requirements, including all relevant laws, regulations, guidelines and standards.

1. Disclaimer

This document has been carefully prepared, but in view of the ongoing development and upgrading of the products described, Sinsegye (Shenzhen) Computer System Co., Ltd. reserves the right to modify and update the document at any time without prior notice. Please note that any changes to the delivered products based on the data diagram and the description in this document are prohibited.

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II. Safety Statement

1. Safety regulations

For your safety, please read the following instructions. You always follow the product-specific safety instructions, which you can find in the appropriate place in this document.

2. Exclusion of liability

All components are available in hardware and software configurations. It is not allowed to modify the hardware or software configuration other than those described in the document, and Sinsegye shall not be responsible for the hardware or software other than that stated in the document.

3. Qualification of personnel

The instructions apply only to trained experts in control, automation and drive technology who are familiar with the applicable national standards.

4. Signal words

The signal words used in the document are classified as follows. To prevent injury and damage to people and property, read and observe the safety and warning notices.

5. Personal injury warning

	☐ 警告 危险的类型 说明不避开危险后果 说明如何避免危险的发生	Warning indicates a potentially hazardous situation that, if not avoided, could result in serious injury or death.
	☐ 注意 危险的类型 说明不避开危险后果 说明如何避免危险的发生	Caution indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury, or damage to the device.
提醒 危险的类型 说明不避开危险后果 说明如何避免危险的发生		Caution indicates a potentially hazardous situation that, if not avoided, may only result in damage to the device.

6. Warning of damage to property or the environment

<table><tr><td>注意</td></tr><tr><td>危险的类型 说明不避开危险的后果 说明如何避免危险的发生</td></tr></table>	注意	危险的类型 说明不避开危险的后果 说明如何避免危险的发生	Environment, device, or data may be damaged.
注意			
危险的类型 说明不避开危险的后果 说明如何避免危险的发生			

7. Product processing information

This information includes, for example, proposals for action, assistance or further information about the products.

Overview

I. Product Concept

Xvisual is a HMI component application developed by in SINSEGYE. It is a SCADA product oriented to configuration editing and human-computer interaction, and can facilitate users and FAE to easily, quickly and efficiently complete visual customization and automatic page control drawing.

It provides an easy-to-use development environment and rich functions that allow configuration engineers to quickly set, test, and deploy HMI visualization applications that fit the current industry. It can be widely applied in business scenarios such as petrochemical industry, metallurgical steel, electric power, photovoltaic, 3C manufacturing, biomedicine and the like, and provides custom HMI applications for users.

It is an ideal choice for creating large distributed industrial automation systems. It can run on servers, embedded computers, and the cloud. It can exchange information between nodes and interact with external databases in real time. It has rich functions to support transmission of dozens of common industrial communication protocols, and has common report, alarm, trend curve, and a variety of scene control. It has common templates in various industries to meet various production monitoring needs of users.

II. Product Features

General	
Operating System	Windows (Win10 and above), Linux
Unlimited point locations, labels and devices	✓
Remote configuration and maintenance of the project	✓

Safety	
HTTPS、SSL	✓
Protection against violent attacks	✓
Use CAPTCHA when logging in	✓
Connection with the monitor	✓
Password encryption	✓
User activity records	✓
Redundancy	
Primary and backup servers, with automatic data synchronization	✓
Central and remote servers, with automatic data synchronization	✓
Filing	
Maximum number of standalone archives	31
Write by cycle	✓
Write by change	✓
Connection	
Industrial Protocol	OPC UA、Modbus、Simulator

Programmability

REST API	✓
Sample module with a source code	✓
Developer Documentation	✓

III. Component Composition

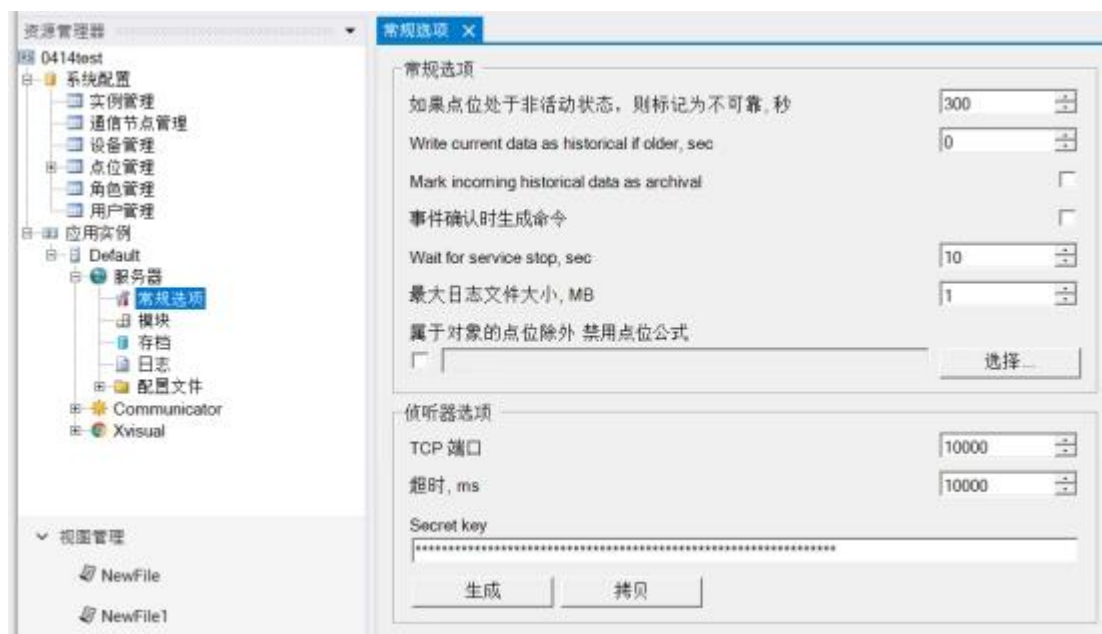
Product Components	Description
Xvisual-Server	Xvisual-Server manages the data archiving, performs mathematical calculations, and provides information to client applications. Xvisual-Server writes data to the primary archive and makes a backup copy at the same time
Xvisual-Communicator	Xvisual-Communicator interacts with the device and transfers the data to the Xvisual-Server application. The communication with the device connected to the system is performed in parallel through multiple lines. Xvisual-Communicator receives current data from the device, archives data and events, and sends commands to the device. This application helps troubleshoot the problems of communication lines and devices.
Xvisual-Webstation	Xvisual-Webstation can be accessed from any computer or tablet PC connected to the organizational network without installing software. Access is managed by the system administrator who defines user permissions.
Xvisual-Agent	Xvisual-Agent transfers configuration between an Xvisual instance and an Administrator application. Xvisual-Agent also provides log files which can be

	displayed in the administrator. The agent runs as a service on a server on which the Xvisual instance controlled by the Agent is installed. Xvisual instance includes all or part of Server, Communicator, and Webstation.
Xvisual-Administrator	An integrated development environment that provides the ability to edit the configuration database, configure major Xvisual applications, server modules, and device drivers.

1、Xvisual-Server

Xvisual-Server manages the data archiving, performs mathematical calculations, and provides information to client applications. Xvisual-Server writes data to the primary archive and makes a backup copy at the same time.

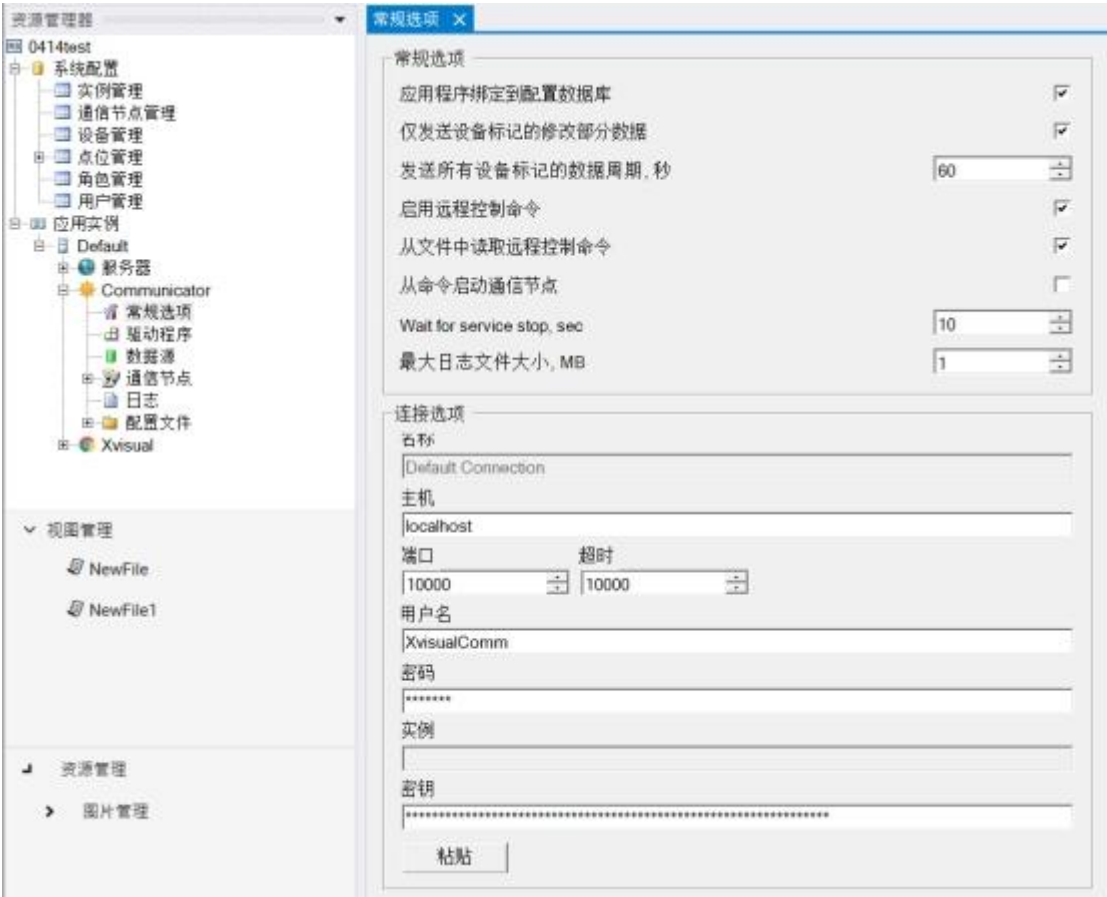
Xvisual-Server works as a service. It has no user interface. The server runs continuously in the background regardless of user login and logout. The graphical shell for Xvisual-Server configuration is built into the Xvisual-Administrator application. In addition, the Xvisual-Server module allows you to extend the function of the server based on customer requirements.



2、Xvisual-Communicator

Xvisual-Communicator interacts with the controller and transfers data to the Xvisual-Server application. The communication with the Xvisual-Communicator connected to the system is performed in parallel through multiple lines. Xvisual-Communicator receives current data from the controller, archives data and events, and sends commands to the controller. This application helps troubleshoot problems of communication lines and devices.

Xvisual-Communicator works as a service, and it has no user interface. The graphical shell for Xvisual-Communicator configuration is built into the Xvisual-Administrator application. The information about the application, communication lines, and each connected device is written to the log file. Xvisual-Communicator is designed for uninterrupted operation. Developers can implement their own device drivers to interact with various controllers.



3、Xvisual-Webstation

Xvisual-Webstation is a web application that displays information and sends commands to a scheduler through a browser. The following are Schedule View of Webstation Application and Table View of Webstation Application.



The screenshot shows the Xvisual-Webstation interface in table view. The table displays a large number of data points across multiple columns. The interface includes a sidebar with navigation options and a main area with a search bar and a list of components. The status bar at the bottom indicates '无事件' (No event).

The user can select a view (table or schedule) and access the date of the archived data. To display the trend, click the item icon in the table or the appropriate element in the schedule. Xvisual-Webstation can be accessed from any computer or tablet PC connected to the organizational network without installing software. Access is managed by the system administrator who defines user permissions.

4、Xvisual-Agent

Xvisual-Agent transfers configuration between an Xvisual instance and an Xvisual-Administrator application. In addition, Xvisual-Agent provides log files which can be

displayed in Xvisual-Administrator. It runs as a service on the server on which the Xvisual instance controlled by Xvisual-Agent is installed. The Xvisual instance includes all or part of Xvisual-Server, Xvisual-Communicator, and Xvisual-Webstation applications.

Xvisual-Agent communicates with Xvisual-Administrator through TCP protocol. Therefore, you can install Xvisual-Administrator on the same computer as Xvisual-Agent or on another computer that is accessible over the network. By default, the TPC port used by Xvisual-Agent is 10002. If remote access is required, the server firewall must allow the port to be used.

Xvisual-Agent has no user interface. In order to check its operation, you need to use a log file.

5、Xvisual-Administrator

Xvisual-Administrator is an integrated development environment that provides the functions to edit and configure major Xvisual applications and Xvisual-Server modules for developing Xvisual projects and viewing the status of the automation system.



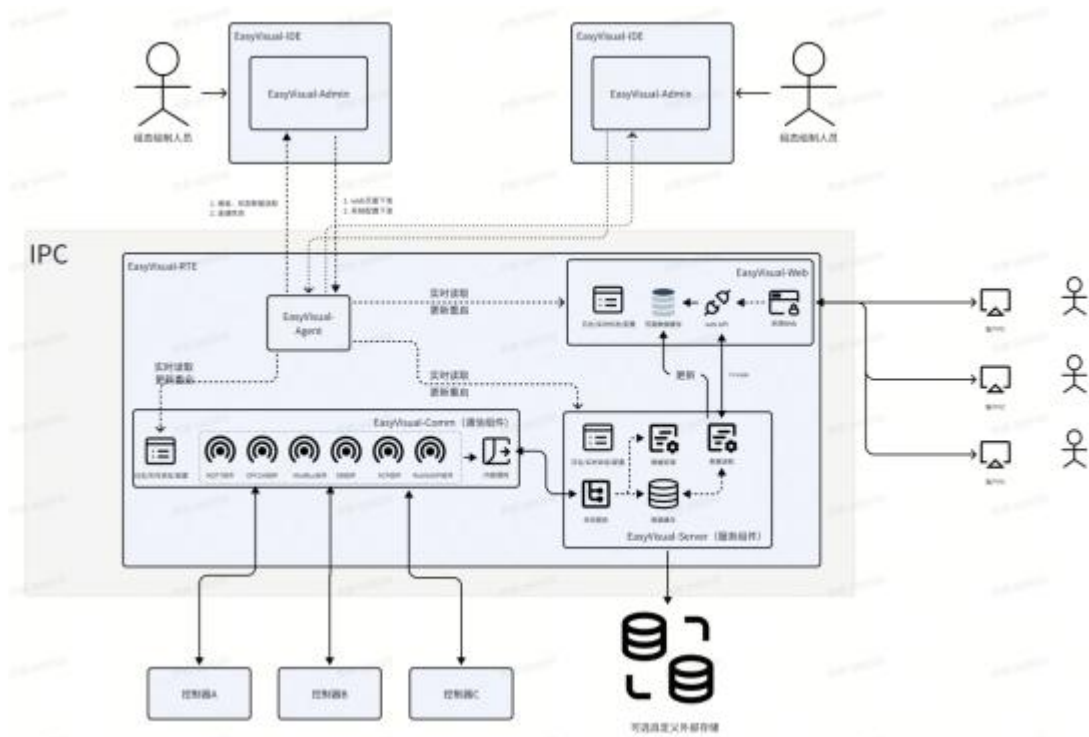
The tools and features of the quickly configured Xvisual-Administrator application are as follows:

- Wizard for creating communication lines, devices, and point locations.
- Point location cloning to minimize manual input.
- A project contains a set of configuration files, mainly in XML format, which facilitates copying the project from one computer to another.

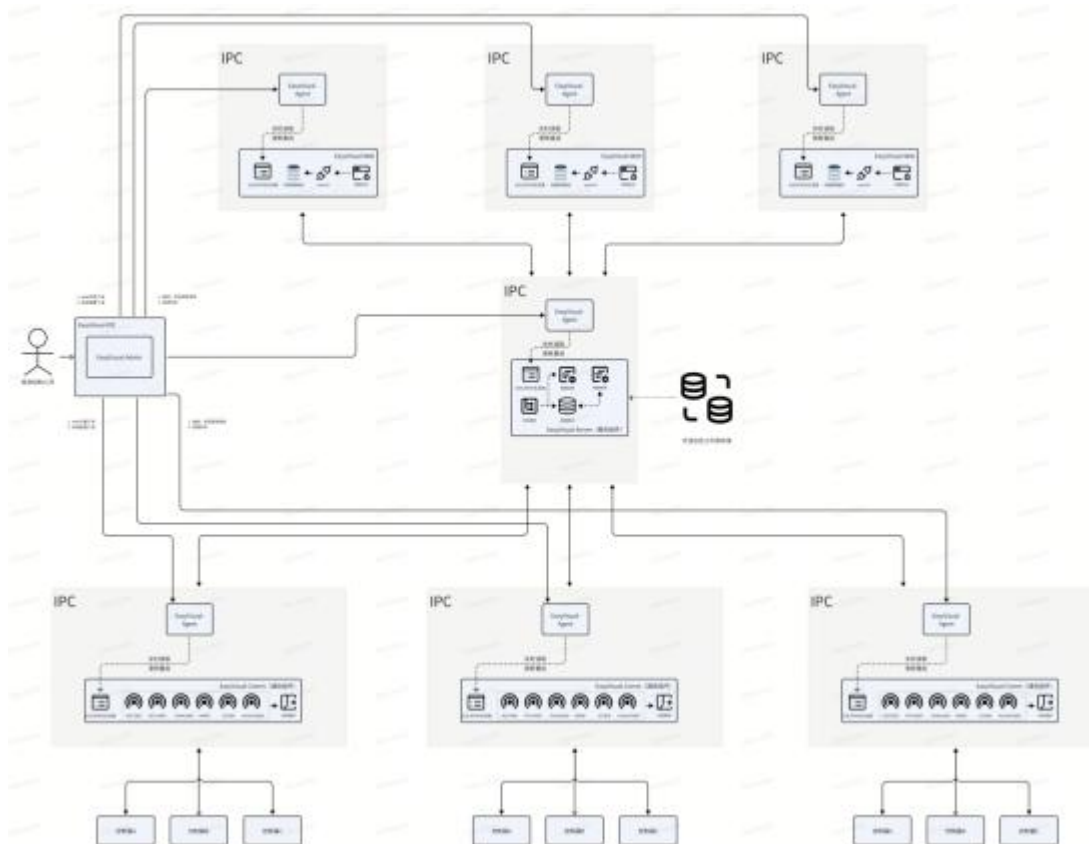
IV. Software Architecture

Xvisual software has a multi-layer structure. The applications (services) contained in the software interact with each other using the TCP protocol. Xvisual can be deployed within IPC or client PC. It can support both stand-alone deployment and distributed deployment.

- Architecture of Product Deployment within the IPC in stand-alone mode



- Architecture of Distributed Deployment Using IPC



Installation and Uninstallation

I. Installation Requirements

1. Requirements for server software

Windows:

- Operating System: Microsoft Windows Server 2016/2019/2022, Microsoft Windows 10/11
- . net Runtime 8.0

Linux:

- Operating System: Ubuntu
- . net Runtime 8.0

2. Requirements for server hardware requirements

The hardware configuration of the server depends on the size of the automation system. The minimum configuration is determined by operating system requirements.

Xvisual can run on physical or virtual machines.

3. Requirements for workstation or tablet PC

The latest Chrome, Firefox, Safari or Edge browsers (viewing with the IE10 kernel below is not supported when viewing pages using the Web) are required.

II. Installation Process

Depending on the version of Xvisual and the operating system used, Xvisual can be installed in different ways, as detailed below.

1. Install IDE on Windows

1.1 Download and run the installation package

Download Setup_Xvisual IDE.exe, the latest installation package for Xvisual IDE, from the official website of SINSEGYE. Double-click to run Setup and select Launch Setup Wizard.

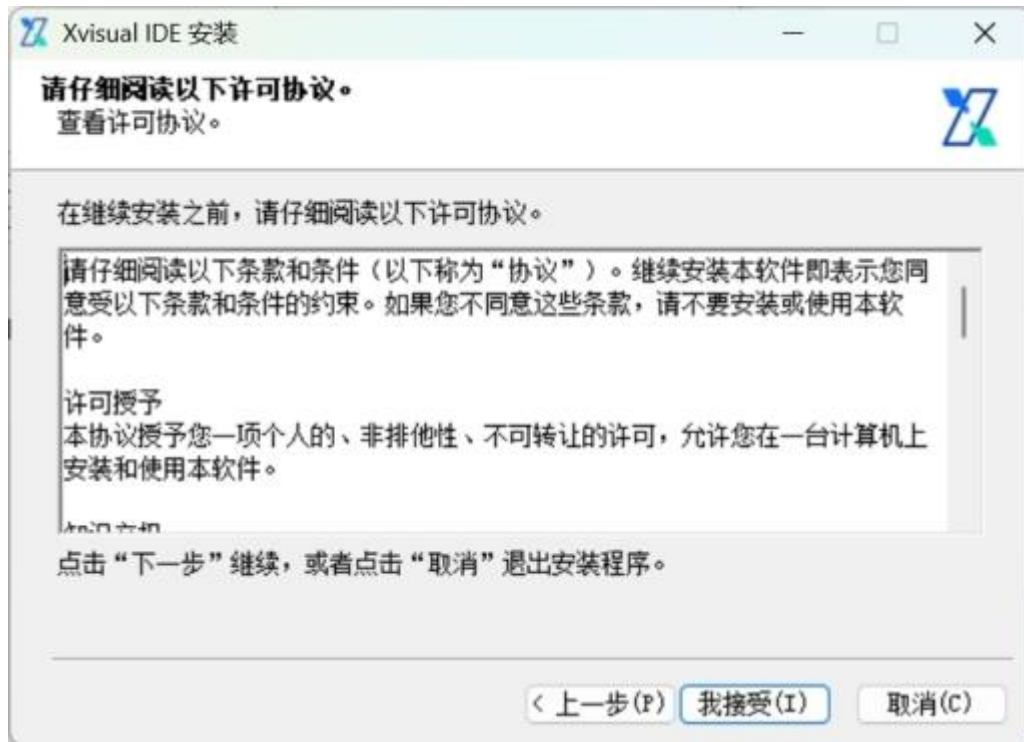


1.2 Administrator permissions

If the system prompts to require administrator permissions to run Setup, please confirm the permission. The Setup of Xvisual IDE may require administrator permissions to write it into system files and registry keys.

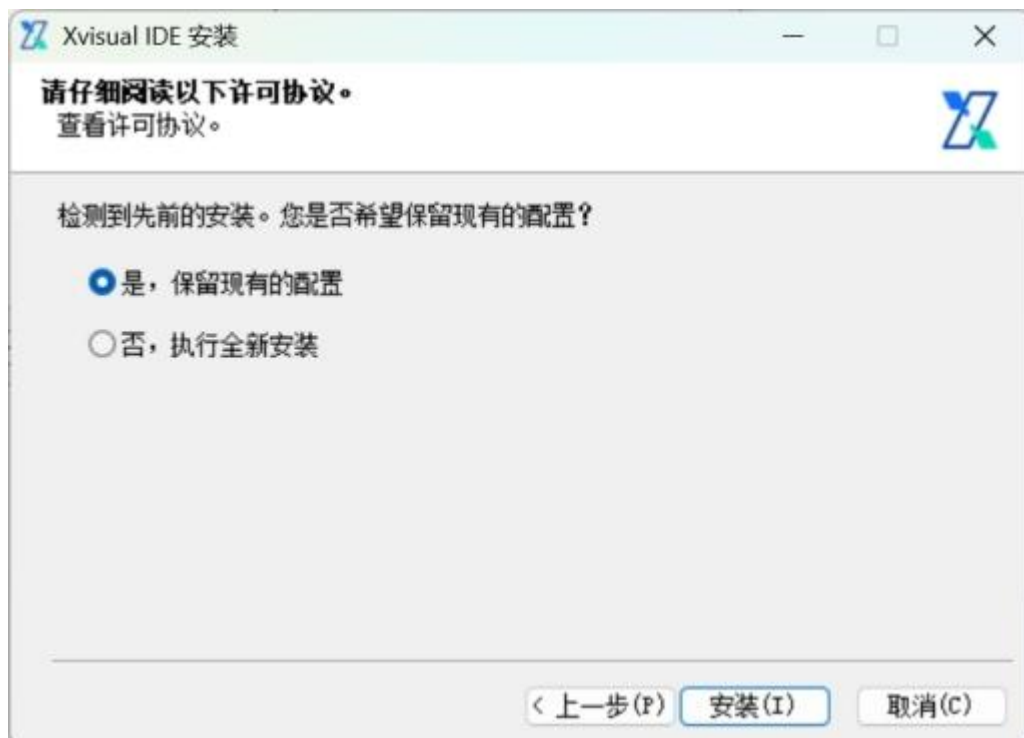
1.3 Installation wizard

In the installation wizard, click the “Next” button. You will see User License Agreement. After reading and agreeing to the terms, click "I Accept" to continue the installation.



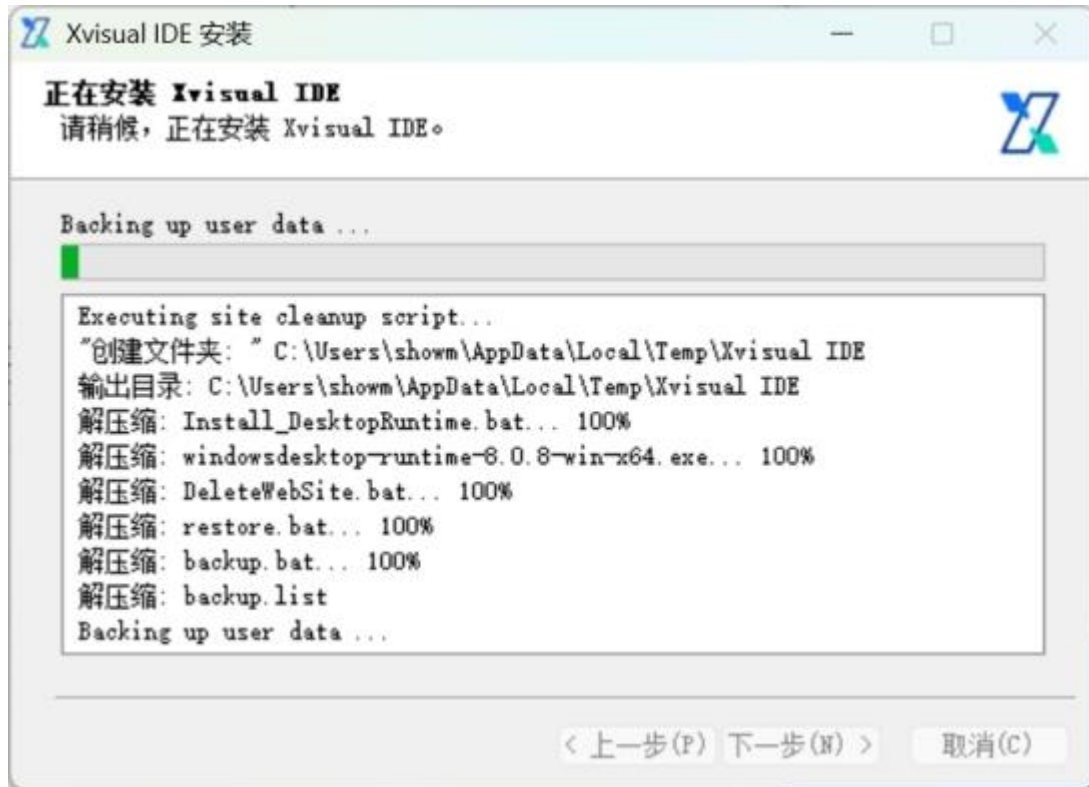
1.3 Will the existing configuration be retained during the installation

If the host has an older version installed but not uninstalled, the following screen will appear. You can choose whether to retain the existing configuration. Select "Yes" to retain the existing configuration during the installation. Select "No" to perform a new installation without retaining the previous configuration.



1.4 Install .NET 8 Desktop Runtime

Xvisual IDE relies on .NET 8 Desktop Runtime to support its application runtime environment. During the installation, the Setup automatically installs or updates to the latest version of .NET 8 Desktop Runtime. This step may take a long time. Please be patient.



1.5 Completion of installation

When the installation is complete, click the “Finish” button and the Xvisual IDE will start automatically. By default, the Setup installs Xvisual IDE to C:\Xvisual\Xvisual IDE. The Setup will automatically create shortcuts to Xvisual IDE in the desktop and start menu. Once the installation is complete, you can quickly launch Xvisual IDE using these shortcuts.



2. Install RTE on Windows

2.1 Download and run the installation package

Download the latest installation package of Xvisual RTE from the official website of SINSEGYE, such as Setup_Xvisual RTE.exe. Double-click to run Setup and select Launch Setup Wizard.



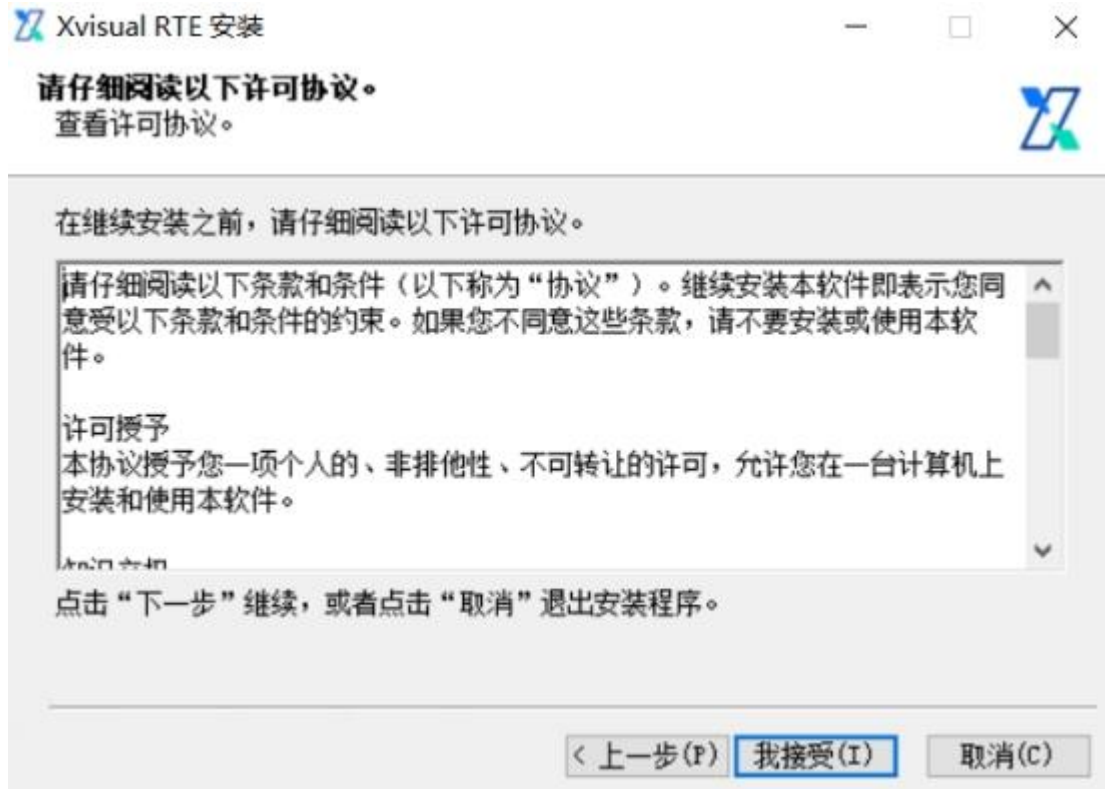


2.2 Administrator permissions

If the system prompts to require administrator permissions to run Setup, please confirm the permission. The Setup of Xvisual RTE may require administrator permissions to write it into system files and registry keys.

2.3 Installation wizard

In the installation wizard, click the “Next” button. You will see User License Agreement. After reading and agreeing to the terms, click "I Accept" to continue the installation.



2.4 Will the existing configuration be retained during the installation

If the host has an older version installed but not uninstalled, you can choose whether to retain the existing configuration. If you choose “No”, you will be prompted to configure the website destination port. If the old interface is not installed, there will be no such an interface.



2.5 Install and configure the dependencies

Xvisual RTE relies on .NET 8 Runtime and ASP. NET Core Hosting to support its application runtime environment. During the installation, the Setup automatically installs and configures the dependencies to the latest version. This step may take a long time. Please be patient.



2.6 Complete the installation and reboot

By default, the Setup installs Xvisual RTE to C:\Xvisual\Xvisual RTE.

2.7 Access Xvisual

When the reboot is complete, the Xvisual RTE service will start automatically. You can access Xvisual RTE in a browser using the server's IP address or the host name, using HTTP port 80 by default. For example:

```
Bash
http://<your-server-ip>/
```

The default login page displays the Xvisual user interface with admin as the default administrator account and Xvisual as the password.



3. Install RTE on Unbutu

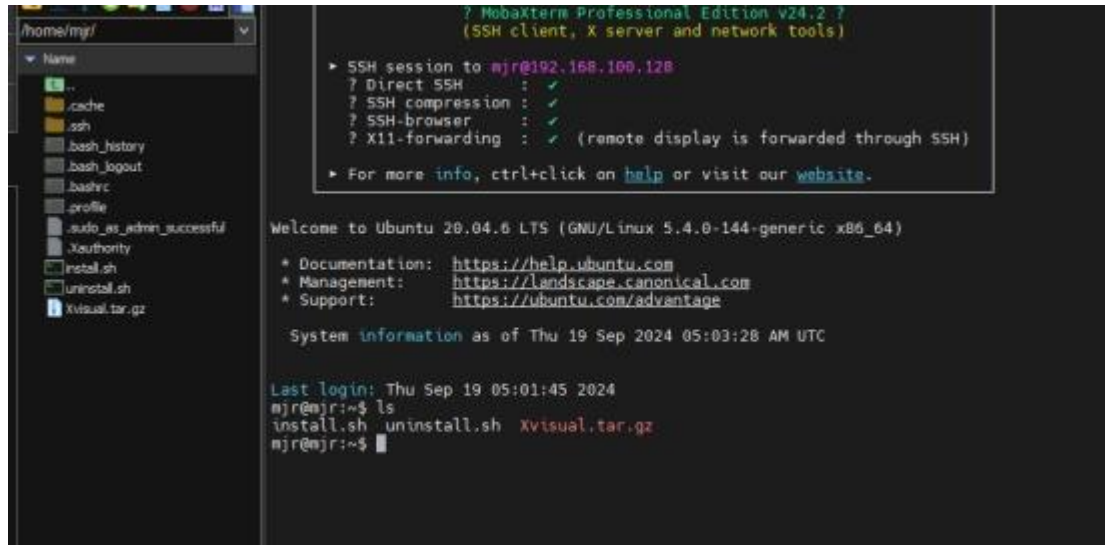
3.1 Download and copy the installation package

Download the latest deployment package and installation script of Xvisual RTE from the official website of SINSEGYE. Copy the downloaded file to any directory on the target server.

3.2 Run the installation script

Open the terminal, switch to the directory where the installation package and script are saved, and use the following command to execute the installation:

```
Bash
sudo bash ./install.sh
```



3.3 Administrator permission operation

The installation script of Xvisual IDE requires root permissions due to the modification of system files and software-dependent installation involved in the installation process. When you run the script, you will be prompted to enter the root password. Make sure you have sufficient permissions to complete the installation.

3.4 Install and configure the dependencies

Xvisual RTE relies on .NET 8 Runtime to support its application runtime environment. During the installation, the Setup automatically downloads and configures the dependencies. .NET 8 Runtime is a core dependency of Xvisual RTE for running related services and applications. The installation of the dependencies may take a long time depending on system performance. Please be patient.

3.5 Complete the installation and reboot

By default, Xvisual RTE is installed to /opt/xvisual.

3.6 Access Xvisual

When the reboot is complete, the Xvisual RTE service will start automatically. You can access Xvisual RTE in a browser using the server's IP address or the host name, using HTTP port 80 by default. For example:

```
Bash
```

```
http://<your-server-ip>/
```

The default login page displays the Xvisual user interface with admin as the default administrator account and Xvisual as the password.

III. Updating of Installation

1. Install IDE on Windows

1.1 Download the latest version of the installation package

Download the latest version of the IDE installation package from the official website of SINSEGYE. The installation package will overwrite the older version.

1.2 Run the new version of the Setup

Double-click to run the new version of the Setup. The subsequent installation process is the same as the initial installation. The Setup automatically detects and updates the existing IDE installation without uninstalling the older version.

2. Install RTE on Windows

2.1 Download the latest version of the installation package

Go to the official website of Xvisual to download the latest version of the RTE installation package. The installation package will overwrite the older version.

2.2 Run the new version of the Setup

Double-click to run the new version of the Setup. The subsequent installation process is the same as the initial installation. The Setup automatically detects and updates the existing RTE installation without uninstalling the older version.

3. Install RTE on Unbutu

3.1 Download the latest version of the software package and installation script

Go to the official website of Xvisual to download the latest version of the Xvisual RTE installation package and the corresponding install.sh installation script.

3.2 Run the new version of the Setup

Similar to the initial installation, switch to the download directory and run the following command to perform the update:

```
Bash
sudo bash ./install.sh
```

The Setup automatically detects and updates existing environment and RTE installations without first uninstalling the older version.

IV. Uninstallation Process

1. Uninstall IDE on Windows

If you want to uninstall Xvisual IDE, you can do so through “Settings” or “Control Panel” of Windows.

On Windows 10/11, go to “Settings” > “Applications” > “Applications and Features”, find Xvisual IDE, and click “Uninstall”.

In earlier versions of Windows, you can find Xvisual IDE through “Control Panel” > “Applications and Features”, and choose to uninstall it.



2. Uninstall RTE on Windows

If you want to uninstall Xvisual RTE, you can do so through “Settings” or “Control Panel” of Windows.

On Windows 10/11, go to Settings > Applications > Applications and Features, find Xvisual RTE, and click “Uninstall”.

In earlier versions of Windows, you can find Xvisual RTE through “Control Panel” > “Programs and Features”, and choose to uninstall it.

3. Uninstall RTE on Unbutu

If you want to uninstall Xvisual RTE, you can do so through the supplied script `uninstall.sh`. The uninstallation of script will delete Xvisual RTE. Execute the following command:

```
Bash
sudo bash ./uninstall.sh
```

The uninstall script will delete `/opt/scada` and the services and configuration files related to Xvisual RTE. If you need to retain your project files or database files, please back them up earlier.

After the uninstallation is complete, NET 8 Runtime remains in the system because it may have been installed for other applications. If they are used only for Xvisual RTE, you can manually uninstall these dependencies:

```
Bash  
sudo apt remove dotnet
```

Quick operation guide

I. Preliminary preparation

1. System and permission checks

- Ensure the operating system meets requirements (Windows Server 2016/2019/2022, Windows 10/11, or Linux-Ubuntu)
- .NET 8 Runtime has been installed (IDE/RTE dependency)

2. Download the software package

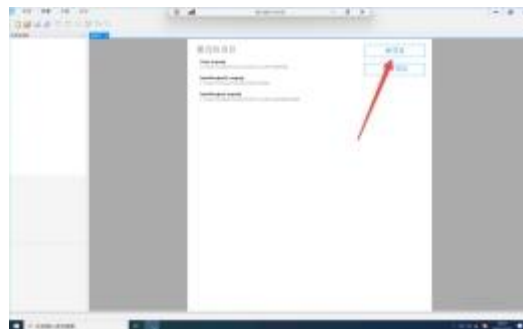
- Go to the onlinehelp official website of SINSEGYE ([OnlineHelp of SINSEGYE](#)) to download the latest version of the Xvisual IDE and RTE installation packages

II. Creating and Configuring Projects

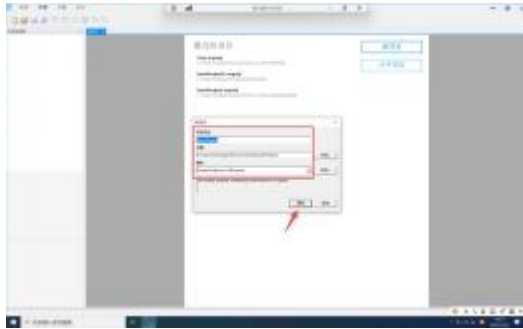
1



2



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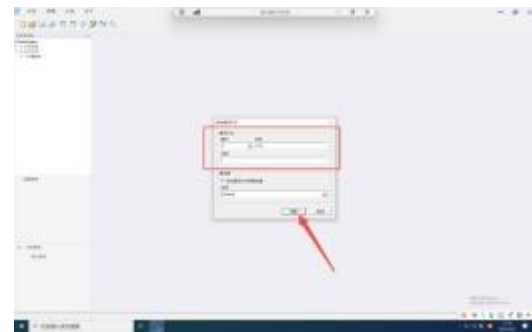
III. Core Configuration Process

1. Create a communication node

1



2



2. Create a device

1



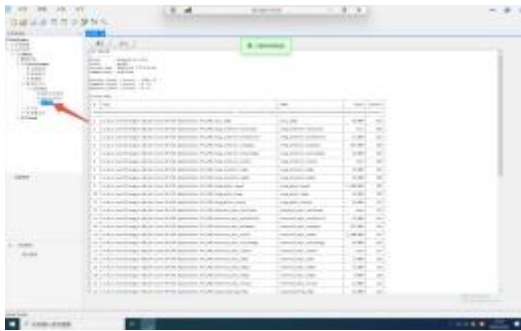
2



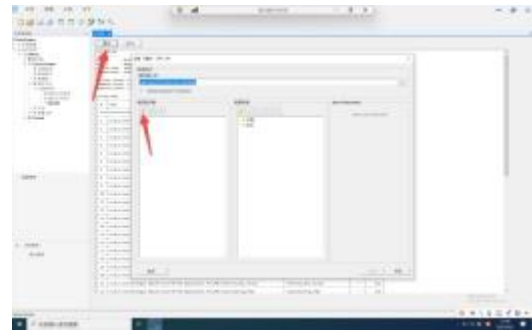
3. Create a point location (Tag)

1

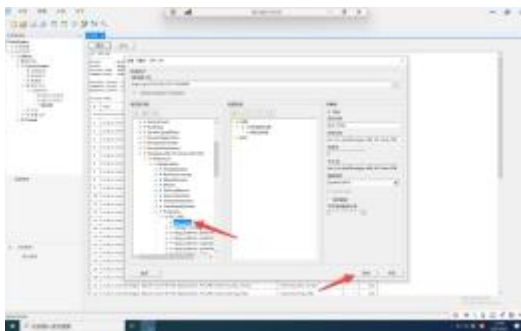
2



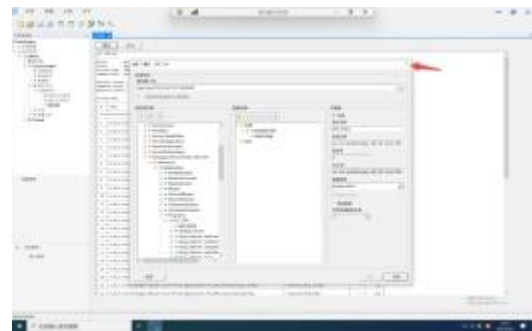
3



4



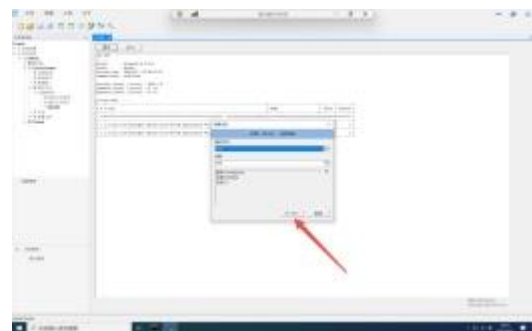
5



6



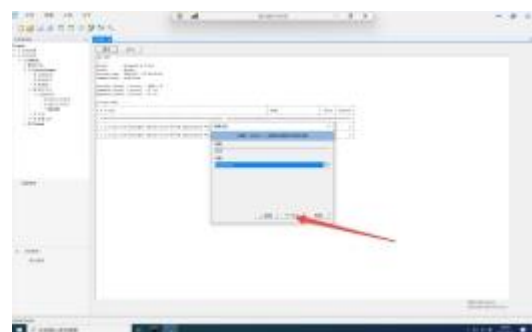
7



8



9

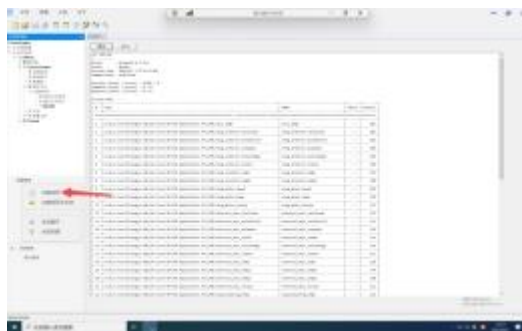


10

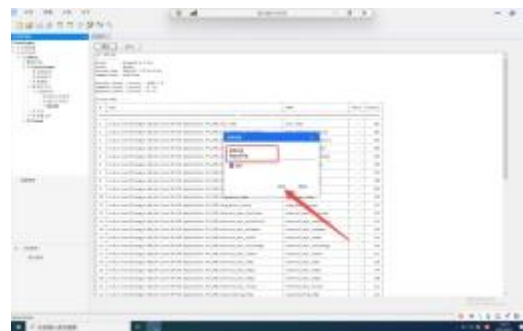


IV. Creating a View

1



2



Right click

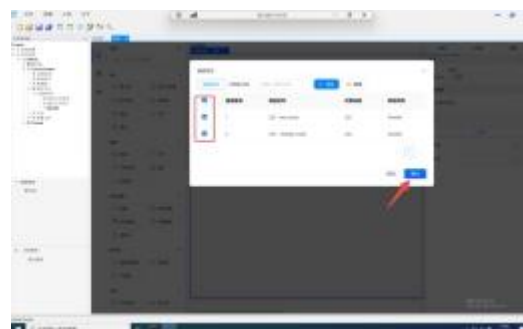
V. Read-write Channel Binding

Channel binding can interact with point locations in Xvisual and PLC

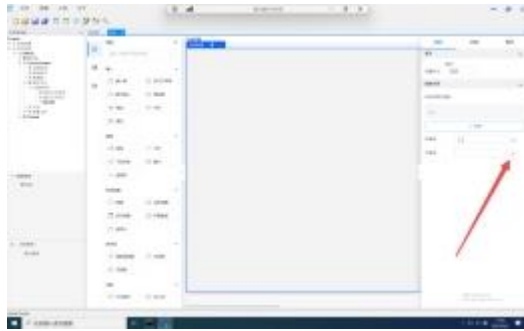
1



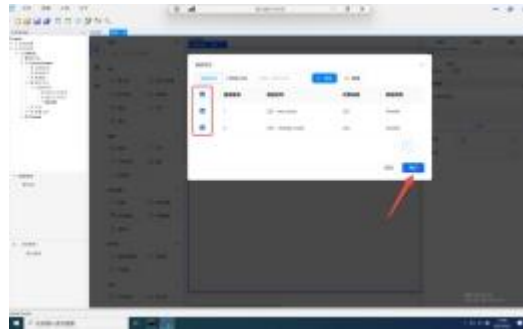
2



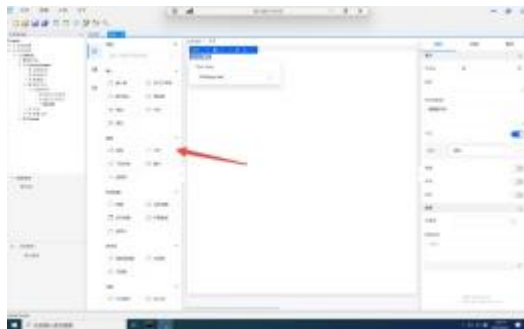
3



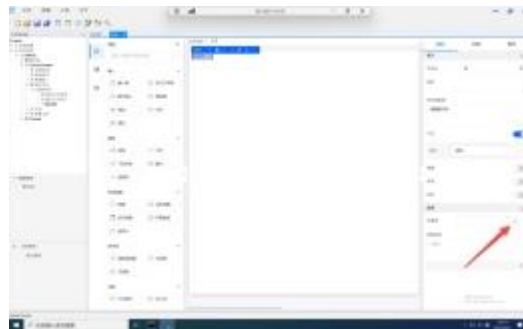
4



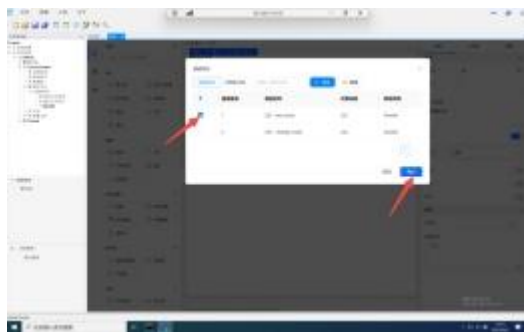
5



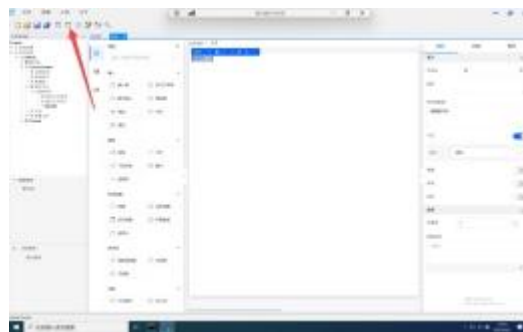
6



7



8



9



10



11

Enter the IP:80 of the publishing device
(e.g. 192.168.110.158:80)

12



VI. Data Display and Conversion

The data read from PLC can be converted into another format

1



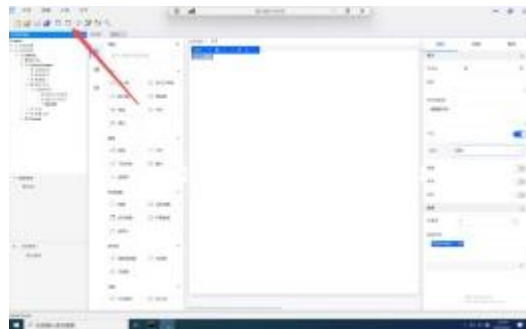
2



3



4

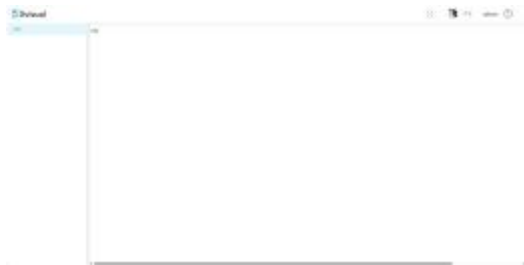


5

6

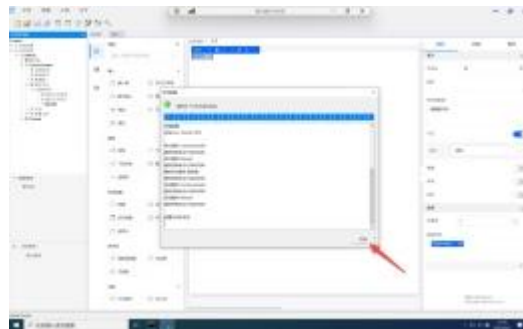


7



Enter the IP:80 of the publishing device
(e.g. 192.168.110.158:80)

8



VII. Data Interception

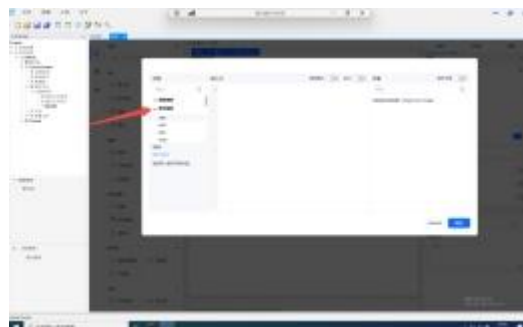
The data obtained from the PLC can be intercepted (demonstrated as follows: keep two decimal places)

1



3

2



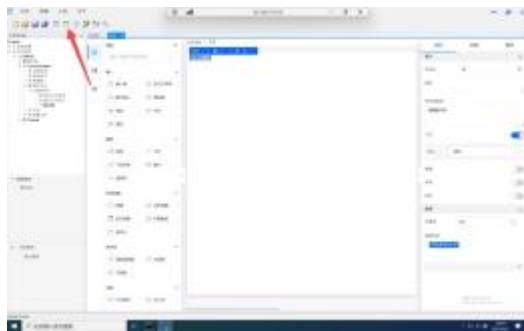
4



5



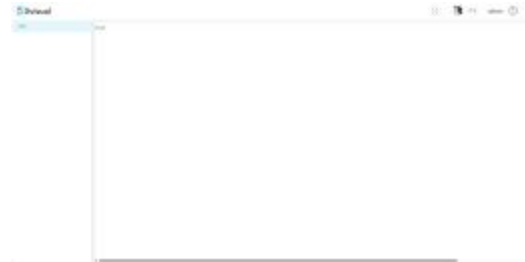
6



7

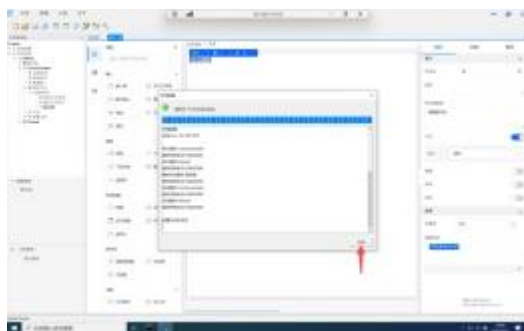


8



Enter the IP:80 of the publishing device
(e.g. 192.168.110.158:80)

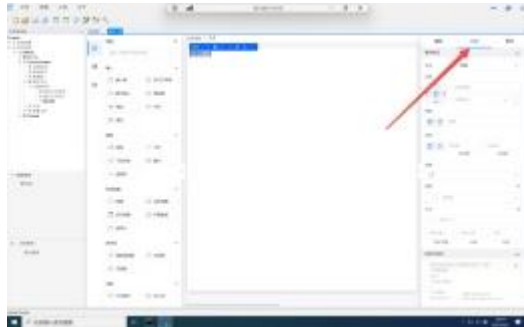
9



VIII. Setting Events

Setting events is a core function of Xvisual, and it can be designed on demand to perform a variety of operations (demonstrated as below: background change)

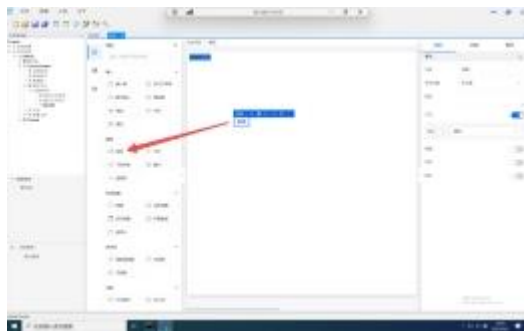
1



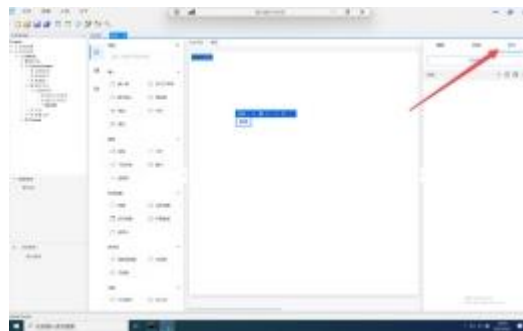
2



3



4



5



6



7



8





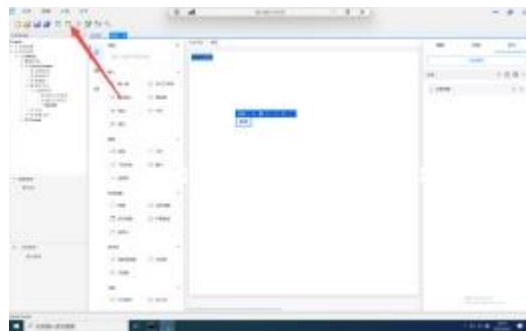
9



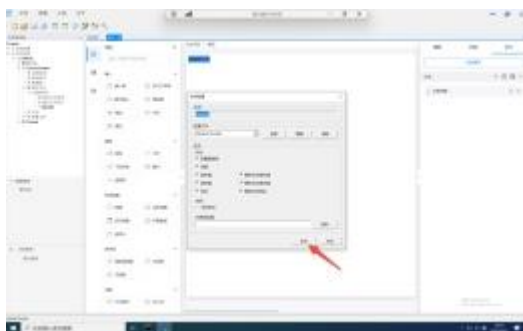
10



11



12



13

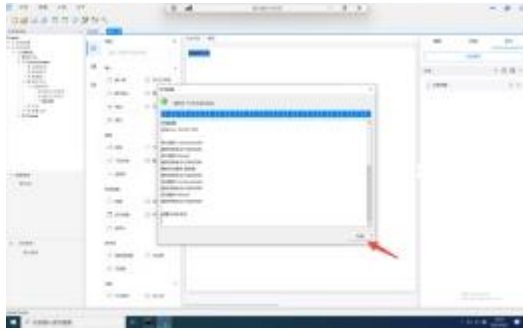


14



Enter the IP:80 of the publishing device
(e.g. 192.168.110.158:80)

15



IX. Presentation Document

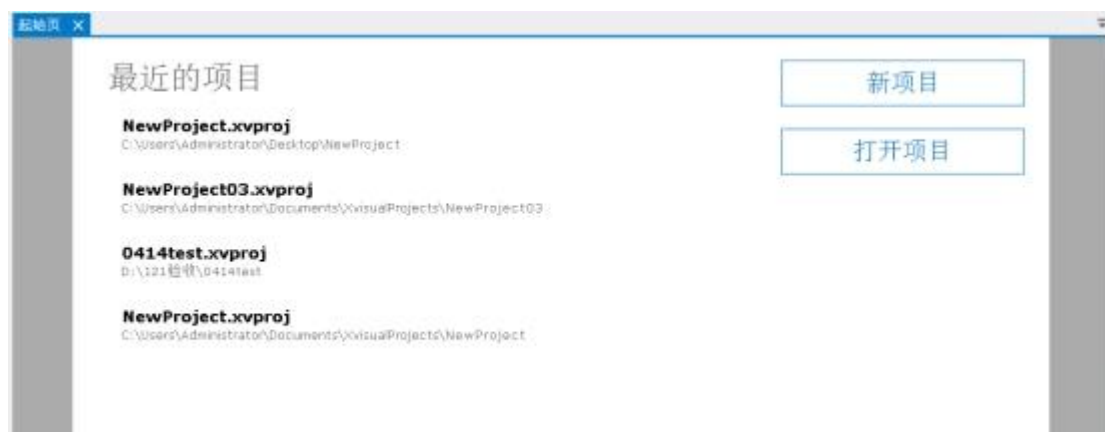
Link to download the presentation document:

Function introduction

1. Project basis

1.1 Project creation

Xvisual is configured on a project basis. A project is a set of different formats of files stored in the project directory. To create and edit a project, please use the Administrator application. When Administrator starts, the “Start Page” will be opened, with a button to create a new project or open an existing project.



Input the name of the project to be added in the pop-up window for creating a new project and select the location where the project is created; Templates define the initial

configuration added to the project, and you can select another existing project as a template for creation of a new project.

新项目

项目名称

NewProject

位置

D:\Xvisual测试\

浏览...

模板

EmptyProject.en-GB.xvproj

浏览...

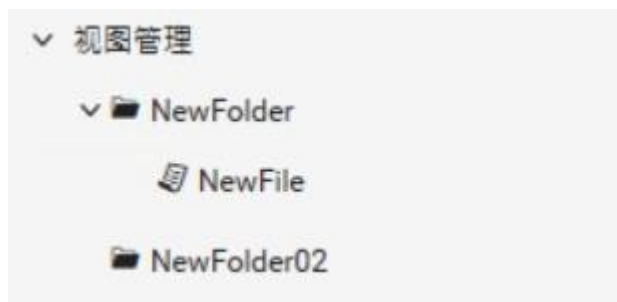
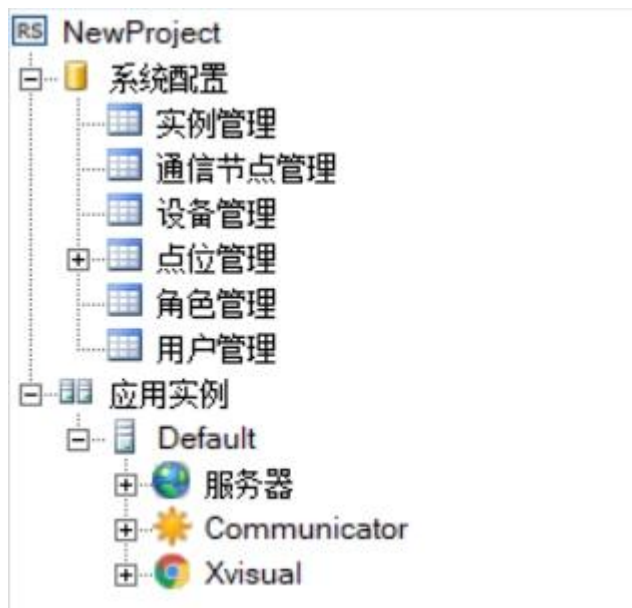
An empty project containing dictionaries in English.

确定 取消

1.2 Project structure

The structure of the Xvisual project is displayed in the Project Explorer, which is located on the left side of the main Administrator window. The project includes the following main parts:

- System configuration, which is used to manage all users and roles as well as all communication nodes, devices, and point locations for system communications
- Application instance, which is used to manage the actually connected device instance of the system
- View management, which is used to manage all component views of the system
- Resource management, including picture management



1.3 General configuration sequence of the project

When starting to use Xvisual, it is recommended that you follow the general configuration sequence described below. After gaining some experience to better understand the dependency relationship between applications, you can change the sequence to improve efficiency.

1.3.1 Creating a project

Create a new project or open an existing project in the Administrator application.



1.3.2 Creating a communication node

After creating or reopening a project, you can use the toolbar or wizard to add a communication node.

4. Use the toolbar to add a communication node.



5. Use the wizard to add a communication node.



6. Add a communication node setting interface.

添加通信节点
✕

通信节点

编号

2

名称

test0401

说明

通信器

☒ 添加通信节点到通信器

实例

Default

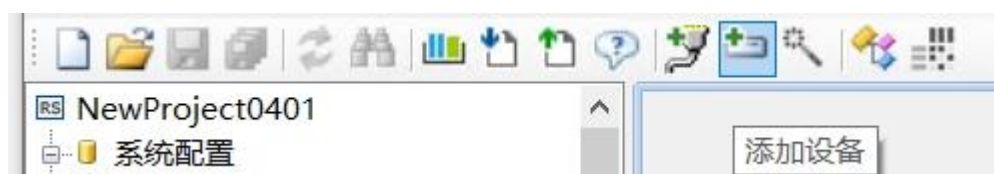
确定

取消

1.3.3 Creating a device

After establishing the communication node, you can use the toolbar or wizard to add a device.

1. Use the toolbar to add a device.



2. Use the wizard to add a device.



3. Add a device setting interface to set the device name and select the device type and the communication node.

添加设备×

设备

编号

1

名称

100.100

代码

设备类型

OPC UA

数字地址

字符串地址或者主机名

通信节点

0402test

说明

通信器

☒ 将设备添加到通信器

实例

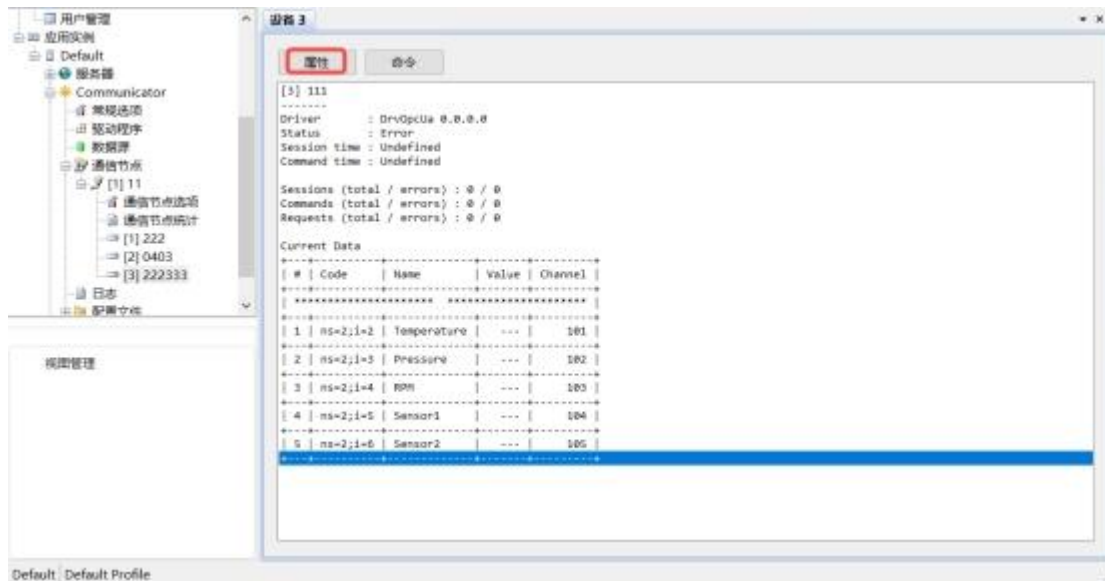
Default

确定

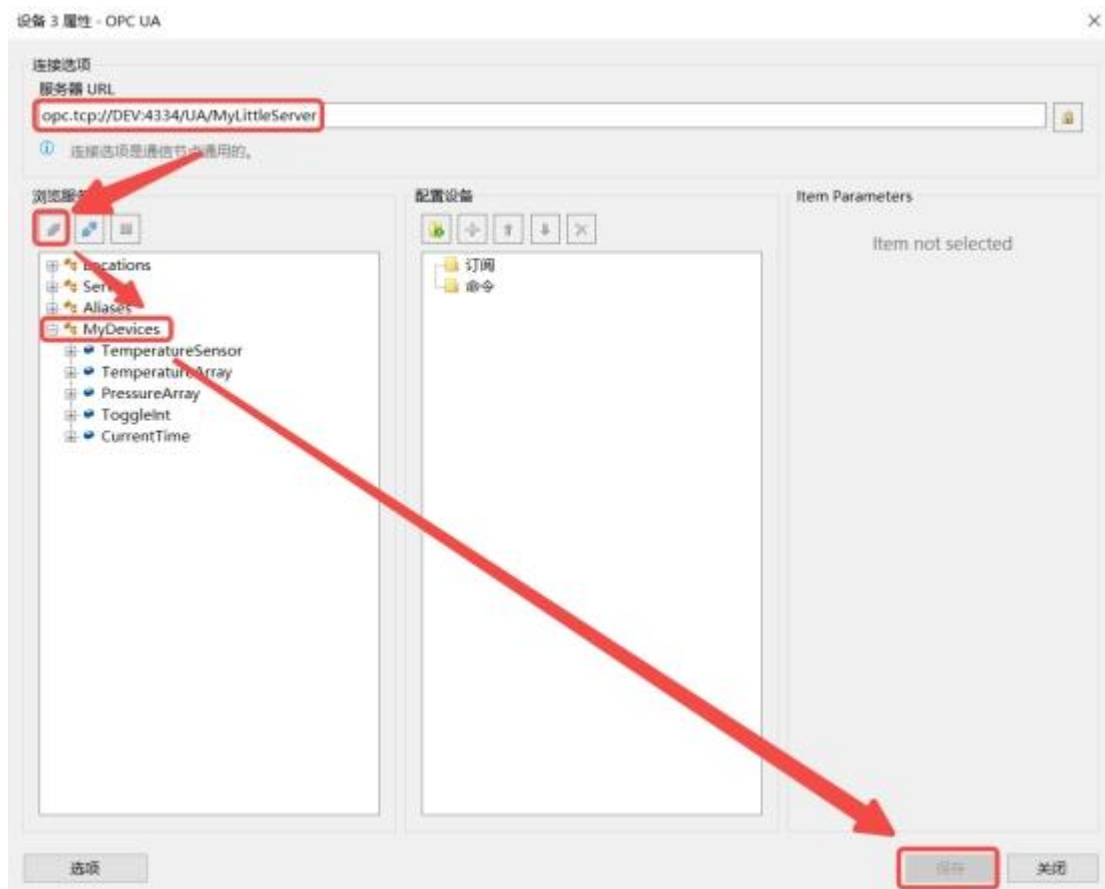
取消

The image shows a software window titled "添加设备" (Add Device) with a close button (X) in the top right corner. The window is divided into two main sections: "设备" (Device) and "通信器" (Communicator). The "设备" section contains several input fields: "编号" (ID) with a spinner box set to 1, "名称" (Name) with a text box containing "100.100", "代码" (Code) with an empty text box, "设备类型" (Device Type) with a dropdown menu showing "OPC UA", "数字地址" (Digital Address) with an empty text box, "字符串地址或者主机名" (String Address or Hostname) with an empty text box, "通信节点" (Communication Node) with a dropdown menu showing "0402test", and "说明" (Description) with an empty text box. The "通信器" section contains a checked checkbox labeled "将设备添加到通信器" (Add device to communicator) and a dropdown menu labeled "实例" (Instance) showing "Default". At the bottom right, there are two buttons: "确定" (OK) and "取消" (Cancel).

4. The device property description interface displays the device status and some recent data; Click the “Property” button to enter the detailed property setting interface.



5. Add an OPC address, add a corresponding URL in the server URL, and then click the “Connect” button. If the connection succeeds, the devices that can transmit data will be displayed. Click the device you want to connect and click “Save”; If the connection fails, you should check the server URL for problems.



1.3.4 Creating a point location

After establishing the device, you can use the toolbar or wizard to add a point location.

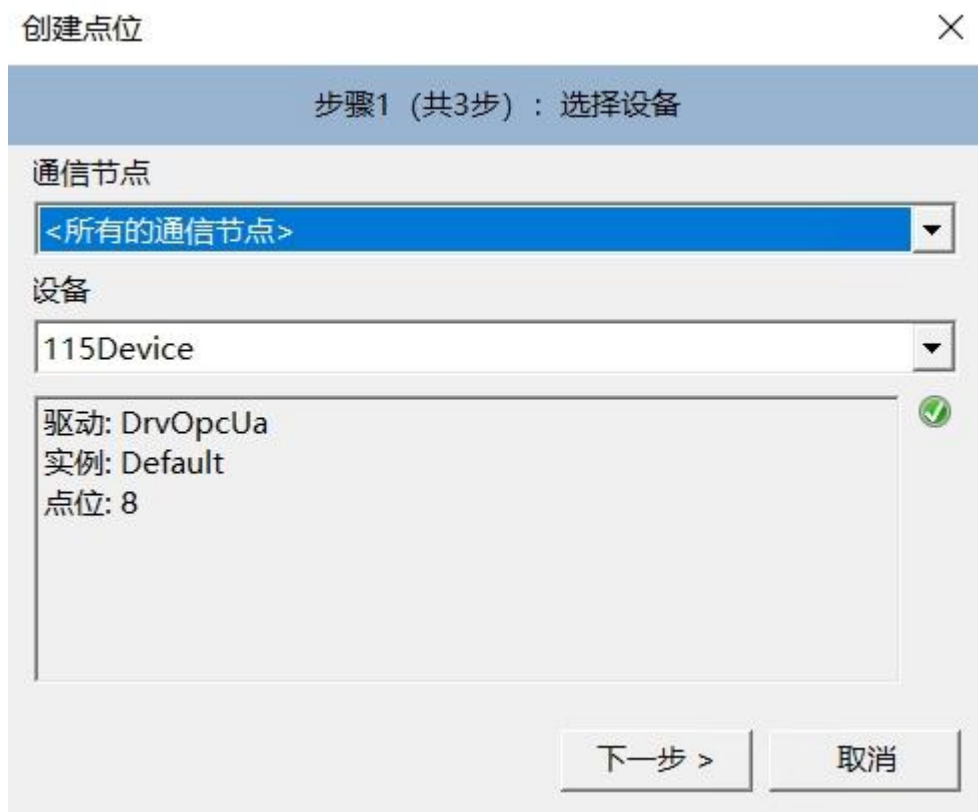
1. Use the toolbar to add a point location.



2. Use the wizard to add a point location.



3. After adding the point location, a pop-up window for creating a point location will appear. Step 1 is set to select a device. You need to select the communication node and device where the new point location is added newly.



4. Step 2 of creating a point location is to select the object of the newly created point location, where the device is the one checked in Step 1. All objects can be viewed in the instance management.

创建点位

步骤2 (共3步) : 选择新创建的点位的对象

设备

115Device

对象

Enterprise

< 返回

下一步 >

取消

5. Step 3 of creating a point location is to assign a point location number, where the device is the one checked in Step 1, and the point location is the point location information that can be transmitted on the device.

创建点位

步骤3 (共3步) : 分配点位号

设备

115Device

点位号

开始

101

结束

108

映射

重置

☒ 预览

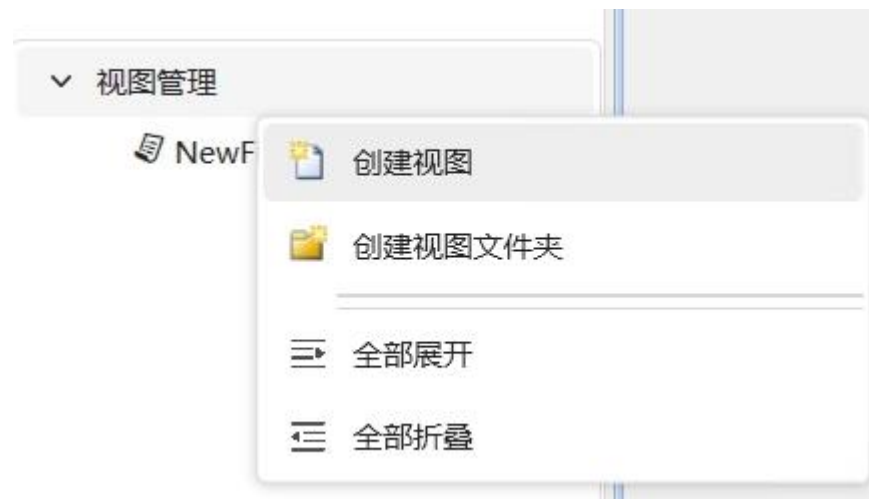
< 返回

创建

取消

1.3.5 Creating a view

After the communication node, the device and the point location are created, a view folder and a view can be created to obtain richer data information; You can also create and edit the view directly after the project is created.



1.3.6 Publishing configuration

1. After completing the above configuration, you can select the “Publish” button in the bar to publish the configured project.



2. In the “Publish Configuration” interface, you can set the instance name and related options.

发布配置

×

实例

Default

配置文件

Default Profile

创建

编辑

删除

选项

包含:

☒ 配置数据库

☒ 视图

☒ 服务器

☒ 通信器

☒ 网站

☒ 重新启动服务器

☒ 重新启动通信器

☒ 重新启动网站

名称:

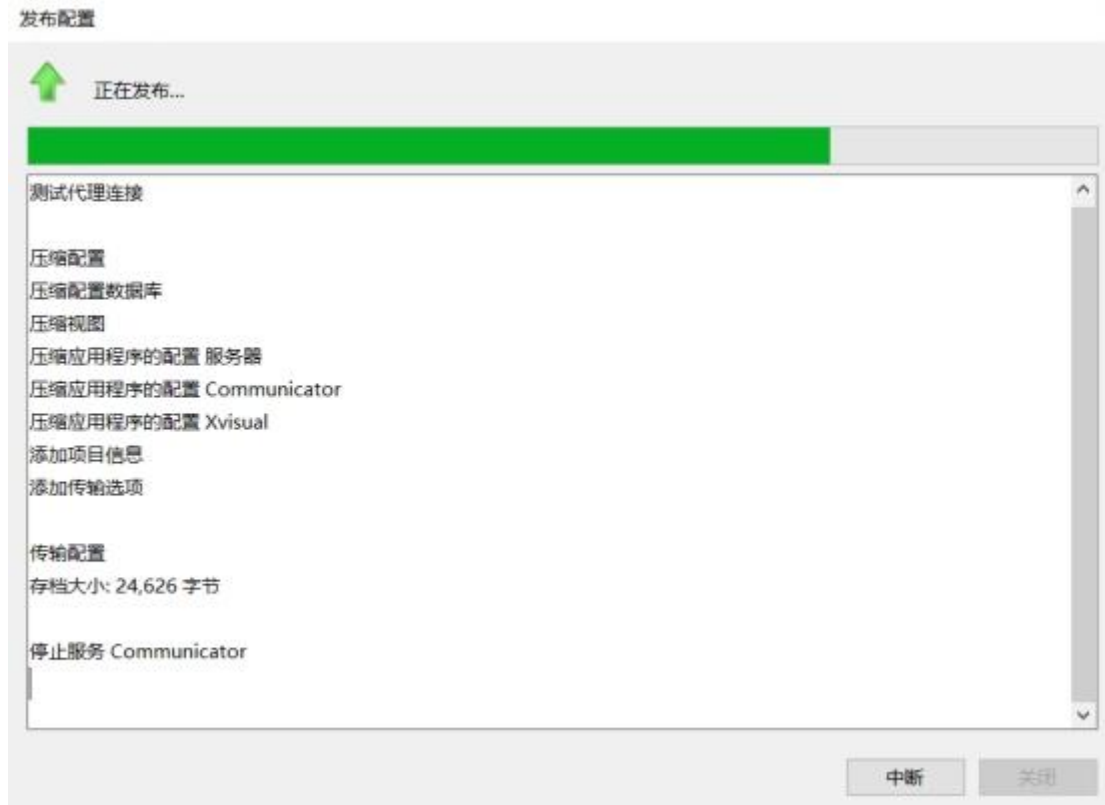
☐ 注册表项

对象筛选器:

选择...

发布

关闭



Explanation:

An instance is a set of Xvisual applications which are installed on the same computer and run the same project.

A single project can contain multiple instances of swap data. The Administrator application can be connected to a remote server, so configuration and control of Xvisual can be done by one engineer using one workstation.

2. Communication node

2.1 Creating a communication node

1. Use the toolbar to add a communication node.



2. Use the wizard to add a communication node.



3. Add a communication node setting interface.

添加通信节点

通信节点

编号 名称

2 test0401

说明

通信器

☒ 添加通信节点到通信器

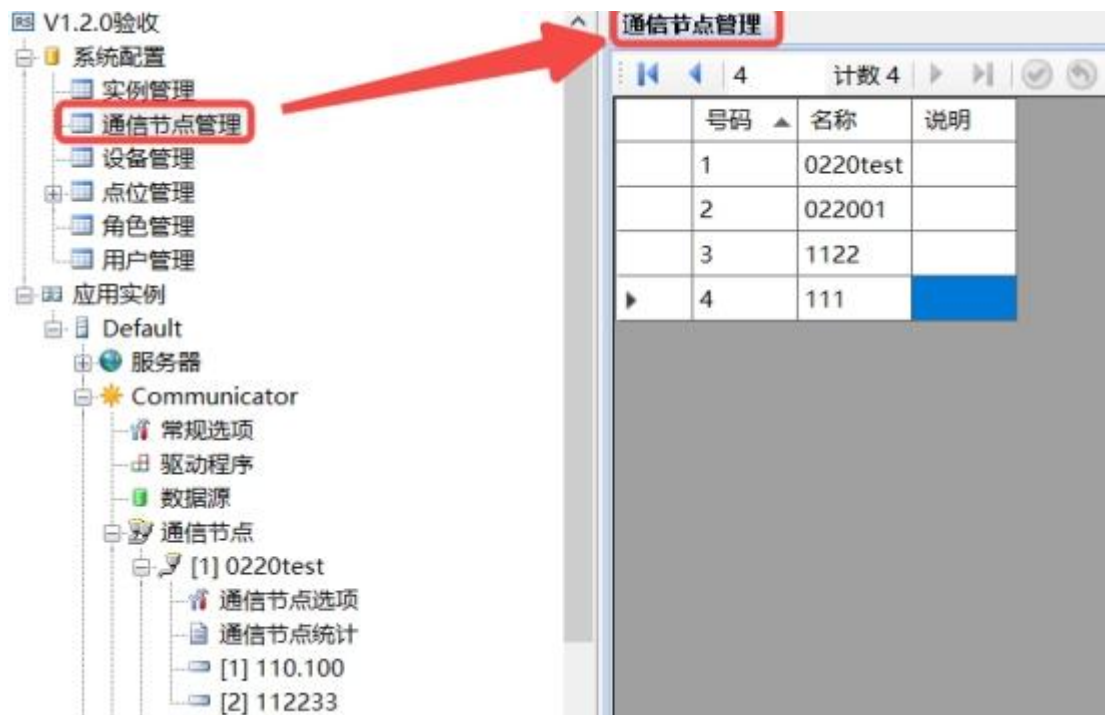
实例

Default

确定 取消

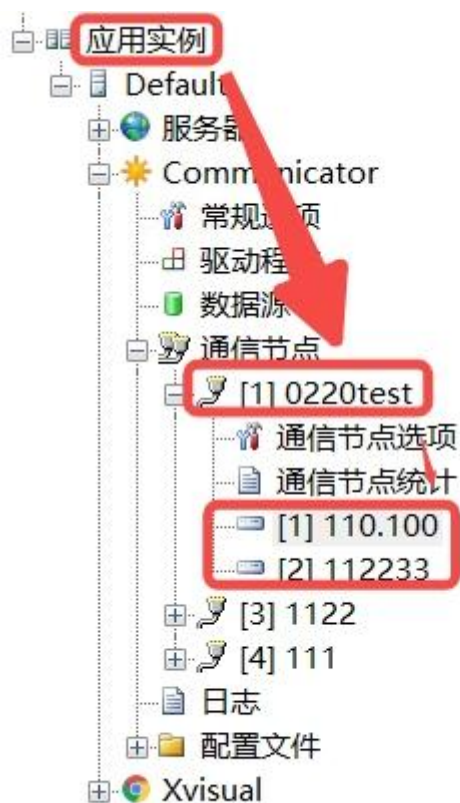
2.2 Communication node management

Click "Communication Node Management" in the system settings to view all existing communication nodes.



2.3 Device management of communication nodes

Click "Communication Node" in the application instance to view the devices under each communication node.

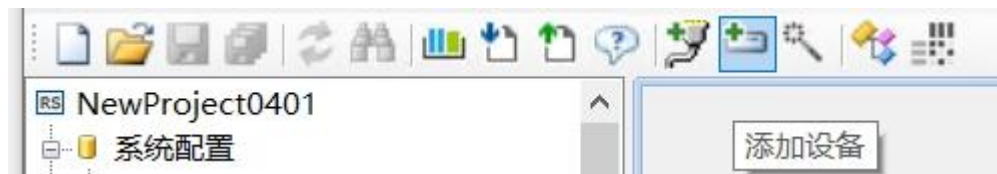


3. Device

3.1 Creating a communication device

After establishing the communication node, you can use the toolbar or wizard to add a device.

1. Use the toolbar to add a device.



2. Use the wizard to add a device.



3. Add a device setting interface.

添加设备



设备

编号

1

名称

100.100

代码

设备类型

OPC UA

数字地址

字符串地址或者主机名

通信节点

0402test

说明

通信器

☒ 将设备添加到通信器

实例

Default

确定

取消

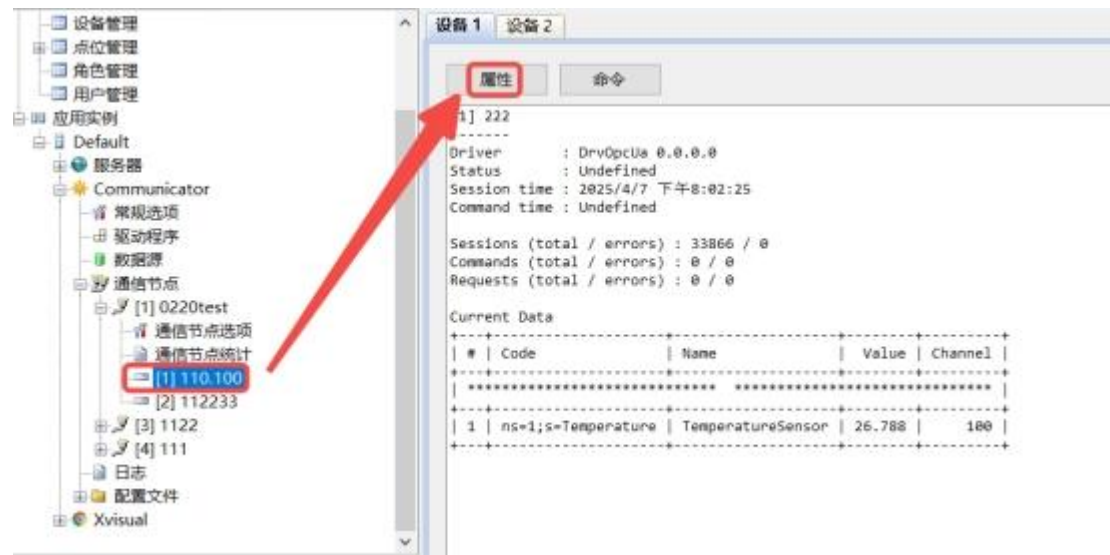
3.2 Device management



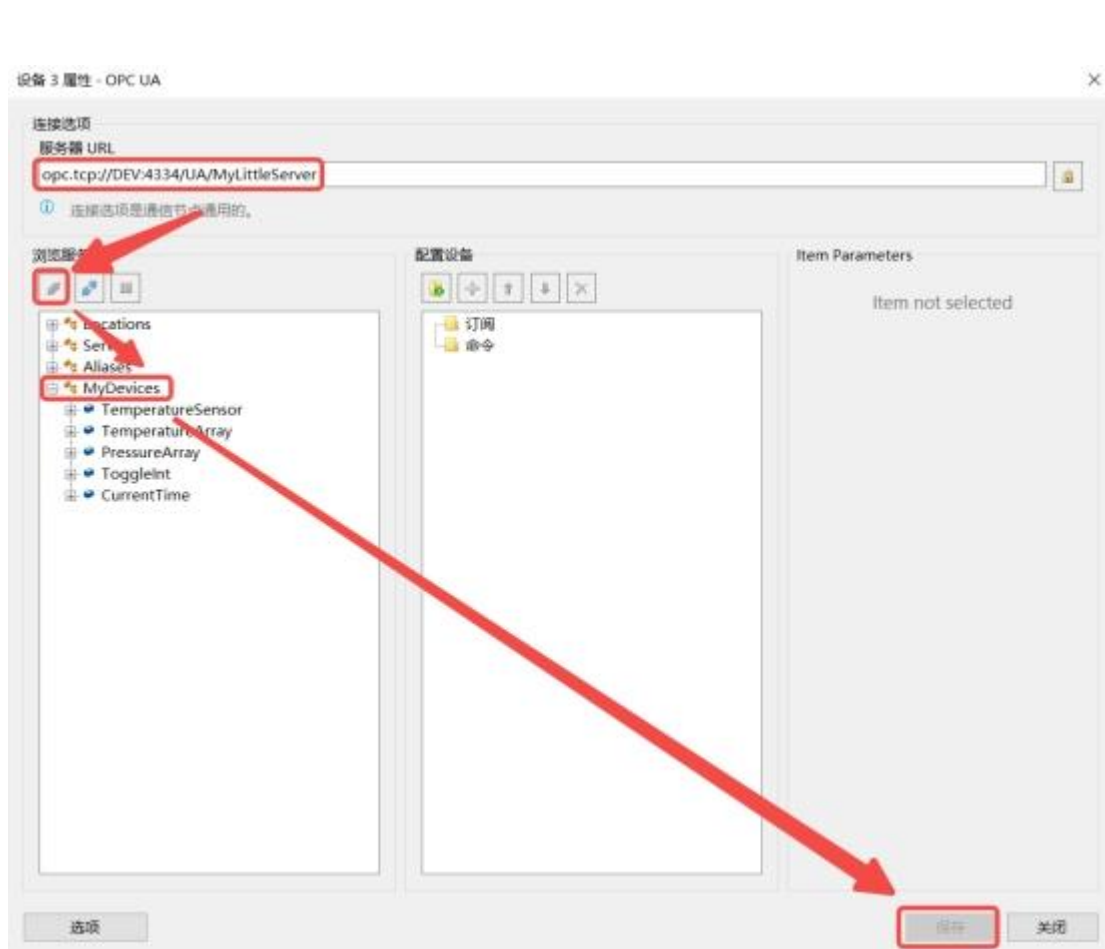
3.3 Configuration properties

3.3.1 OPCUA device

1. In the application instance, click the OPCUA device under "Communication Node" to view the basic parameters of the device. Click the "Property" button to enter the property setting interface.

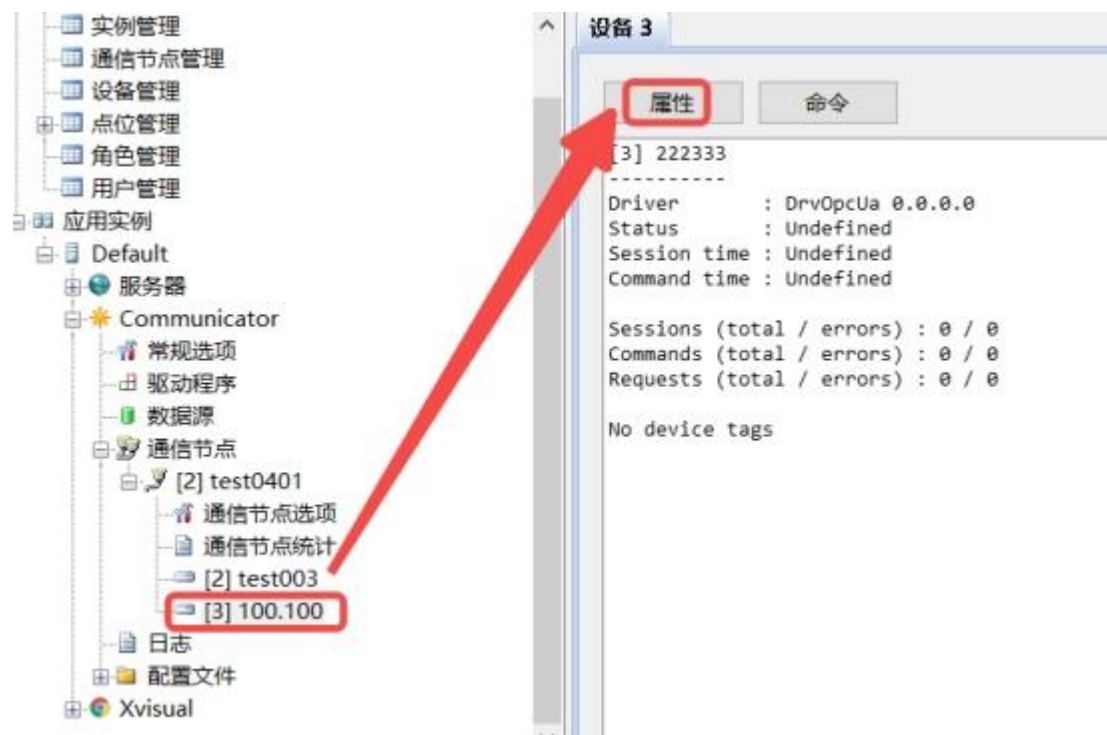


2. Input the server URL in the property setting interface, click the "Connect" button, select the device to be monitored, and click the "Save" button to save the data.

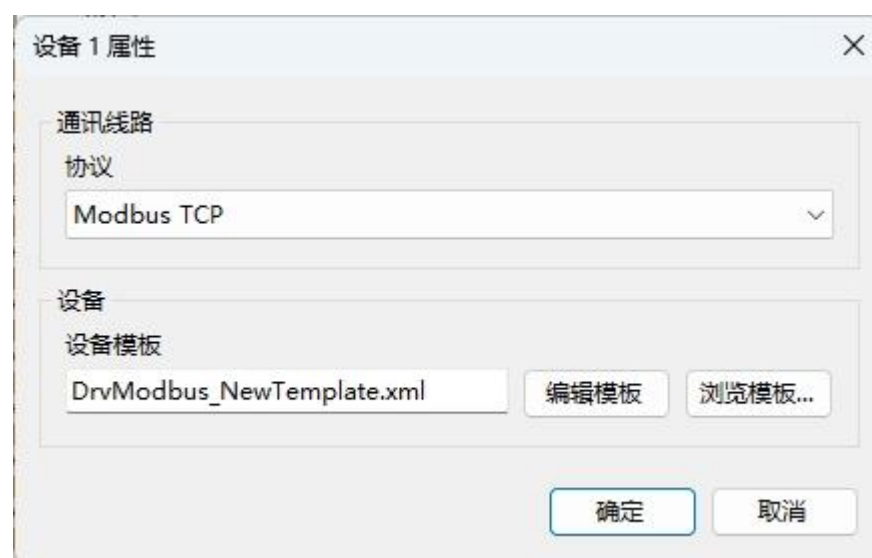


3.3.2 MODBUS device

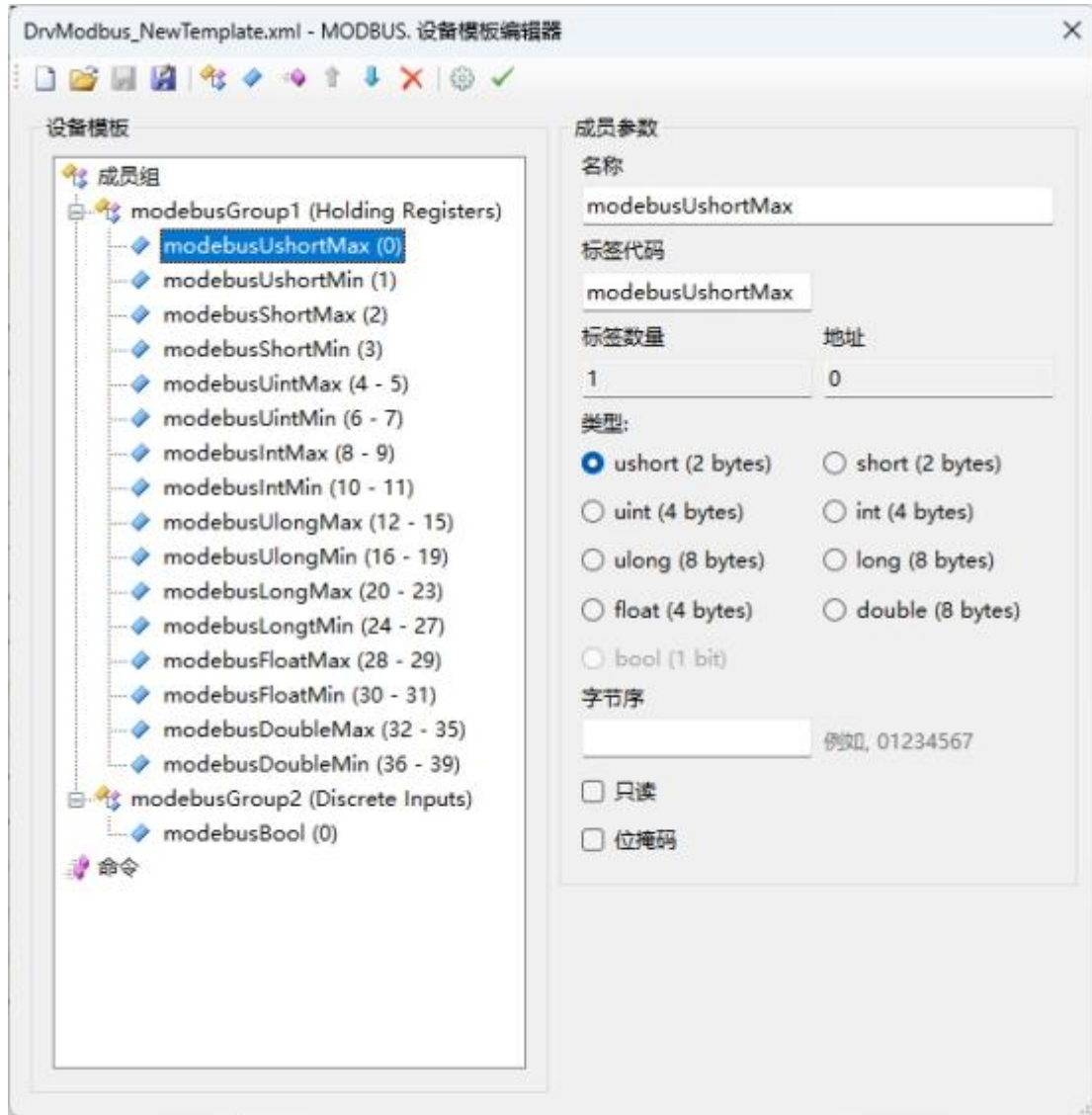
1. In the application instance, click the MODBUS device under "Communication Node" to view the basic parameters of the device. Click the "Property" button to enter the property setting interface.



2. Select Modbus TCP protocol and edit the template.



3. Add members in the MODBUS Device Template Editor.



4. Point location

A point location is an entity with number and name, and it is related to both of current and historical data and events. Synonyms in other systems: variable and tag.

4.1 Creating a point location

After establishing the device, you can use the toolbar or wizard to add a point location.

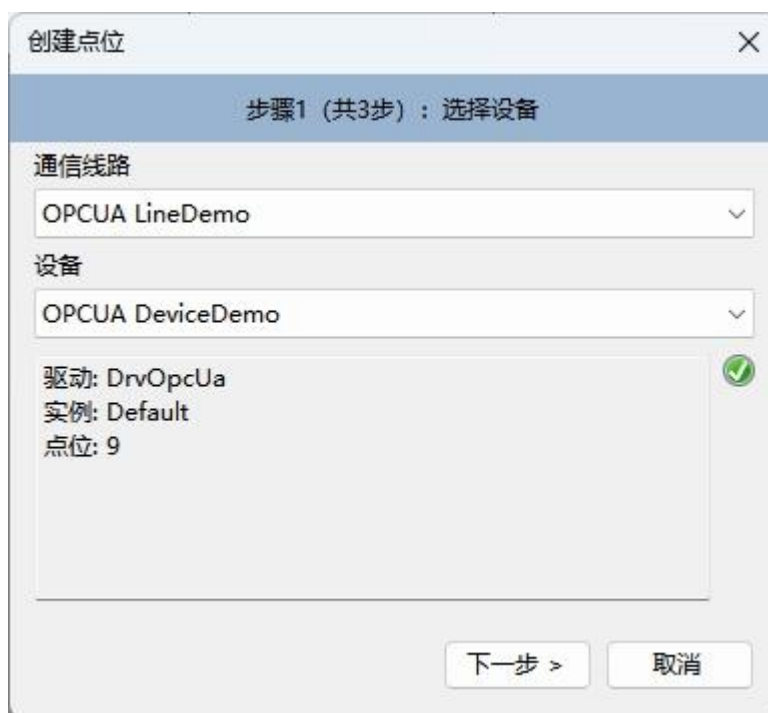
1. Use the toolbar to add a point location.



2. Use the wizard to add a point location.



3. After adding the point location, a pop-up window for creating a point location will appear. Step 1 is set to select a device. You need to select the communication node and device where the new point location is added newly.



4. Step 2 of creating a point location is to select the object of the newly created point location, where the device is the one checked in Step 1. All objects can be viewed in the instance management.

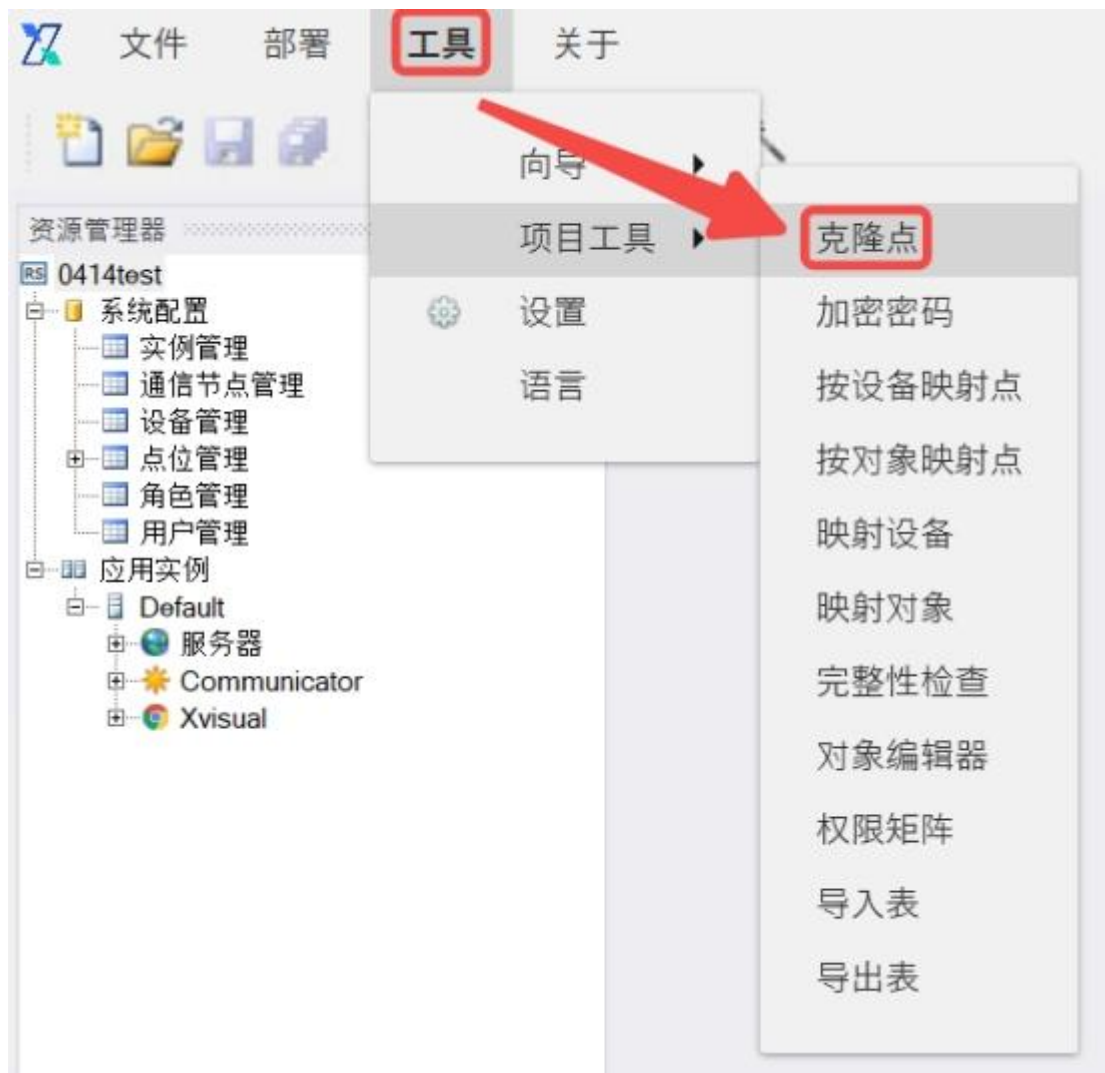
The dialog box is titled '创建点位' (Create Point Location) and shows '步骤2 (共3步) : 选择新创建的点位的对象' (Step 2 of 3: Select the object for the newly created point location). It contains two dropdown menus: '设备' (Device) with 'OPCUA DeviceDemo' selected, and '对象' (Object) with 'Enterprise' selected. At the bottom are three buttons: '< 返回' (Return), '下一步 >' (Next), and '取消' (Cancel).

5. Step 3 of creating a point location is to assign a point location number, where the device is the one checked in Step 1, and the point location is the point location information that can be transmitted on the device.

The dialog box is titled '创建点位' (Create Point Location) and shows '步骤3 (共3步) : 分配点位号' (Step 3 of 3: Assign point location number). It contains the same '设备' (Device) dropdown with 'OPCUA DeviceDemo' selected. Below it is a section for '点位号' (Point Location Number) with '开始' (Start) and '结束' (End) labels. The '开始' field contains 'p01' and the '结束' field contains '909'. To the right of these fields are two buttons: '映射' (Map) and '重置' (Reset). At the bottom left is a checked checkbox labeled '预览' (Preview). At the bottom right are three buttons: '< 返回' (Return), '创建' (Create), and '取消' (Cancel).

4.2 Cloning a point location

Cloning a point location allows you to quickly create a similar point location based on the existing point locations. You can access the cloning tool from Tools > Project Tools menu in the Administrator application.



You can specify the source and destination point location number on the clone form, or select new objects and devices for the cloned point locations.

克隆点位
×

源点位编号

开始

结束

1

1

目标点位编号

开始

结束

1

1

选项

替换对象

<保持不变>

替换设备

<保持不变>

☐ 在公式里替换点位号

克隆

关闭

4.3 Setting of point location characteristics

Open the point location management, select the point location to be modified in the table, and then click the button or double-click the table row to open the point location property pop-up window. Edit the point location in the point location property sheet. Open the point location management, select the point location to be modified in the table, and click the button or double-click the table row.

- The generic page allows you to edit the name, data type, and data length of a point location. If no data type is specified, the Double type will be used. If a data type is specified and is greater than 1, the Server application will create several point locations with the series numbers. Point location types are described in a table with the same name. The tag code string field is used to connect point locations and device tags in the communicator.

点位属性

×

通用限制存档事件

☒ 激活

编号

1

Name

115Device - RPM

Code

数据类型

数据长度

点位类型

Input/output

对象

1

Enterprise

设备

1

115Device

标签号

标签代码

ns=2;i=4

确定取消

2. The restriction page allows you to set a restriction record to be used by one or more point locations. In the second case, the restriction is marked as shared. If the “bind to channels” option is enabled when the restriction is created, the actual value of the restriction is the point location number specified when the restriction is created.

点位属性

×

通用限制存档事件

限制

创建

☐ 仅显示共享限制

极低

低

高

极高

死区

确定

取消

3. The archive page allows you to specify the archive where the data are stored. If no archive is selected, i.e. the archive mask is empty or equal to 0, the point location data will be stored in the default archive.

点位属性

通用 | 限制 | 存档 | 事件

十进制数据

0

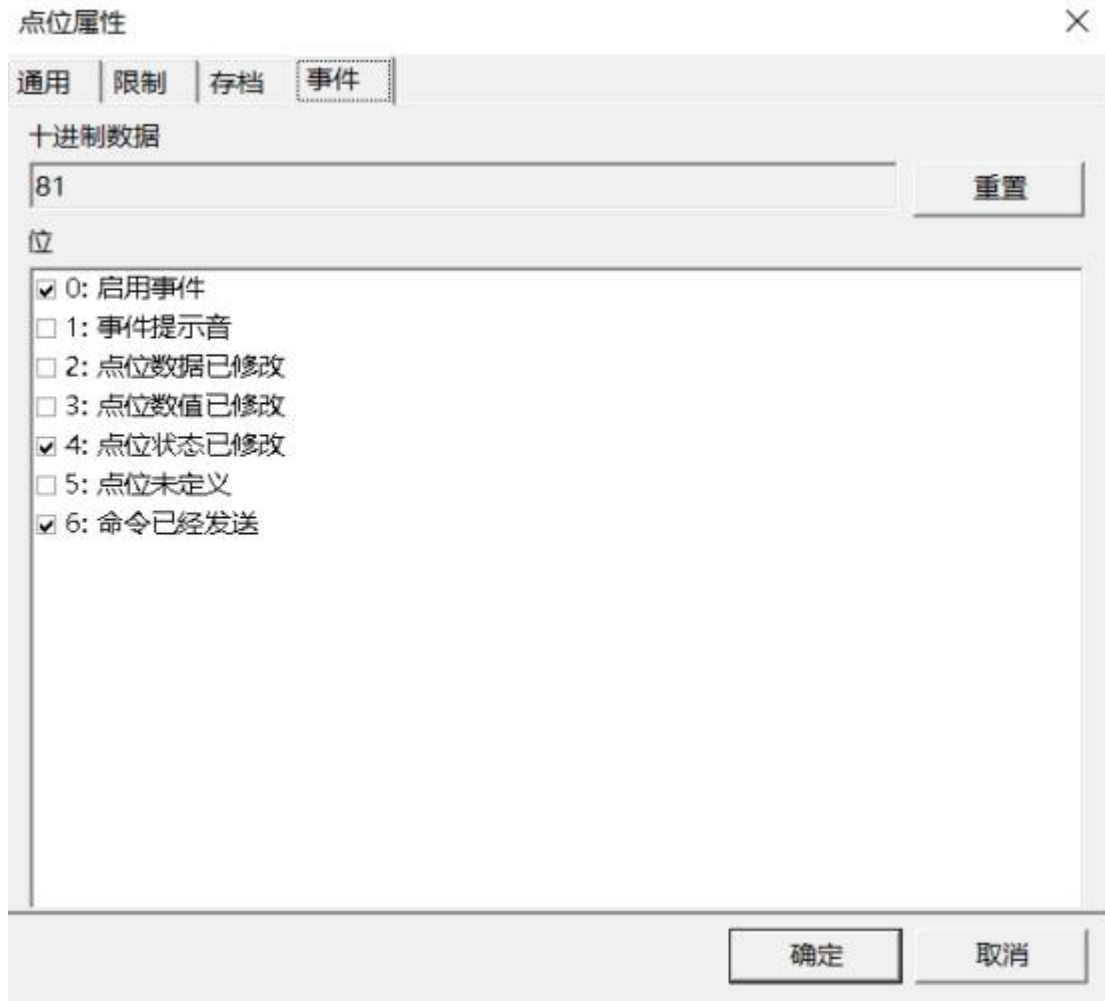
重置

位

- ☐ 0: Current data
- ☐ 1: Minute archive
- ☐ 2: Hourly archive
- ☐ 3: Daily archive
- ☐ 4: Event archive
- ☐ 5: Current data copy
- ☐ 6: Minute archive copy
- ☐ 7: Hourly archive copy
- ☐ 8: Daily archive copy
- ☐ 9: Event archive copy
- ☐ 16: Custom archive

确定 取消

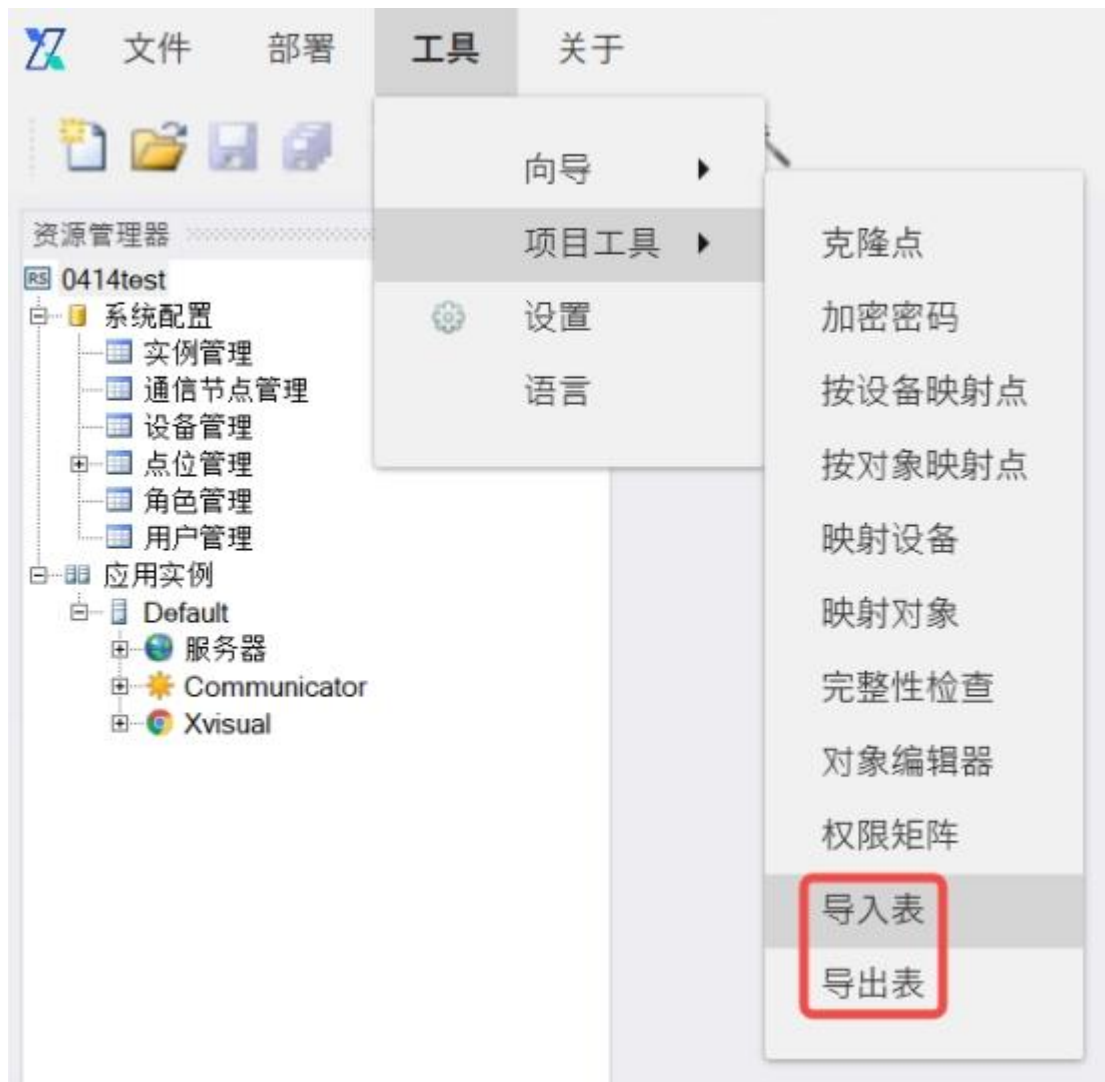
4. On the “Events” page, set the criteria for recording point location related events in the event archive.



4.4 Import and export tables

Tools for importing and exporting the configuration database tables make it easier to use the work done previously from other projects. You can access the import and export tools from Tools > Project Tools menu in the Administrator application. It supports DAT, XML, and CSV table formats. If you need to limit the range of imported and exported data, specify the start id and end id.

1. You can access the import and export tools from Tools > Project Tools menu in the Administrator application



2. Import table interface.

导入表

×

表

点位管理

源文件

选择...

源IDs

开始

☐ 0

结束

☐ 0

目标 IDs

开始

☐ 0

结束

0

导入

关闭

3. Export table interface.

导出表

×

表

点位管理

格式

Runtime tables (*.dat)

IDs

开始

☐ 0

结束

☐ 0

导出

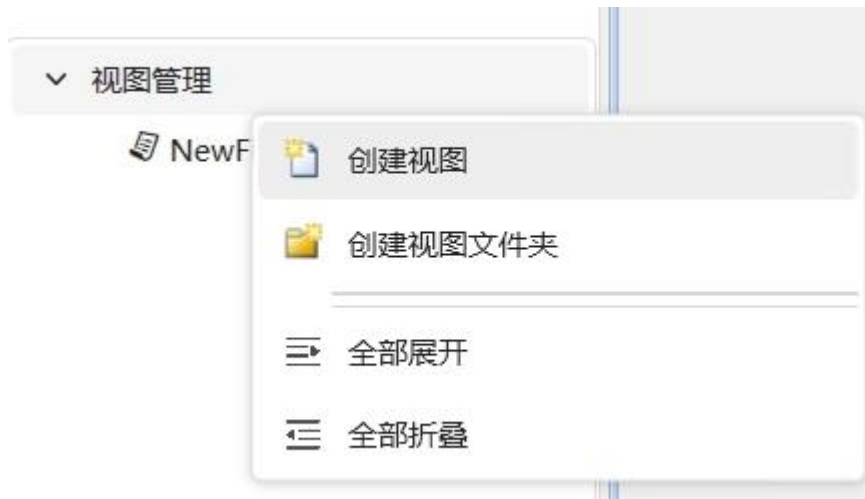
关闭

5. View

A view is a representation of data in the Webstation application.

5.1 Creating a view

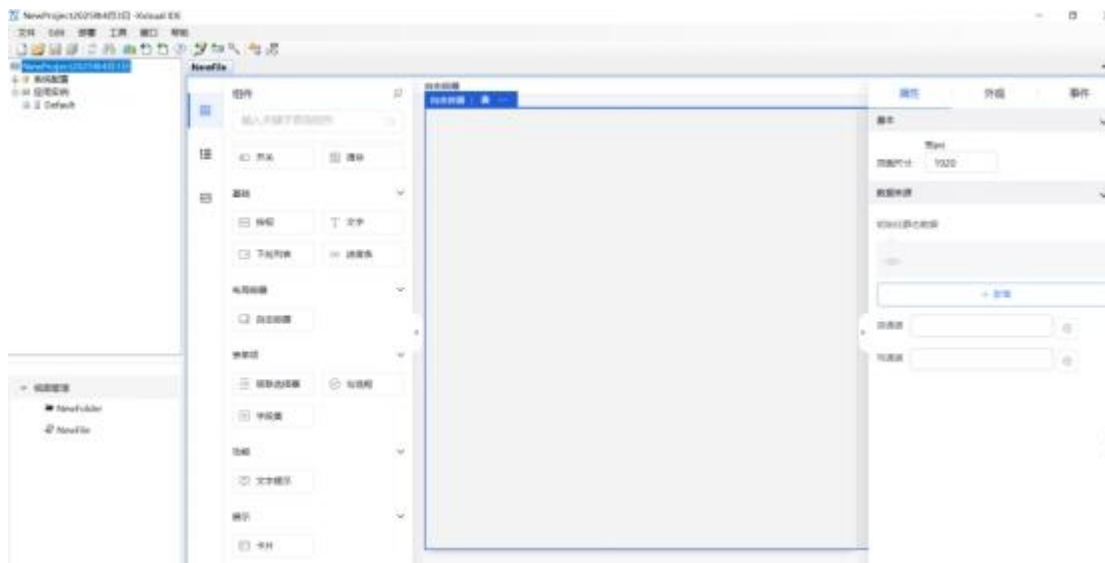
1. The view file is located in the “Views” section of the project. To create a file or folder in the “Views” section, right-click the Explorer node in the section and select “New file” or “New folder” from the context menu that appears.



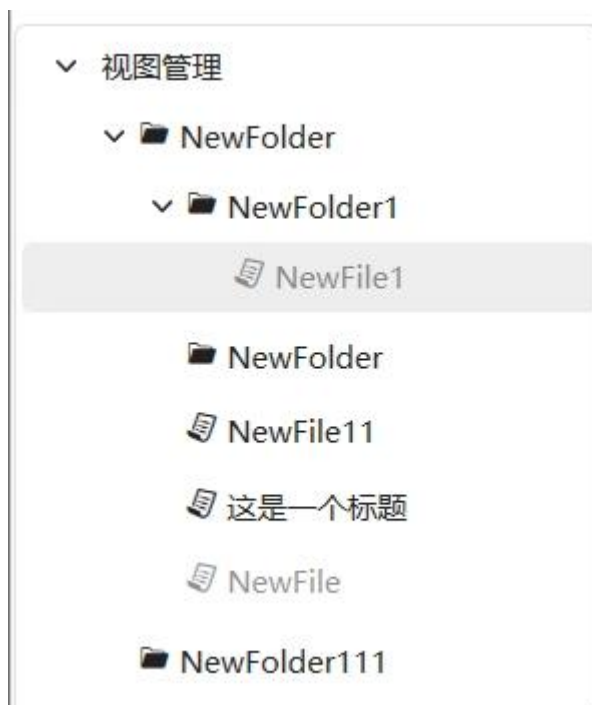
2. To create a file, right-click to select “New File” in the pop-up menu, enter the file name, and finally click “Save” in the pop-up “New” window. If the entered view name already exists, “File name already exists in the destination folder” will be prompted.



3. After saving successfully, the web-editor drawing area of the new page will be automatically displayed in the canvas area on the right side of the system. Enter the web-editor drawing area, and select the self-owned container component, of which the property information, appearance information and custom events are displayed in the right area.

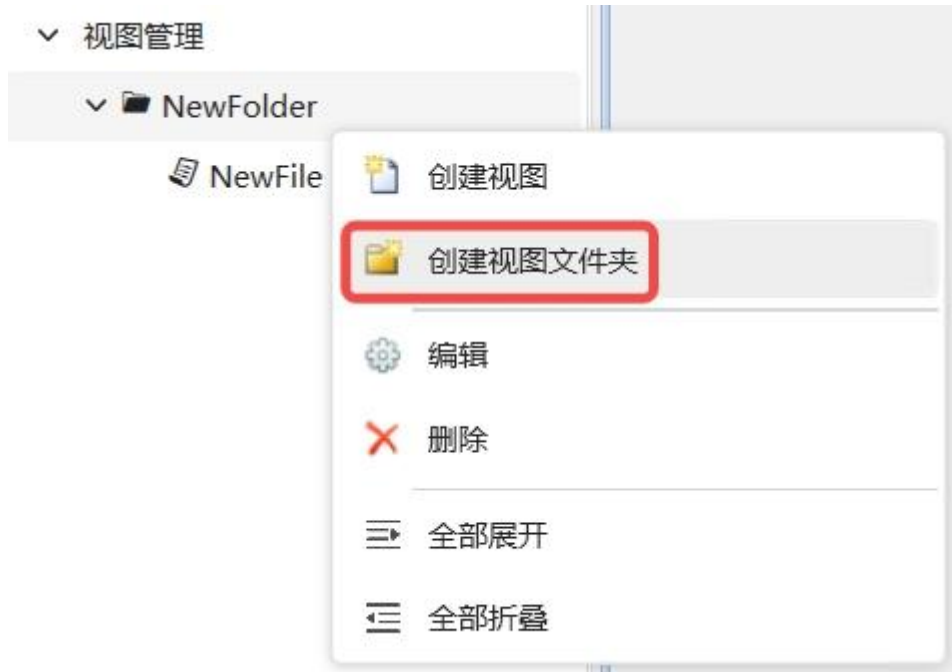


5.2 View file management



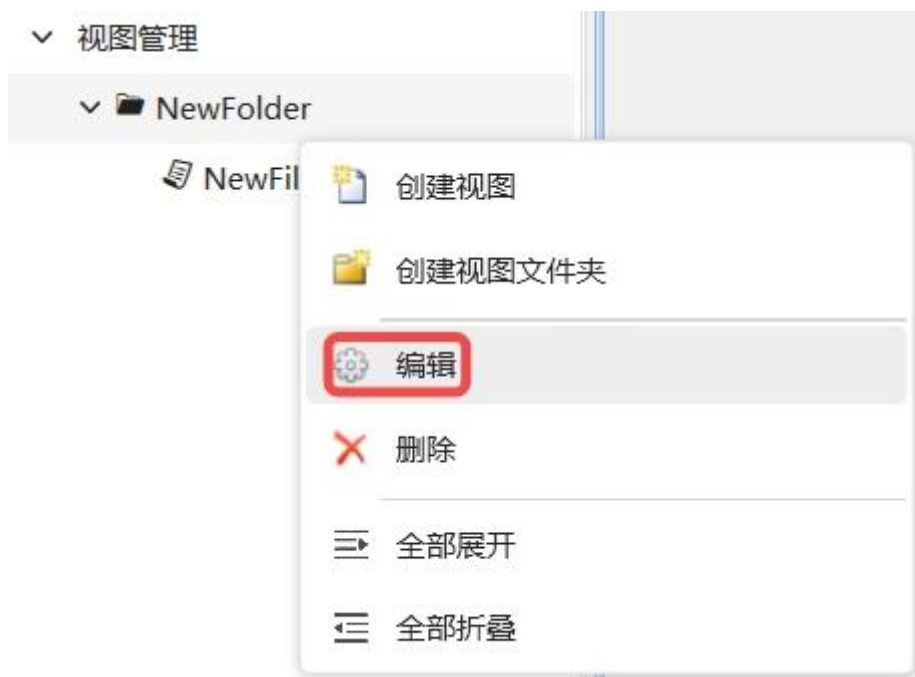
5.2.1 Folder addition

Select "New Folder" to pop up the view configuration window. The user can input the folder name, set the display property, and click Save.



5.2.2 Folder editing

1. To rename the view file relative to a view or view folder in the Views section of the project, right-click the Explorer node in this section and select “Edit” from the context menu that appears.



2. After selecting "Edit", the system will pop up the Rename window, where you can modify the file name and status and click “Save” to complete the adjustment of the view

name. It should be noted that the view you are drawing cannot be renamed because it is already open. You need to close it before renaming it.



5.3 Configuration Editor

5.3.1 Overall layout

XVisual page view – the Configuration Editor consists of three parts: component tree (left), content area (middle), and property bar (right).



5.3.2 Component tree

The Component Tree, located on the left side of the editor, displays the flat pattern by default, and you can also click the “Collapse” button. The Component Tree contains three parts: component list, view structure and source code.



5.3.2.1 Components

Component refers to the basic constituent unit used to draw drawings, and is an officially provided basic function library.

General operating procedures

Step 1: Drag the component to the drawing

1. In the Configuration Editor, click the component button on the left panel to switch to the component tab page.
2. Find the component to be drawn in the components, and then drag it to the canvas.

Step 2: Configure the component

1. Configure the properties: Select the component dragged to the canvas, and in the right panel, enter the component property information.
2. Configure the data binding: In the data tab page of the Properties panel, select the channel and set the data conversion content.
3. Configure the component appearance: In the right panel, click "Appearance" to switch to the setting page of component appearance to set the appearance style of the component, or write the custom css code to set the style.
4. Configure the component time: In the right panel, click "Event" to switch to the setting page of the component event, click to select the event trigger type, and write the event trigger content and execution result.

Within the component page, users can use fuzzy search to quickly locate the desired components. When searching, you only need to input the keyword of the component, click Query icon or press Enter to search for the relevant component.



5.3.2.2 View structure

The view is the structure of all the components used in the Configuration Editor, and the use method is as follows:

1. In the Configuration Editor, click the component button on the left panel to switch to the component tab page.
2. Click the "View Tree" button to switch to the view tab page.



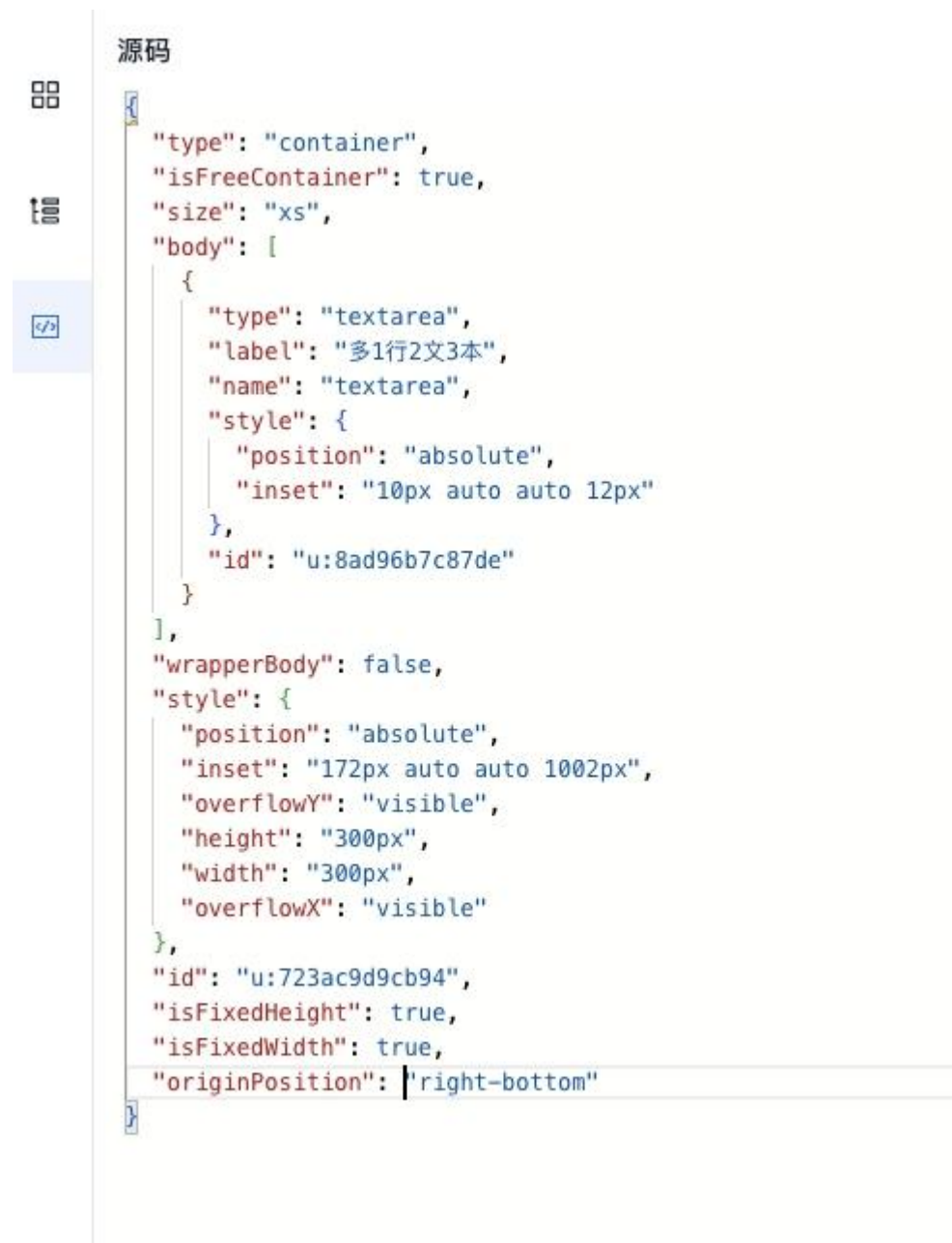
Within the view structure page, users can use fuzzy search to quickly locate the desired component element. When searching, you only need to input the keyword of the component, click Query icon or press Enter to search for the relevant component.



5.3.2.3 Source code

The source code is the source code structure of all components used in the Configuration Editor. The use method is as follows:

1. In the Configuration Editor, click the component button on the left panel to switch to the component tab page.
2. Click the “Source Code” button to switch to the source tab page.
3. Click the specified component to view the source code content of the component.



In the source code page, you can directly adjust the property content of the component according to business requirements, such as component name and font size. You can also

copy the source code and import the content into other drawing pages. The Configuration Editor source code of Xvisual is in json format, and all page contents are finally nested into a json file.

5.3.3 Canvas

The canvas is the operating area for drawing sheets and icons, in the central area of the Configuration Editor. All components will be displayed and placed in the canvas area. A scroll bar will appear when the canvas size exceeds the editor size area.

To clear the canvas with one click, the specific method is to select the outermost component (free container) and click the “Delete/Empty” button to clear all components in the page.



5.3.4 Property bar

The property bar is located on the right side of the editor and displays the flat pattern by default, and you can also click the “Collapse” button. The property bar contains three parts: property, appearance and event.

5.3.4.1 Properties

The property information includes the basic information and data information of the component. The property information is inconsistent in different components. See the description in the component list for details.

属性

外观

事件

基本

▼

宽(px)

页面尺寸: 1920

数据来源

▼

初始化静态数据

<空>

+ 新增

读通道

1,2,3,4,5,6,7,8

写通道

5.3.4.2 Appearance

The appearance information includes the basic appearance information, background information, position dimension information and custom style of the component. The appearance content settings are inconsistent in different components. See the description in the component list for details.

属性

外观

事件

表单项

隐藏标题

宽度

标题样式

位置

宽度

文字

字体大小

字体字重

字体行高

字体

字体字重

行高

字体

边距

外边距

内边距

外边距

内边距

选择框样式

状态

5.3.4.3 Events

An event is a user-defined trigger event operation that reaches a certain execution condition, such as a value change in the input box, acquisition of focus, and loss of focus. Pop-up, custom js and instruction issuing operations can be performed within the time. As shown in the figure below:



General operating procedures

Step 1: Add event

1. In the Configuration Editor, select the component to be designed, and click "Event" in the property bar on the right to switch to the event column.
2. Click “Add Event”, select the trigger condition for the event, and create this type of the event.

Step 2: Add event content

1. Select the time to be added, and click “Create” to pop up the time creation page.
2. Select the trigger time type and fill in the event content.

Event Type

Event types can be divided into pop-up messages, component messages, and other types. Pop-up messages include: open the pop-up window, close the pop-up window, and message reminder; Component messages include: component visibility, component availability, component presentation status, variable assignment and background adjustment; Other types include custom js and instruction issuing.

Open the pop-up window

The main operation of this event is to open the pop-up window that supports complex interaction design.



When selecting a pop-up window, if no pop-up window has been created, you can create a new pop-up window, or select other pop-up windows that have been created in the page. The operation of pop-up window creation is the same as that of other components in the page.





Pop-up windows support complex interaction such as nested buttons, input boxes, charts, and tables. The component properties, appearance and event settings in the pop-up window are the same as those in the external page. Pop-up properties can set the pop-up method, title, dragability, shortcut key closing, etc.

属性

外观

事件

基本



组件名称

弹出方式

- ☒ 弹窗
- ☐ 抽屉
- ☐ 确认对话框

标题

未命名弹窗

展示关闭按钮



点击遮罩关闭



可按 Esc 关闭



隐藏按钮区



左下角展示报错消息



左下角展示loading动画

☒

是否可拖拽

☐

数据映射

☒

原始数据打平

☐

Key

Value



这是必填项

+ 新增

表单验证失败

The appearance settings and event settings in the pop-up window are the same as those of other components.

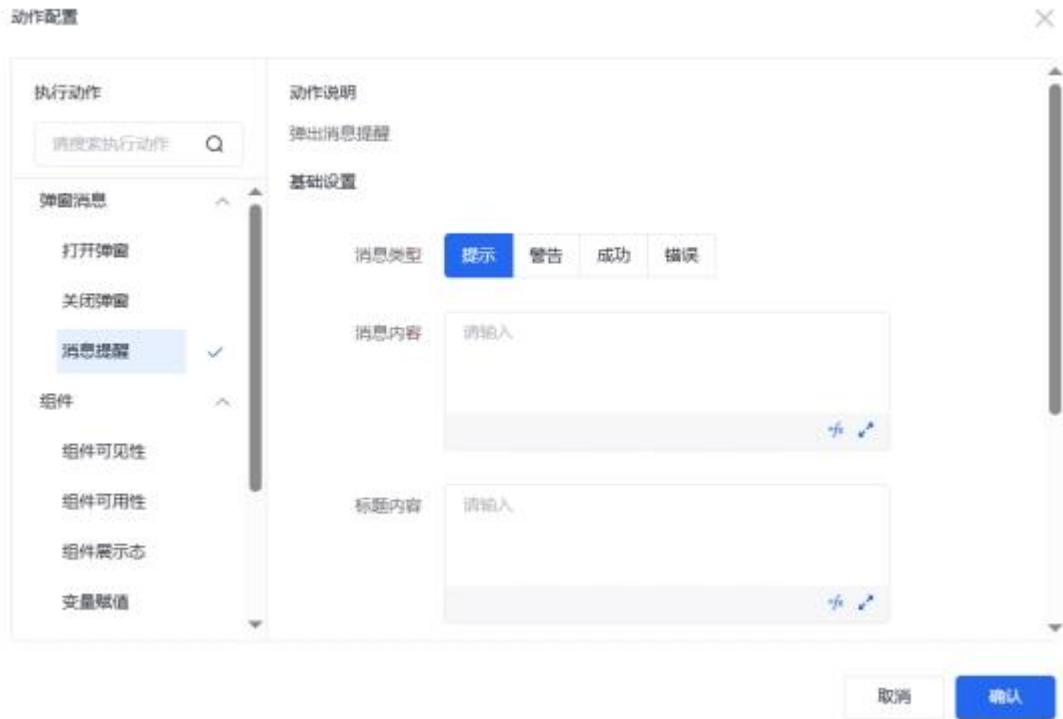
Close the pop-up window

There is no need to do any other settings to close the pop-up window, just create the event.



Message reminder

For message reminder, you need to set the message type and fill in the message content. The message content supports filling of custom functions. Please refer to the Custom Function Operation section for details of filling in the custom functions. Meanwhile, style information such as pop-up position and duration can be set. Title content also supports custom functions.



Component visibility

It can control the display or hide state of the selected component and the conditions under which the state is maintained.



Component availability

You can control the enabled or disabled state of the selected component and the conditions under which the state is maintained.

动作配置 ✕

执行动作

请搜索执行动作 Q

关闭弹窗

消息提醒

组件

组件可见性

组件可用性 ✓

组件展示态

变量赋值

背景调整

位置调整

动作说明

控制所选的组件的启用/禁用

基础设置

目标组件 请选择

条件 ☒ 静态 ☐ 表达式

启用/禁用 ☒ 启用 ☐ 禁用

取消

确认

Component presentation state

You can control the selected component to be in a form input or static state.



Variable assignment

The data value of the target component or variable can be updated by the action of a component variable, a page parameter, or a memory variable.



Background adjustment

You can set the background property of the selected component and the color of the background.



Location adjustment

You can set the location property of the selected component, and select the component to be located and its specific location.



Custom js

In custom js content, users can write function content by themselves, such as performing amplitude on a specified object, adjusting a certain object style, and the like. However, when using a value change method or an operation that may trigger loop nesting, you need to pay attention to the change of the invoked component to avoid loop recursion problems.

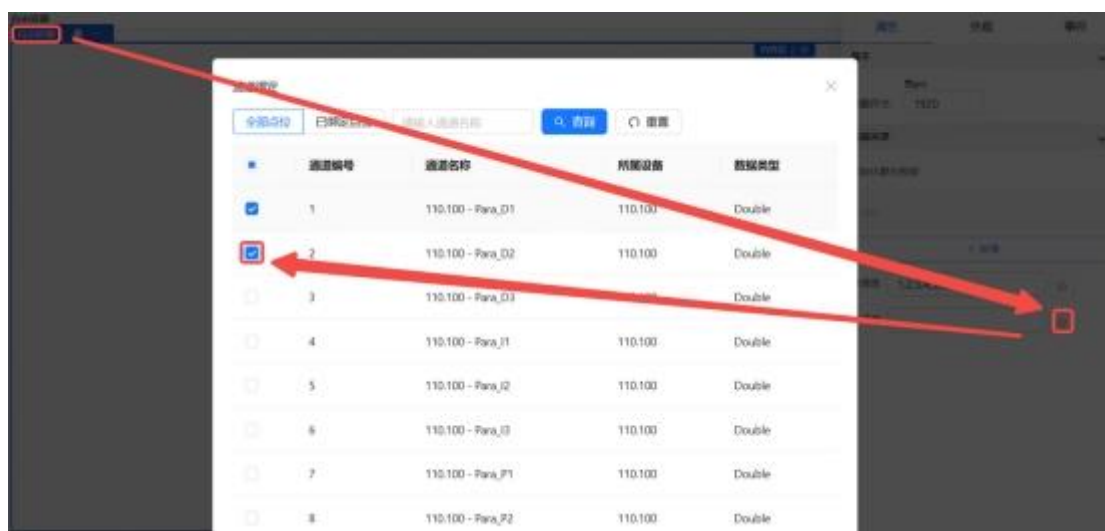
The data in the function is the global data object in the current Configuration Editor page. When you need to use a custom key object, you only need to enter the data.key and set the value. If you want to use a number from another page, you can enter the field name for the component and use it. If you want to use a channel bound within the component, you can use ".\$channel number" after the component field name.



Instruction issuing

In the instruction issuing content, the user can complete the downward data issuing. The issued data can be customized or specific values can be set. The specific operation steps are as follows:

1. Before setting the issuing of data to the channel, you need to check the target channel in the property interface of the free container property bar.



2. Open the "Event" pop-up window, and select to issue the command.



3. Click “Add Issuing Point Location”, click “Channel Selection” to pop up the “Channel Selection” pop-up window, check the channel to issue the corresponding instruction in the pop-up window, and click “OK”. The “Command” pop-up window only shows the list of write channels that have been added by the free container in the current page. Only one channel can be selected in the pop-up window. At the same time, one channel can only be bound once in the same instruction issuing operation.



4. Enter the issuing value or the custom function to set the issuing value. Click the “Custom Function” button to enter the custom function setting page.



5. Select a command mode and set the command delay time.

The delay in instruction issuance is set as the fixed mode by default, and the delay time is 0. A fixed value or custom function mode is supported for instruction content and instruction time. When the instruction delay time is set as the custom function mode, the user needs to ensure that the final output value of the function is an integer.

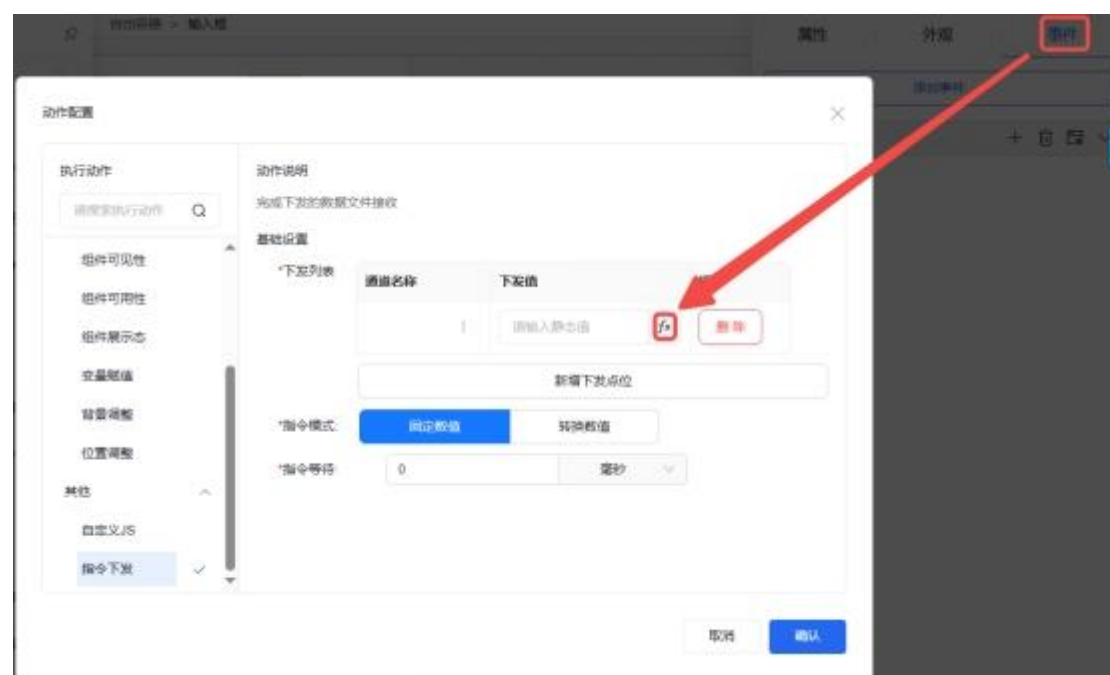
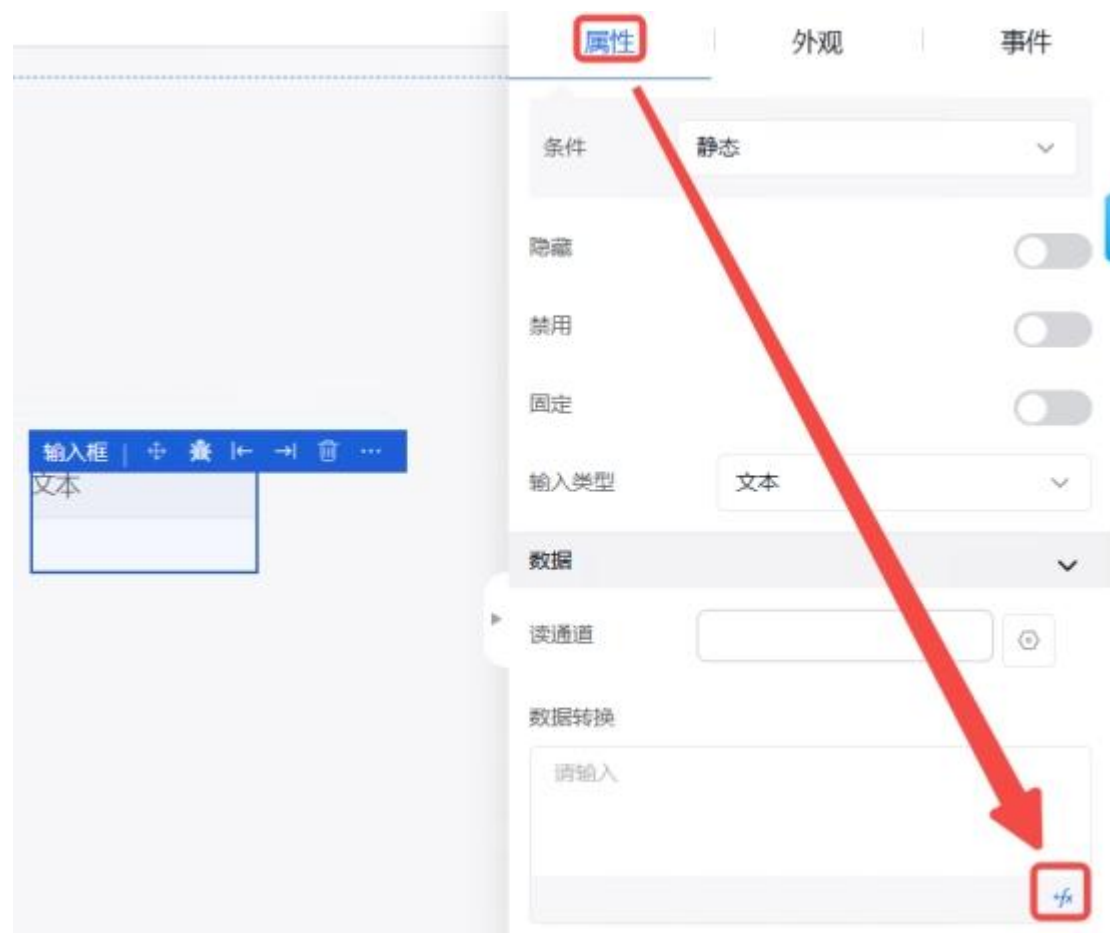
The instruction issuing time unit includes millisecond, second and minute, and the specific time range is as follows:

- When the millisecond is selected, the minimum value shall not be less than 100 and the maximum value shall not exceed 1000. If it is not within the range and the input box loses the focus, the nearest value will be displayed (if 99 is entered, 100 will be displayed; if 1001 is entered, 1000 will be displayed)
- When the second is selected, the minimum value shall not be less than 1 and the maximum value shall not exceed 60.
- When the minute is selected, the minimum value shall not be less than 1 and the maximum value shall not exceed 60.

5.3.4.4 Custom functions

Custom functions can facilitate user-defined data processing and conversion. Custom functions support logical function, mathematical function, text function, date function, array,

encoding, and other operations. Custom functions appear in the section of properties or events that needs to be converted.



The operation steps of the functions are as follows:

1. Open the “Custom Function” pop-up window and select the function to be set.
2. Enter the variable in the function that needs to adjust the current layer.
3. Click “OK”, and the data will be reflected to the content text box of the custom function.

The variables used in custom functions can use the set key value in the free container, the component field name in the current page, or the currently bound channel. When using a key value or component field name, enter the field name or key directly. If the channel is used, \$ shall be added before the channel number. For example, if channel number 2 is used, \$2 shall be used.



5.4 Component list

组件 ☆

输入关键字查询组件



输入 ▼

 输入框

 多行文本框

 数字输入

 复选框

 单选

 开关

 滑块

基础 ▼

 按钮

 文字

 下拉列表

 图片

 进度条

布局容器 ▼

 容器

 自由容器

 状态容器

 折叠面板

 选项卡

5.4.1 Text input box

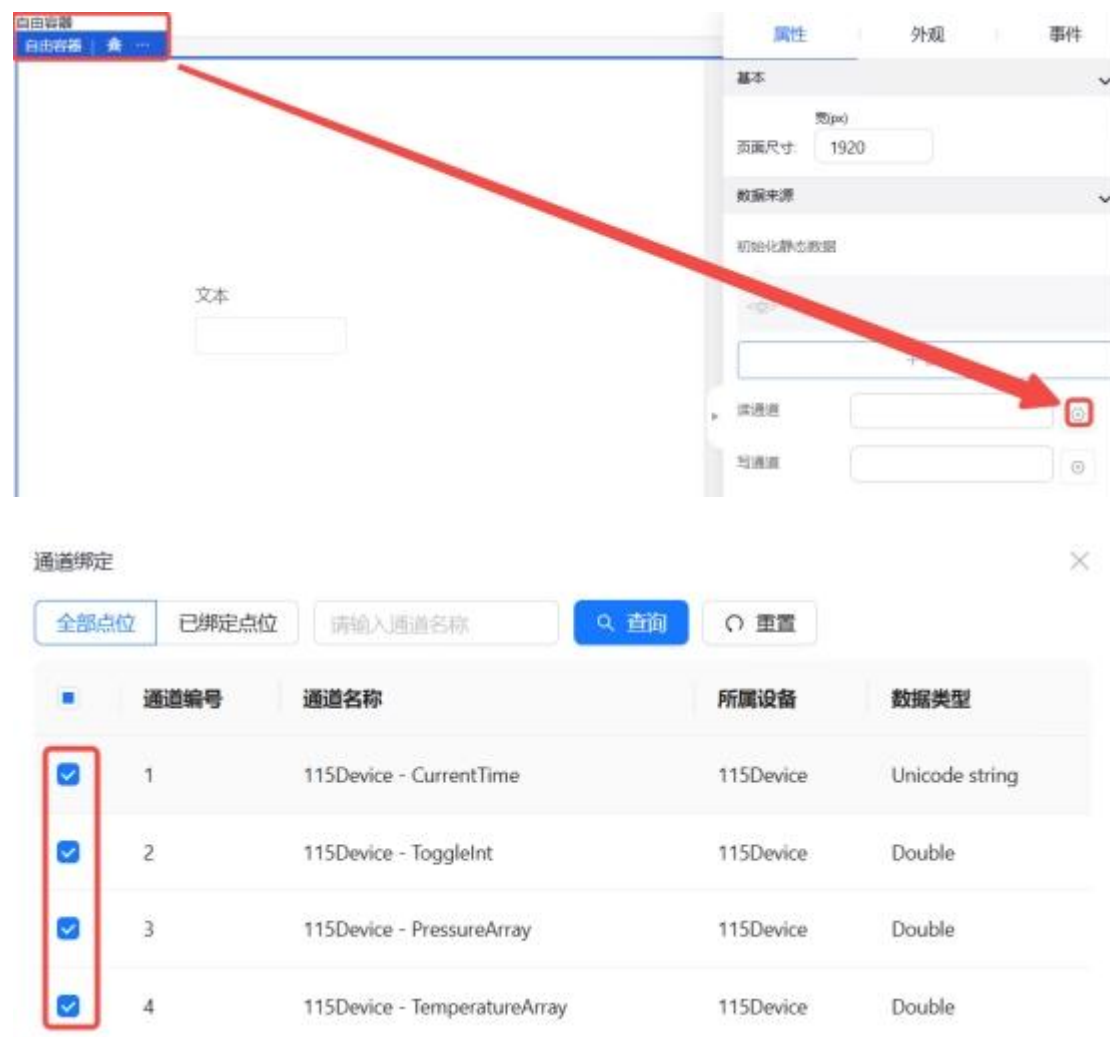
It is a form component for radio text input and presentation. The input types include text, email, password and URL. When an email is selected, the input is automatically started to verify whether it meets the support of email rules. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Field name
		Title
		Description
		Visible
		Disable
		Fixation
	Data	Read channel
		Data conversion
Events	Change in value	
	Getting focus	
	Loss of focus	
	Verification succeeded	
	Verification failed	

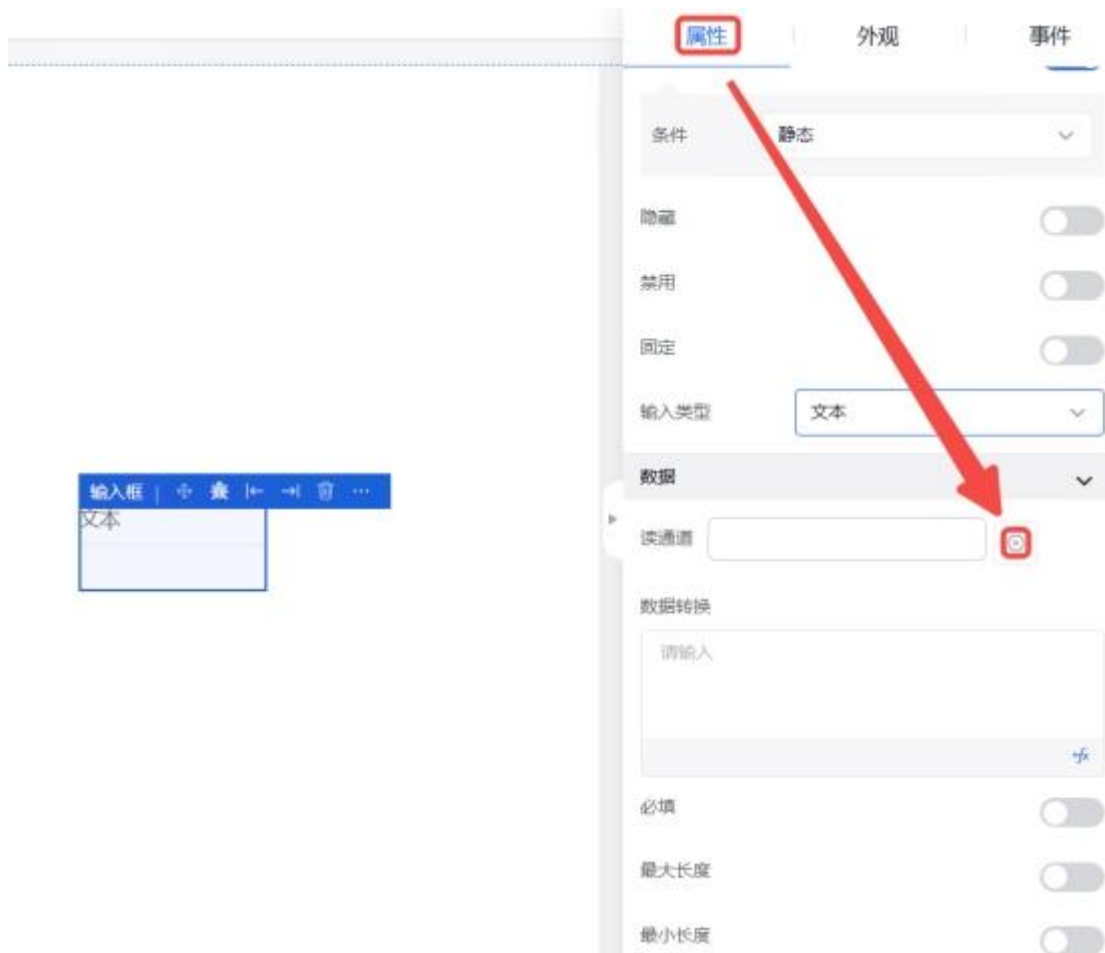
Operation instance

5.4.1.1 Read channel binding

1. Before the read channel binding of an individual component (such as a text input box) is performed, the read channel binding of the free container in which the individual component is located needs to be completed



2. Click on the "Text Input Box" and click the "Edit" button for the read channel in the "Data" section in the "Properties" section of the property bar.



3. Select the channel to be read in the “Channel Binding” interface and confirm it.

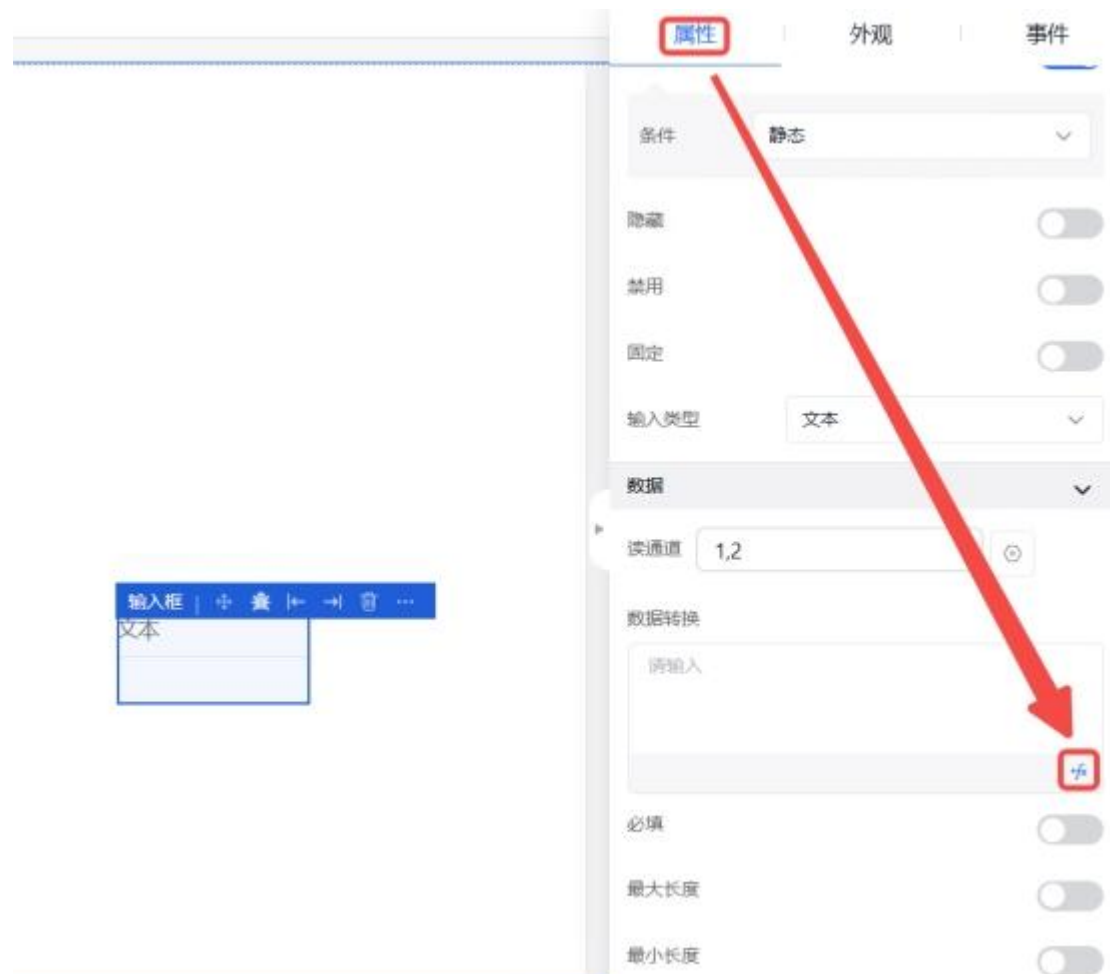
通道绑定

全部点位 已绑定点位 请输入通道名称 查询 重置

	通道编号	通道名称	所属设备	数据类型
☒	1	110.100 - Para_D1	110.100	Double
☒	2	110.100 - Para_D2	110.100	Double
☒	3	110.100 - Para_D3	110.100	Double
☒	4	110.100 - Para_I1	110.100	Double
☐	5	110.100 - Para_I2	110.100	Double
☐	6	110.100 - Para_I3	110.100	Double

5.4.1.2 Data conversion

1. Click the “Text Input Box”, and click the fx button for data conversion in the “Data” section in the “Properties” section of the property bar to enter the function setting page or directly click the blank part to edit the data conversion format.

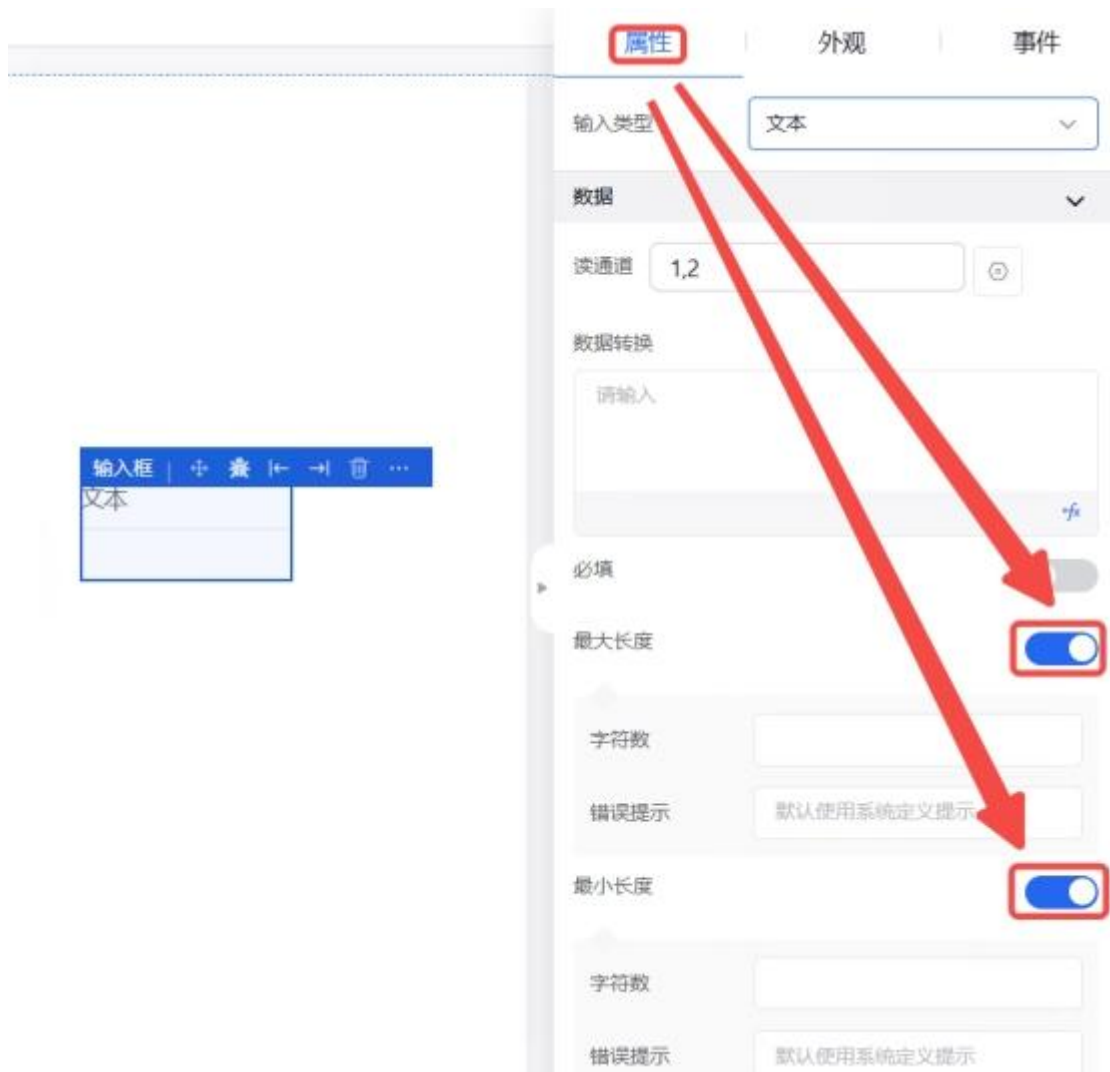


2. In the function setting interface, select the function type you want on the left side, and select the variables to participate in the function on the rear side.



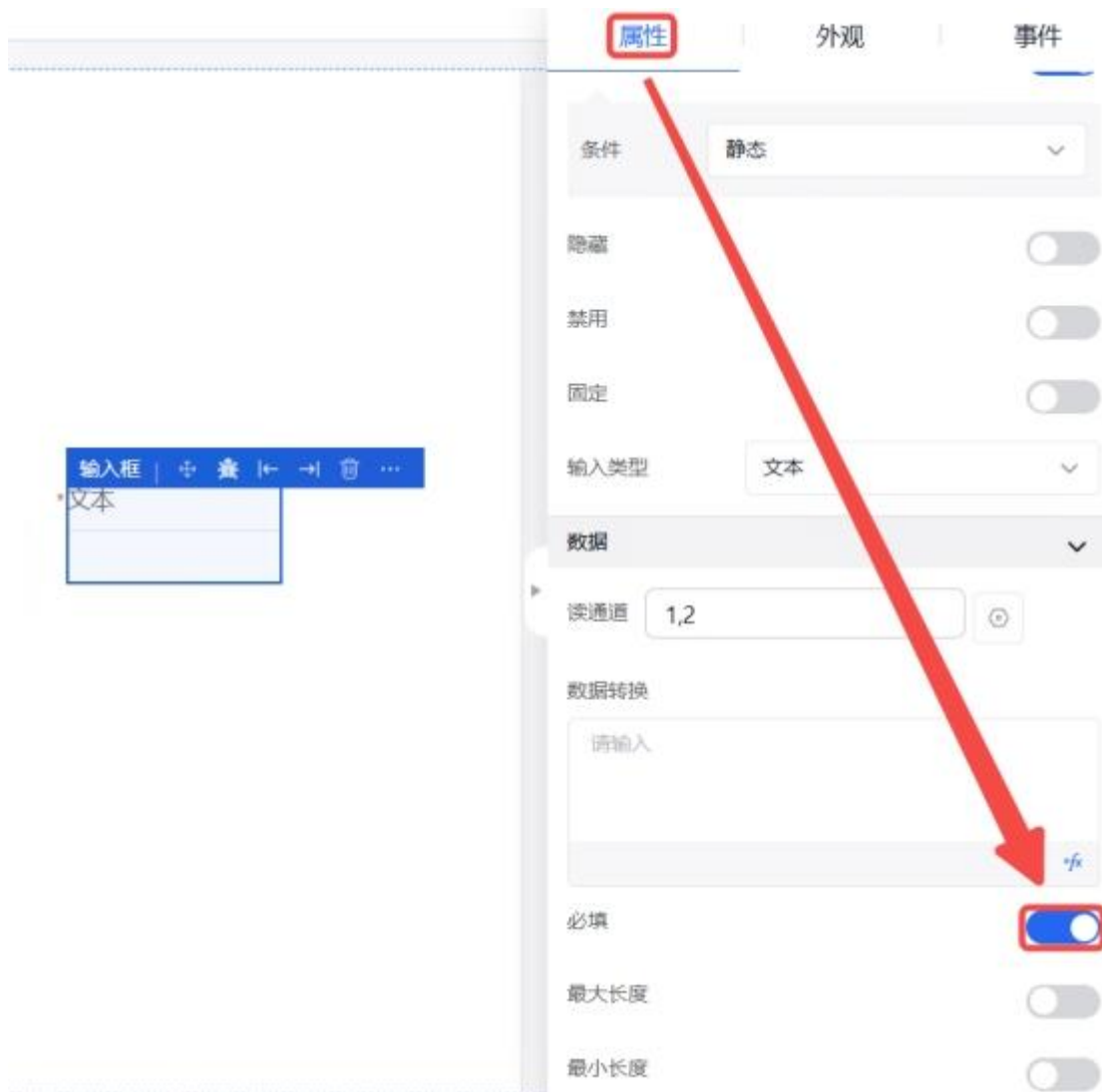
5.4.1.3 Maximum length & minimum length

1. Click the “Text Input Box” and click the button with the maximum length or minimum length of the “Data” section in the “Properties” section of the property bar to edit the limit of the number of characters and the error prompt.



5.4.1.4 Required

1. Click the “Text Input Box” and click the required button in the “Data” section in the “Properties” section of the property bar.



5.4.2 Multiline text box

It is a text component for line feed. It supports properties and events, as shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Field name
		Title
		Default value

		Remove the blank space before the first word
		Counter
		Maximum number of words
		Title Hint
		Control Hint
		Placeholder Hint
		Description
		Data Entry
	Data	Read channel
		Data conversion
	Status	Visible
		Hide
		Disable
		Fixation
		Delete the field when hidden
		Static presentation
		Read only
	Check	Required
		Minimum length

		Maximum length
Events	Change in value	
	Getting focus	
	Loss of focus	
	Verification succeeded	
	Verification failed	

Operation instance

5.4.2.1 Read channel binding

Refer to the read channel binding operation of text input box in 5.4.1.

5.4.2.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.1.

5.4.3 Digital input box

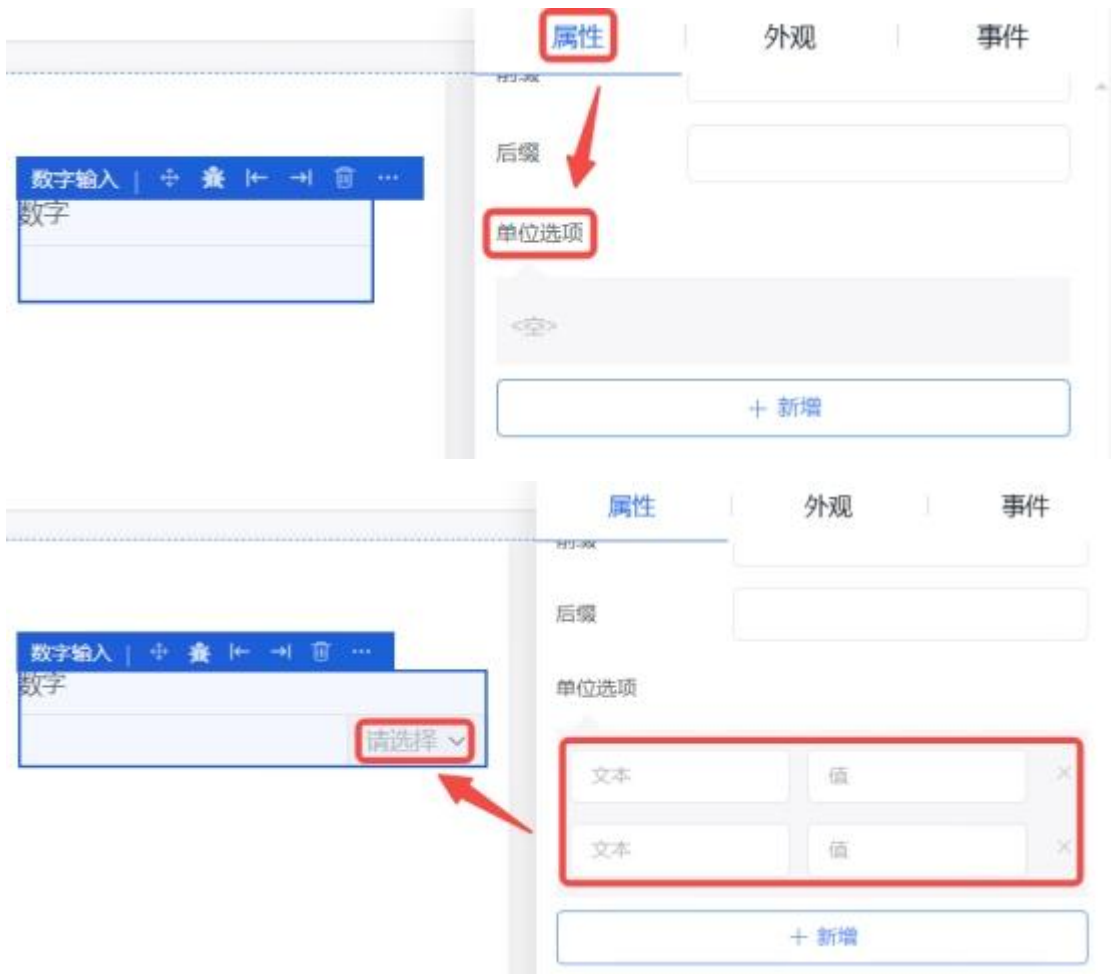
It is a component for requiring digital input. It supports the settings of maximum and minimum values as well as step size and precision. It supports properties and events, as shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Field name
		Title
		Keyboard events
		Thousand separator
		Default value

		Minimum value
		Maximum value
		Step size
		Number of decimal places
		Prefix
		Suffix
		Title Hint
		Control Hint
		Placeholder Hint
		Description
		Data Entry
	Status	Visibility
		Hiddenness
		Prohibitibility
Events	Change in value	
	Verification succeeded	
	Verification failed	
	Getting focus	
	Loss of focus	

5.4.3.1 Add Option

1. Click the “Digital Input Box” to edit the unit option in the “Properties” section of the property bar.



5.4.4 Check box

It is a check box component for configuring multiple check boxes with options or dropping down options with source. It supports properties and events, as shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Field name
		Title

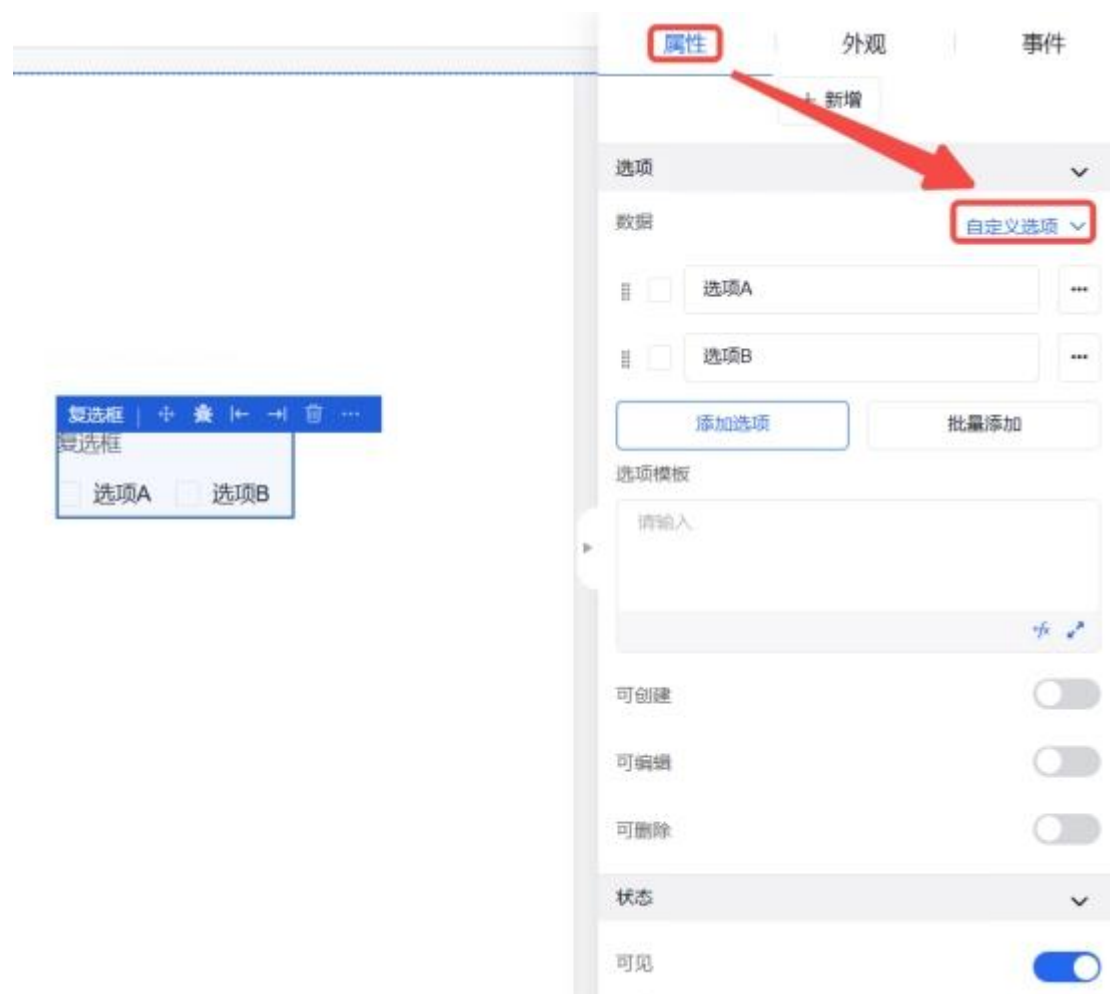
		All selectable
		Stitching value
		Stitching symbol
		Title Hint
		Control Hint
		Description
		Data Entry
	Data	Data
		Option template
		Creatable
		Editable
		Deletable
	Status	Visible
		Hide
		Delete the field when hidden
		Static presentation
		Disable
	Check	Required
Events	Change in value	

	Confirm add
	Confirm edit
	Confirm delete
	Verification succeeded
	Verification failed

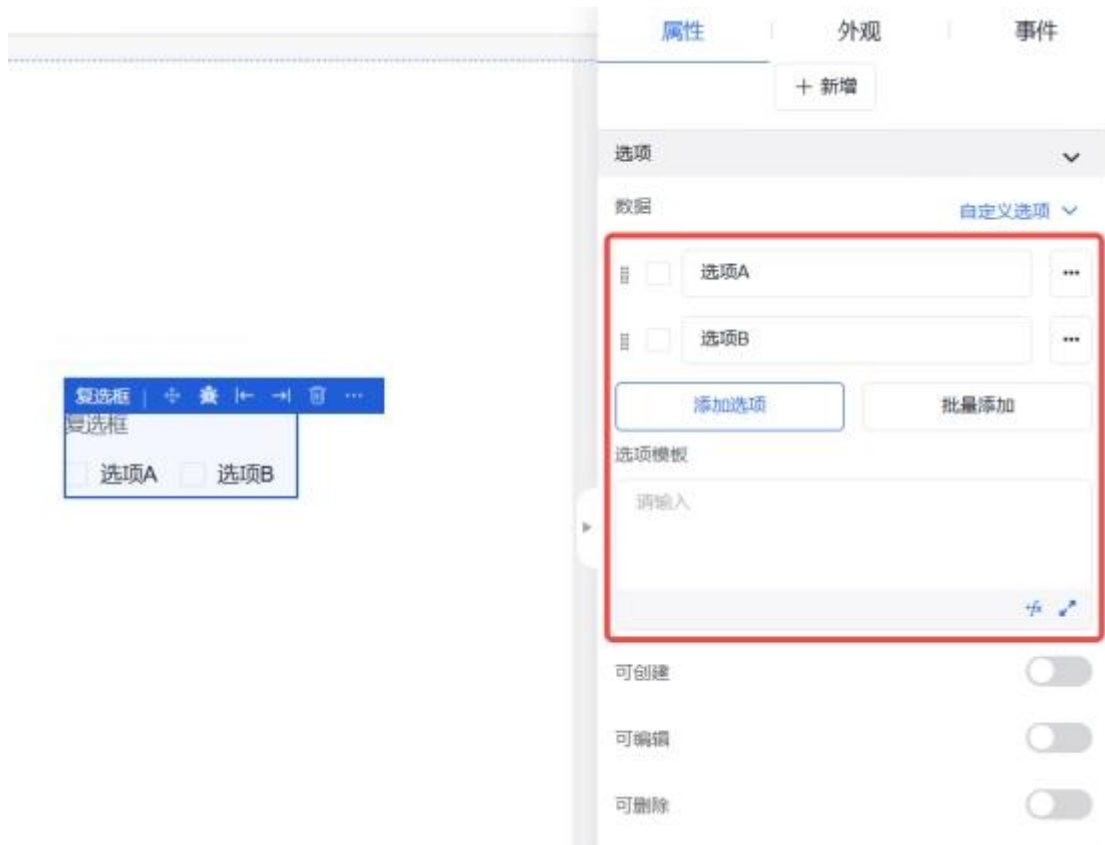
Operation instance

5.4.4.1 Custom options

1. Click “Check box” and then click the custom options in the “Data” section in the “Properties” section of the property bar.

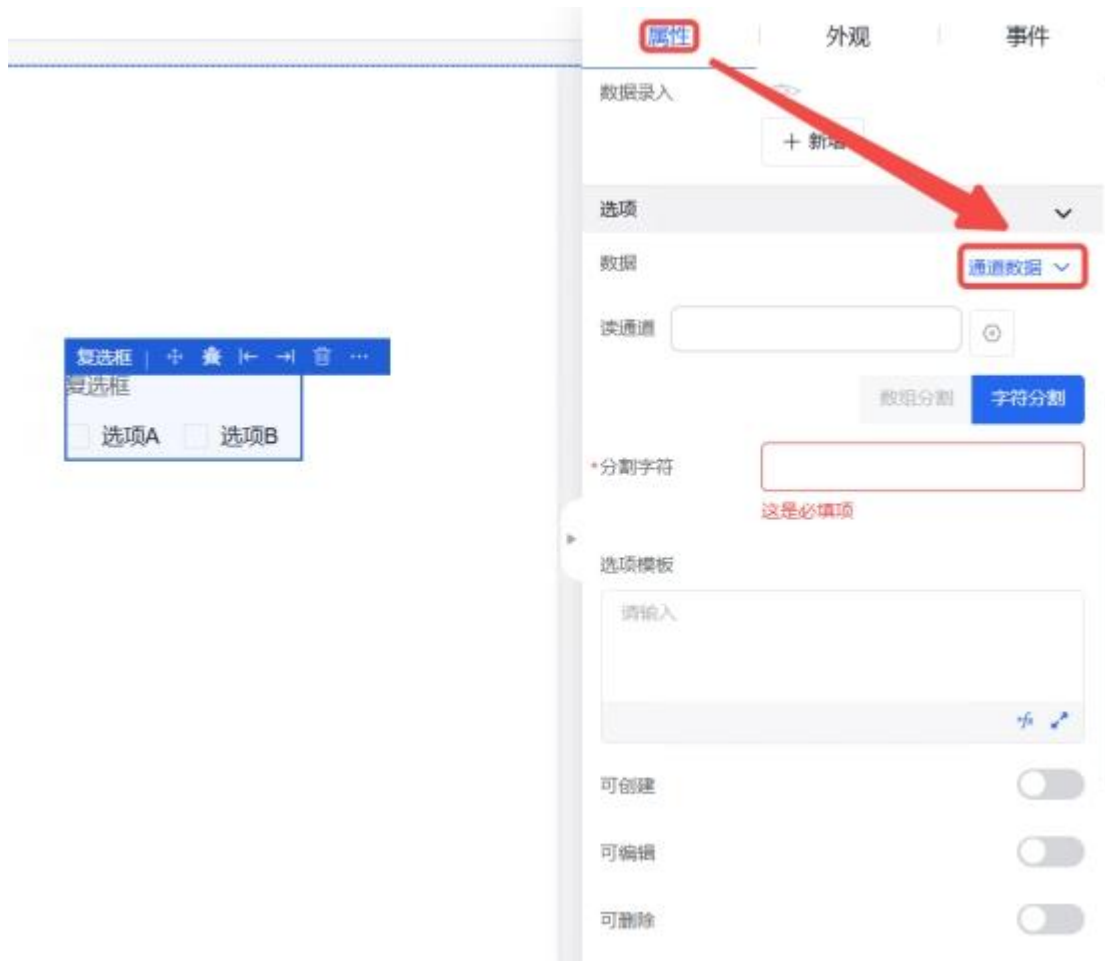


2. Edit the content and number of the check box. The option cannot be repeated. If it is repeated, "Current option is repeated, please re-enter" will be prompted.



5.4.4.2 Channel data

1. Click the “Check Box”, and then click the channel data in the “Data” section in the “Properties” section of the property bar to bind the point location of the channel data (refer to the read channel binding of the text input box); When the character segmentation option is selected, the segmentation character option is required. The channel return string is split according to the specified segmentation character, and the split completion element is backfilled to the drop-down list.



5.4.4.3 New options

1. Click the "Creatable" button to add "New Options" in the display section. The user can edit the options freely.



5.4.5 Radio

It is a radio component for configuring options with options or dropping down options with source. It supports properties and events, as shown in the following table:

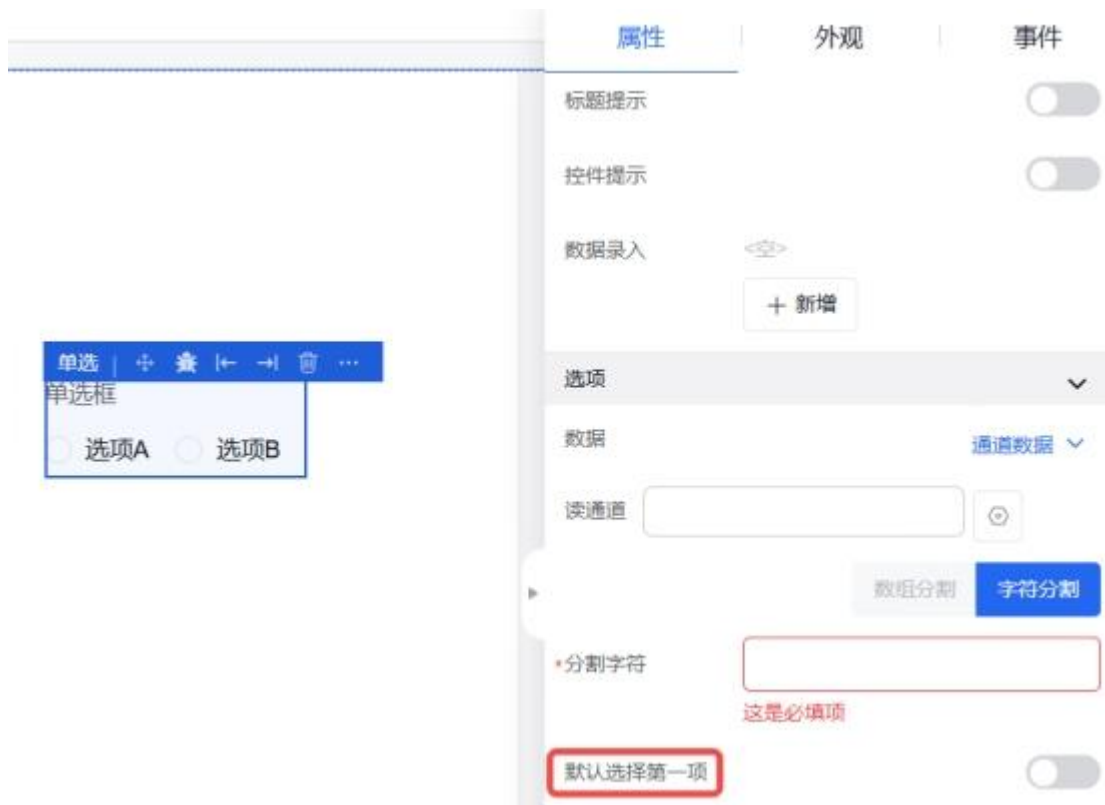
Type	Content	Name
Properties	Basic	Reference position
		Field name
		Title
		Default value
		Title Hint

		Control Hint
		Data Entry
	Data	Data
		The first option is selected by default
	Status	Visible
		Hide
		Delete the field when hidden
		Static presentation
		Disable
	Check	Required
Events	Change in value	
	Verification succeeded	
	Verification failed	

Operation instance

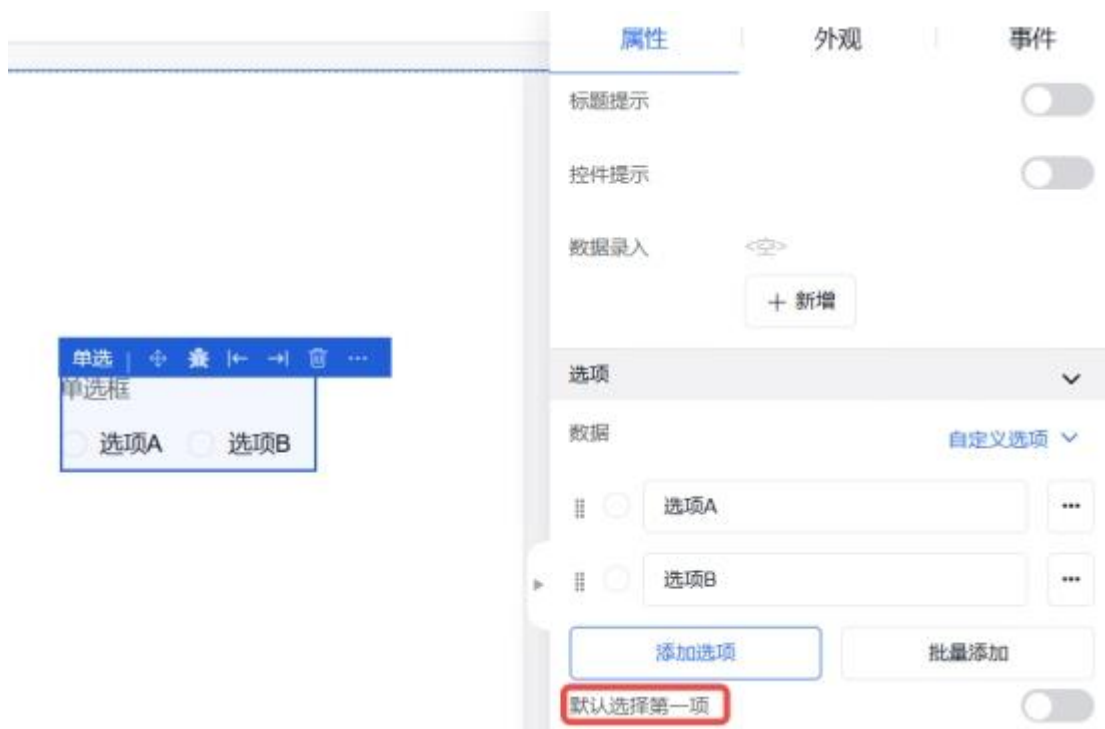
5.4.5.1 Custom options

Refer to the custom operation of the check box in 5.4.4 and add "The first option is selected by default".



5.4.5.2 Channel data

Refer to the channel data operation of check box in 5.4.4, and add “The first option is selected by default”.



5.4.6 Switches

Switch control. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Description
		Visibility
		Fixation
		Disable
Data	Read channel	
	Data conversion	
	Fill in text	
	Value format	
Events	Click	
	Verification succeeded	
	Verification failed	

Operation instance

5.4.6.1 Read channel binding

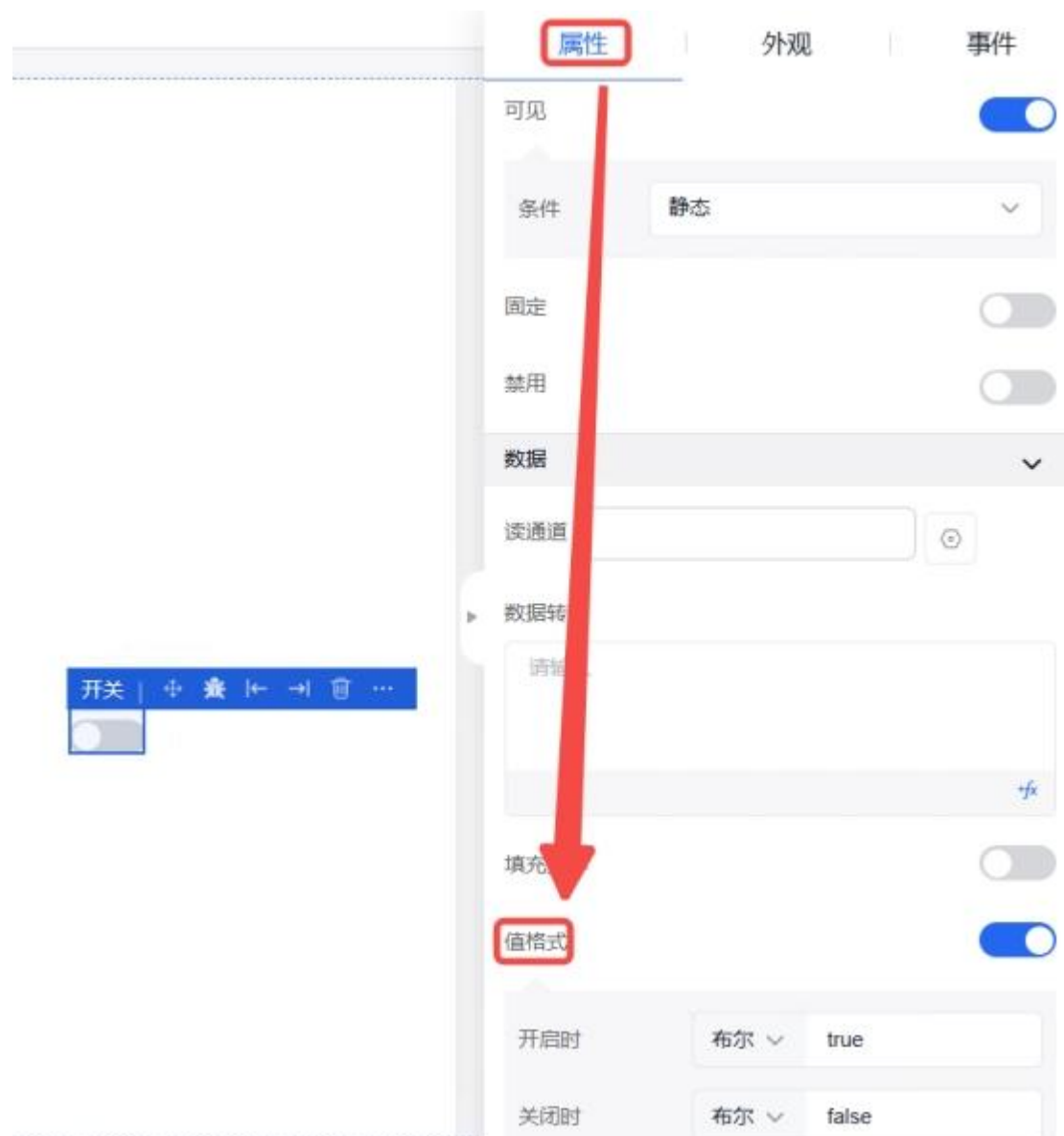
Refer to the read channel binding operation of text input box in 5.4.1.

5.4.6.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.1.

5.4.6.3 Value format

Click "Switch" and click the value format of the "Data" section in the "Properties" section of the property bar to set the parameter format when the switch is changed. The default is Boolean type.



5.4.7 Sliders

Select a value or a range of sliding components. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position

		Field name
		Title
		Double-slider
		Default value
		Minimum value
		Maximum value
		Step size
		Number of decimal places
		Unit
		Value label
		Importable
		Data Entry
Track		Partitioning
		Subscript
		The first option is selected by default
Data		Read channel
		Data conversion
Status		Visible
		Hide

		Delete the field when hidden
		Static presentation
		Disable
	Check	Required
Events	Change in value	
	Getting focus	
	Loss of focus	
	Verification succeeded	
	Verification failed	

Operation instance

5.4.7.1 Read channel binding

Refer to the read channel binding operation of text input box in 5.4.1.

5.4.7.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.1.

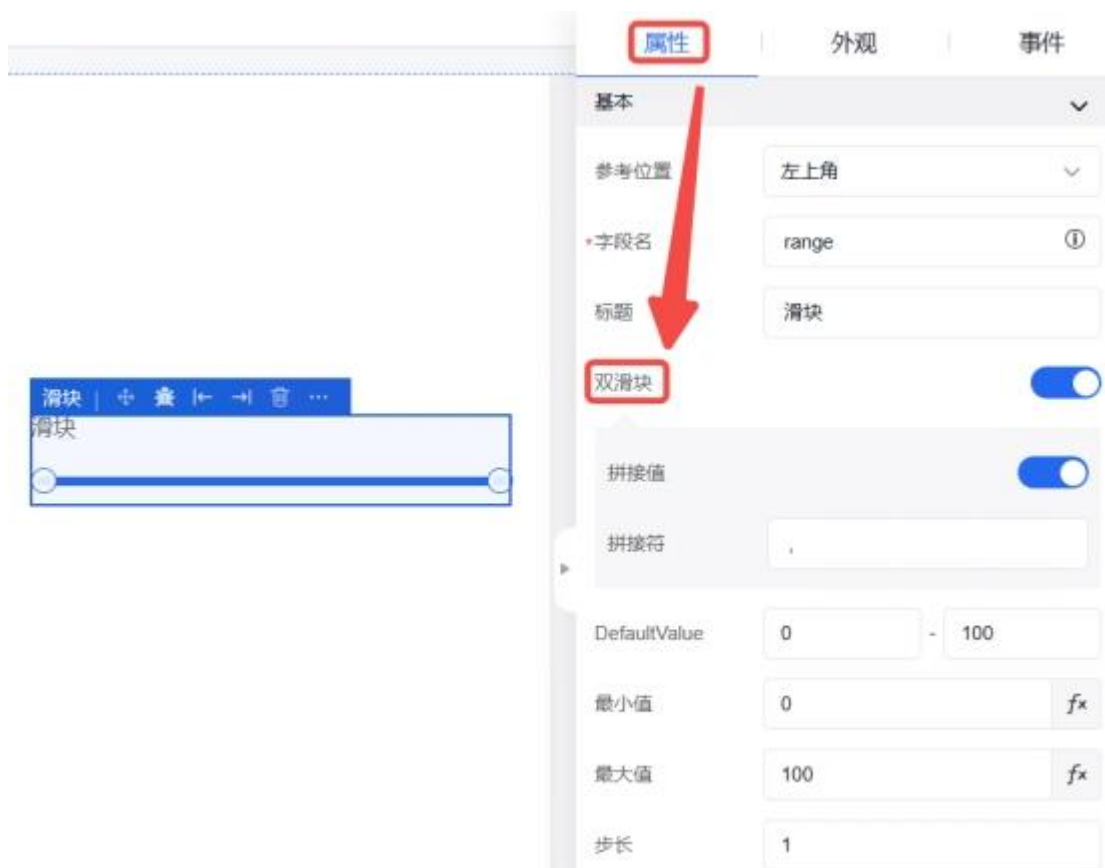
5.4.7.3 Slider partitioning

Click "Slider" and click the partitioning of the "Track" section in the "Properties" section of the property bar to set the partitioning of the slider progress bar according to the business.



5.4.7.4 Double-slider

Click “Slider” and click the double-slider in the “Data” section in the “Properties” section of the property bar to set the sliders at both ends of the progress bar at the same time.



5.4.8 Buttons

Button components that configure different presentation styles and click behaviors. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Content
		Reference position
		Description
		Visible
		Disable
		Fixation
Events	Click	
	Mouse enter	
	Mouse leave	

Operation instance

5.4.8.1 Read channel binding

Refer to the read channel binding operation of text input box in 5.4.1.

5.4.8.2 Data conversion

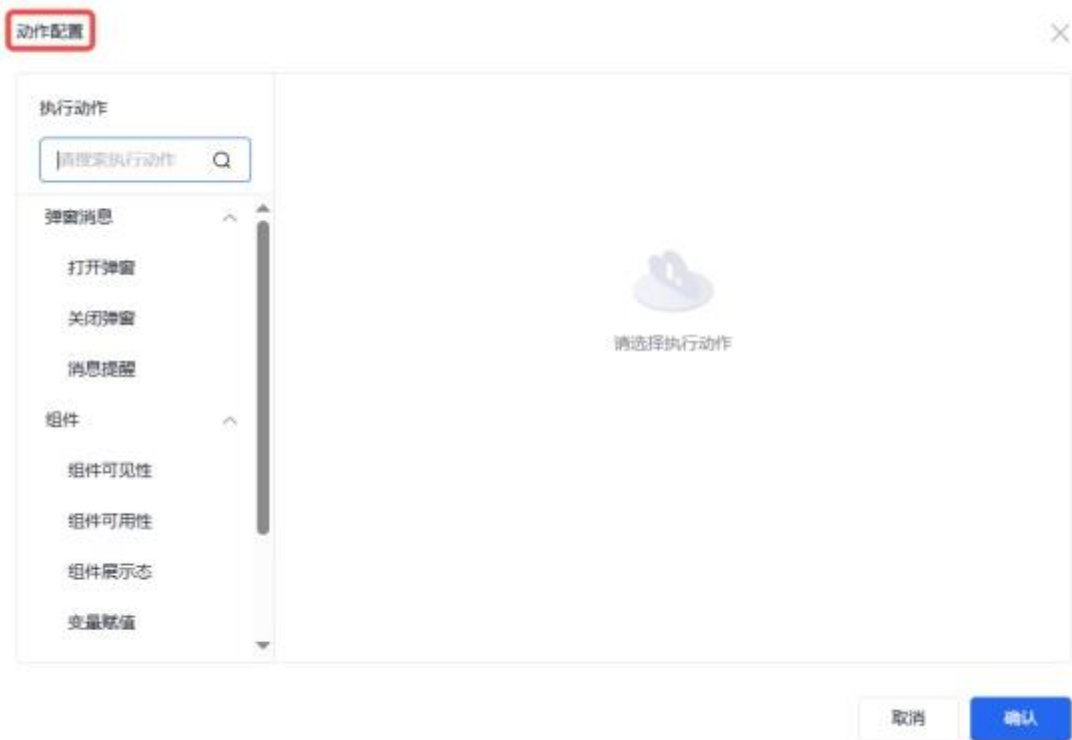
Refer to the data conversion operation of text input box in 5.4.1.

5.4.8.3 Click event

1. Click the “Button”. The “Click” event is present by default in the “Event” section of the property bar.



- Click “+” to enter the action configuration page.



5.4.9 Text

It is to present texts or paragraphs, and it supports text components in which template syntax can be used to associate dynamic data. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Description
		Fill in false data
		Visible
		Disable
		Fixation
Events	Change in value	
	Mouse enter	
	Mouse leave	

5.4.9.1 Read channel binding

Refer to the read channel binding operation of text input box in 5.4.1.

5.4.9.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.1.

5.4.9.3 Maximum length

Refer to the maximum length operation in the text box in 5.4.2.

5.4.10 Drop-down box

It supports multiple selection and input prompt. You can use source to get the drop-down box of the option. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Title
		Removable
		Retrievable
		Multi-selectable
		Title Hint
		Control Hint
		Placeholder Hint
	Options	Data
		The first option is selected by default
		Option template
		Creatable
		Editable
		Deletable
	Advanced	Option value check
		Virtual list threshold
		Option height
	Status	Visible

		Hide
		Delete the field when hidden
		Static presentation
		Disable
	Check	Required

Operation instance

5.4.10.1 Custom options

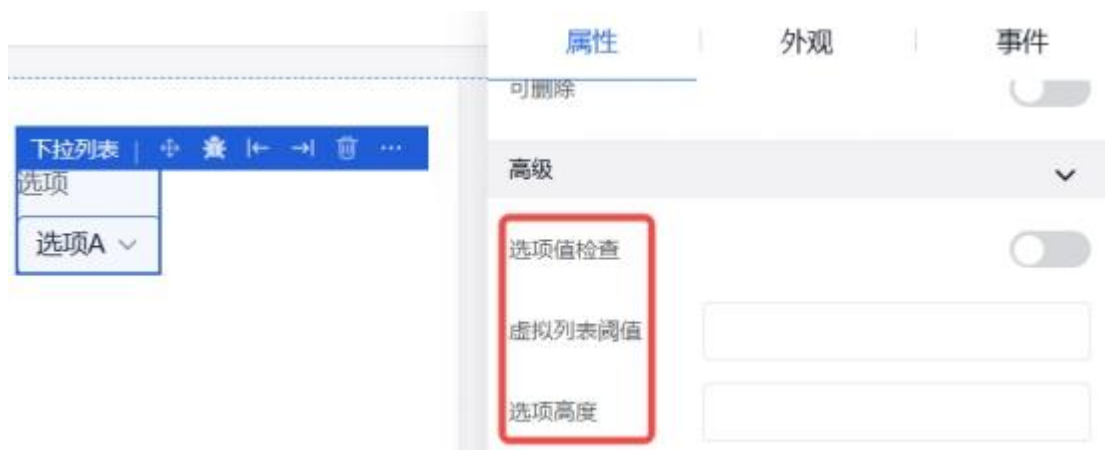
Refer to the custom option operation of check box in 5.4.4.

5.4.10.2 Channel data

Refer to the channel data operation of check box in 5.4.4.

5.4.10.3 Option setting

Click “Button” to check the content of the option value and set the number and height of options in the advanced section of “Properties” section of the property bar.



5.4.11 Progress bar

It can be used to display percentage or current progress, and has no event function. The properties and events are shown in the following table:

Type	Content	Name
------	---------	------

Properties	Basic	Field name
		Reference position
		Type
		Progress value
		Description
		Visible
		Disable
		Fixation
	Data	Read channel
		Data conversion
		Default progress bar
Appearance		Dimensions
		Line width
		Display animation
		Style
		Color

Operation instance

5.4.11.1 Read channel binding

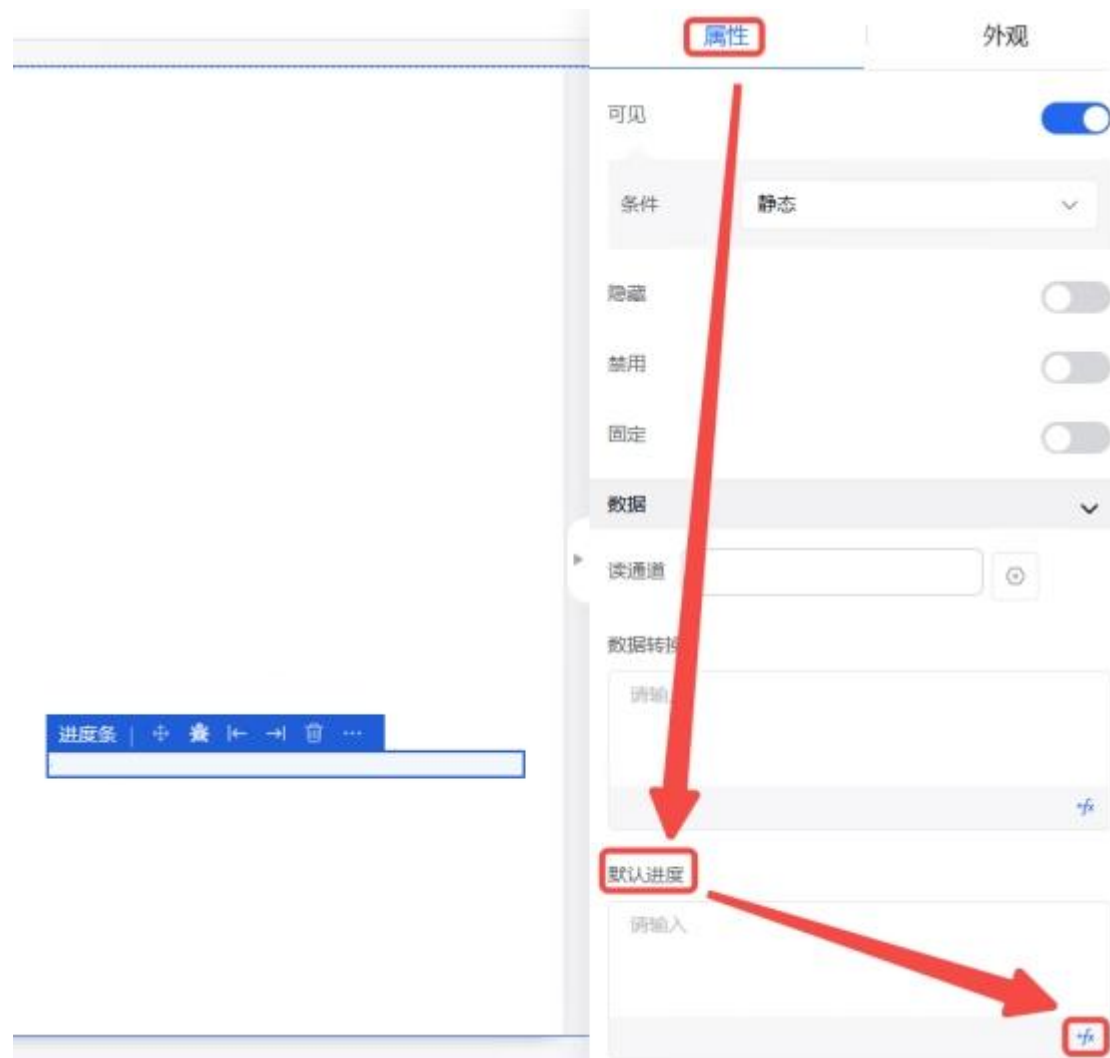
Refer to the read channel binding operation of text input box in 5.4.1.

5.4.11.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.4.

5.4.11.3 Default process

Click the "Progress Bar" to set the default process in the “Data” section of the "Properties" section of the property bar. You can directly input parameters or functions.



5.4.12 Picture

It is a picture component that supports static setting of picture address. Name can be configured to associate with variable. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Picture title
		Picture description

		Display mode
		Width
		Height
		Picture file
		Placeholder graphic
		Amplification limit
		Reduction limit
	Status	Visibility
		Hiddenness
Events	Click	
	Mouse enter	
	Mouse leave	

Operation instance

5.4.12.1 Read channel binding

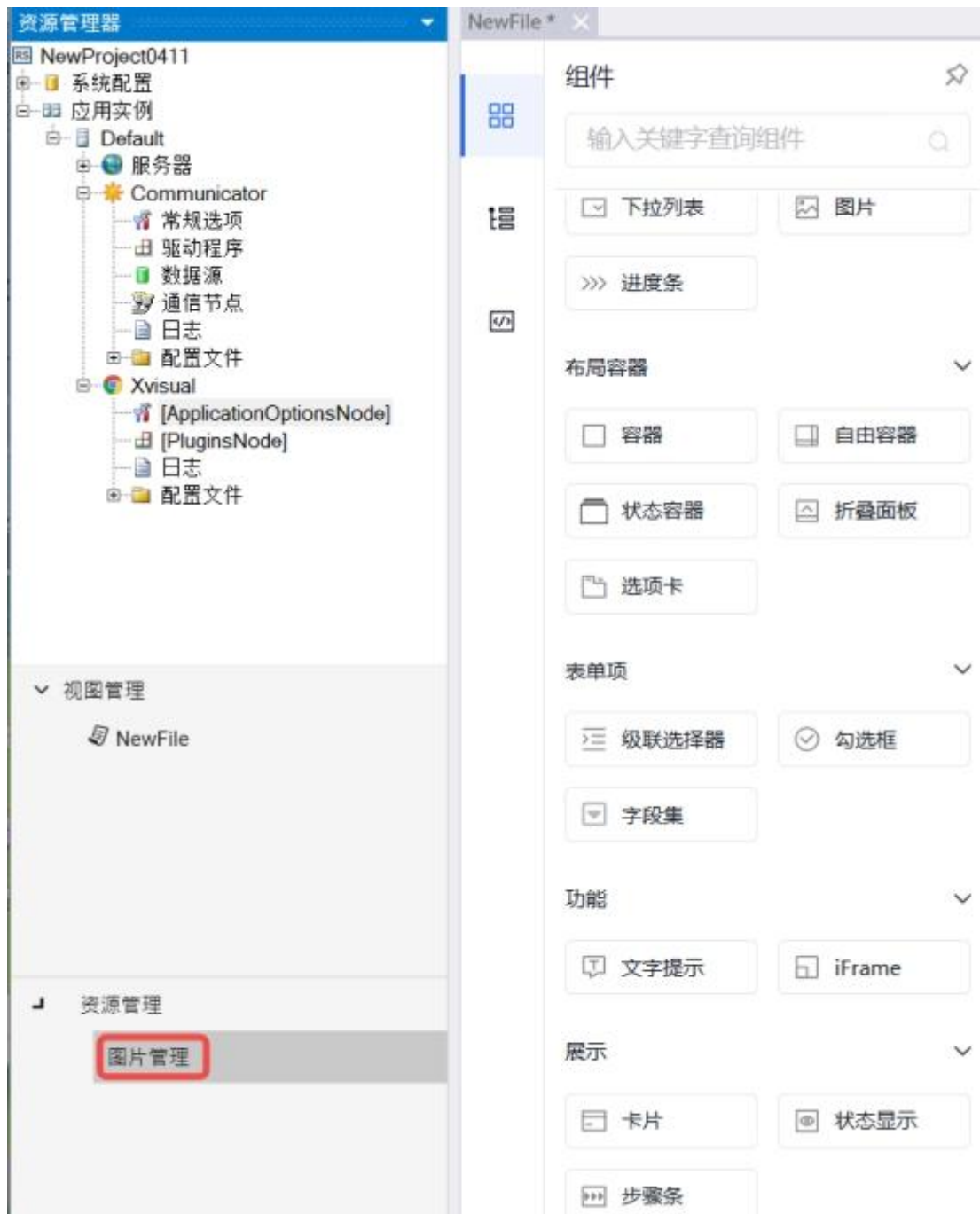
Refer to the read channel binding operation of text input box in 5.4.1.

5.4.12.2 Data conversion

Refer to the data conversion operation of text input box in 5.4.1.

5.4.12.3 Picture management

1. Click to enter the picture management in the resource management.



2. Click “Add Picture Resource” to add pictures, or add pictures in batches.

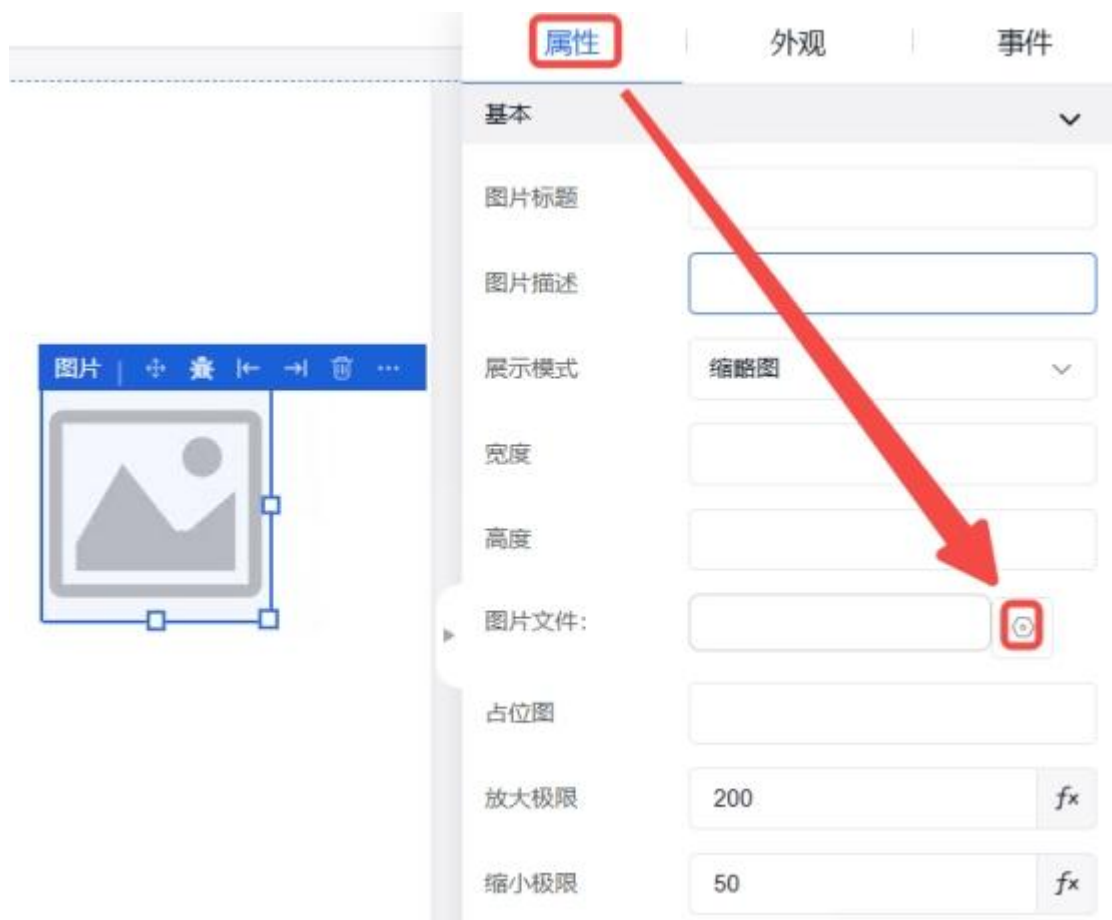


3. After the picture is added successfully, it will be displayed in the icon management.



5.4.12.4 Use of pictures

1. Click the “Picture” component and click the “Edit” button on the picture file in the “Properties” section of the property bar.



2. Enter the picture selection interface, select the picture you want to upload, and click “Save” to complete the operation.



3. Upon completion of the upload, the picture component content in the view is updated to the picture of the user's choice.



5.4.13 Container

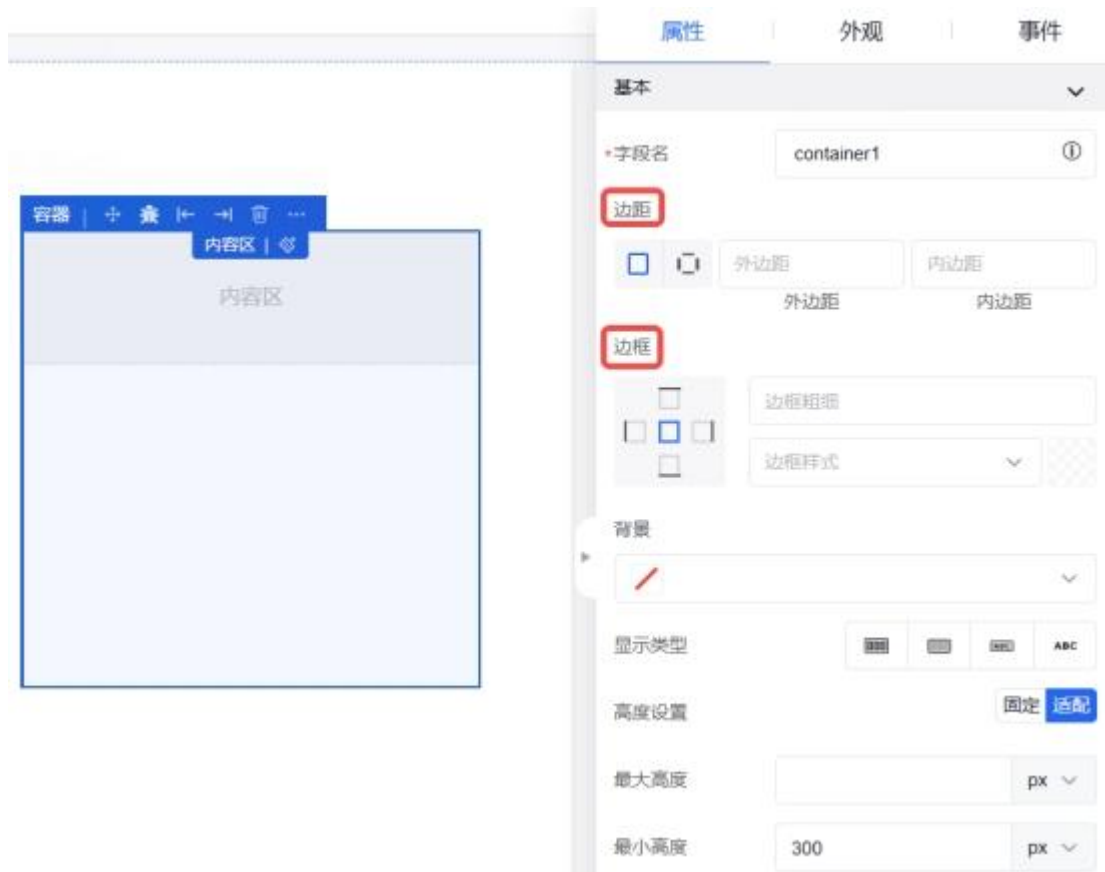
It can place multiple components together. The properties and events are shown in the following table:

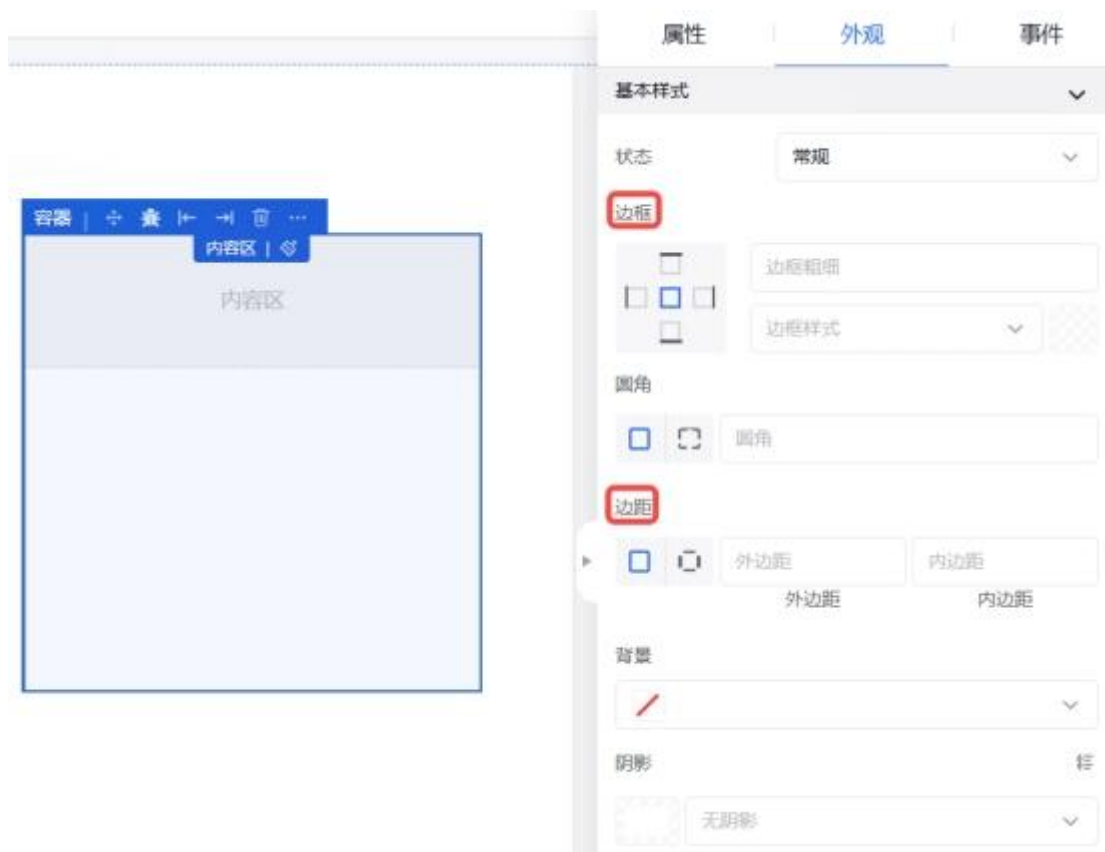
Type	Content	Name
Properties	Basic	Field name
		Margin
		Border
		Background
		Height configuration
		Height
		Vertical content overflow
		Width setting
		Width
		Horizontal content overflow
		Alignment
	Status	Visibility
		Hiddenness
Events	Click	
	Mouse enter	
	Mouse leave	

Operation instance

5.4.13.1 Margin and border settings

Click “Container” to set “Margin” and “Border” in the “Properties” and “Appearance” sections of the property bar.





5.4.14 Free container

It supports direct sub-elements and supports to adjust the position by dragging. The properties and events are shown in the following table:

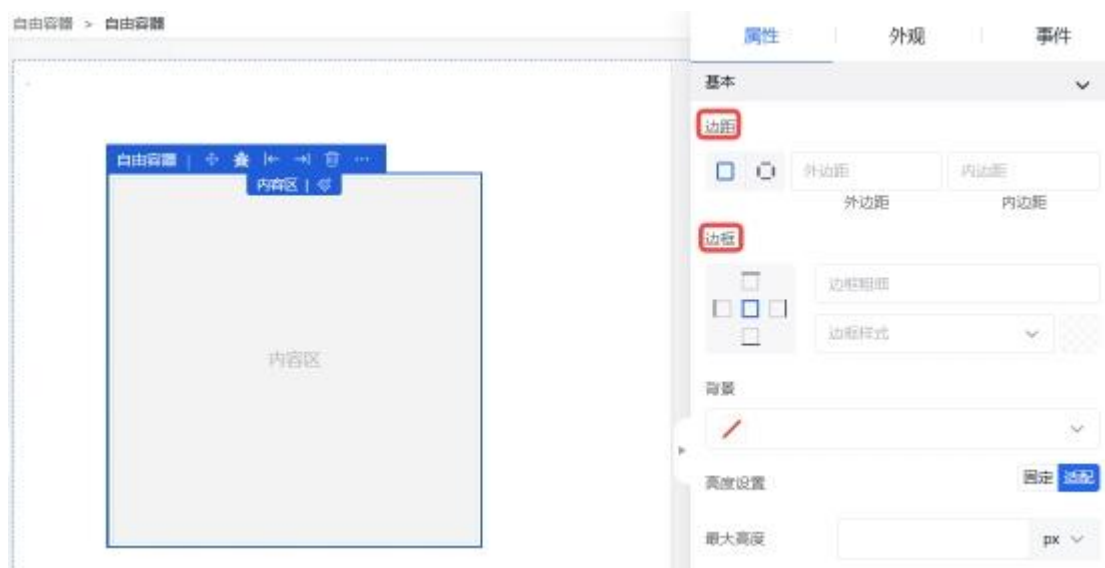
Type	Content	Name
Properties	Basic	Margin
		Border
		Background
		Height configuration
		Height
		Vertical content overflow

		Width setting
		Width
		Horizontal content overflow
		Alignment
	Status	Visibility
		Hiddenness
Events	Click	
	Mouse enter	
	Mouse leave	

Operation instance

5.4.14.1 Margin and border settings

Click “Free Container” to set “Margin” and “Border” in the “Properties” and “Appearance” sections of the property bar.





5.4.15 Status container

It is used for conditional rendering of components according to the status, and it facilitates the design of multi-state components. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Status List
	Status	Visibility
		Hiddenness
Events	Click	
	Mouse enter	
	Mouse leave	

Operation instance

5.4.15.1 Status setting

Click "Status Container" to modify and add status in "Properties" in the property bar.



5.4.16 Folding panel

It can be used for carrying out classified storage when the information amount is large and the classification is more. The properties and events are shown in the table below

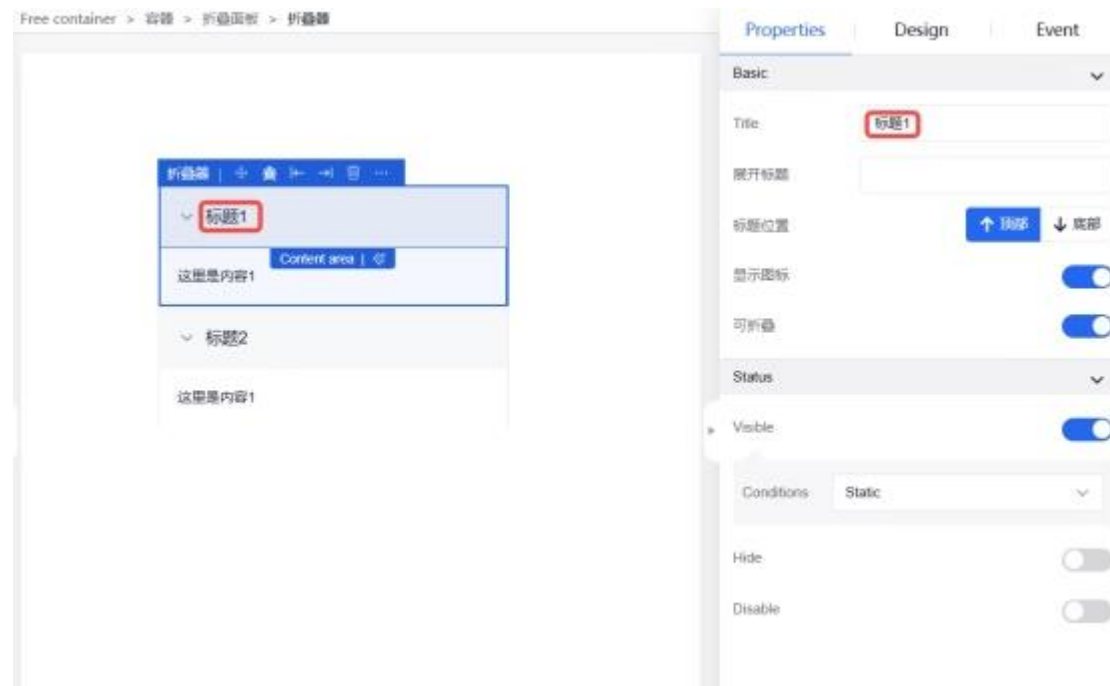
Type	Content	Name
Properties	Basic	Field name
		Reference position
		Icon location
		Custom icon
		Accordion mode
		Panel management
Events	Folding state change	
	Folder expansion	

Folder collapse

Operation instance

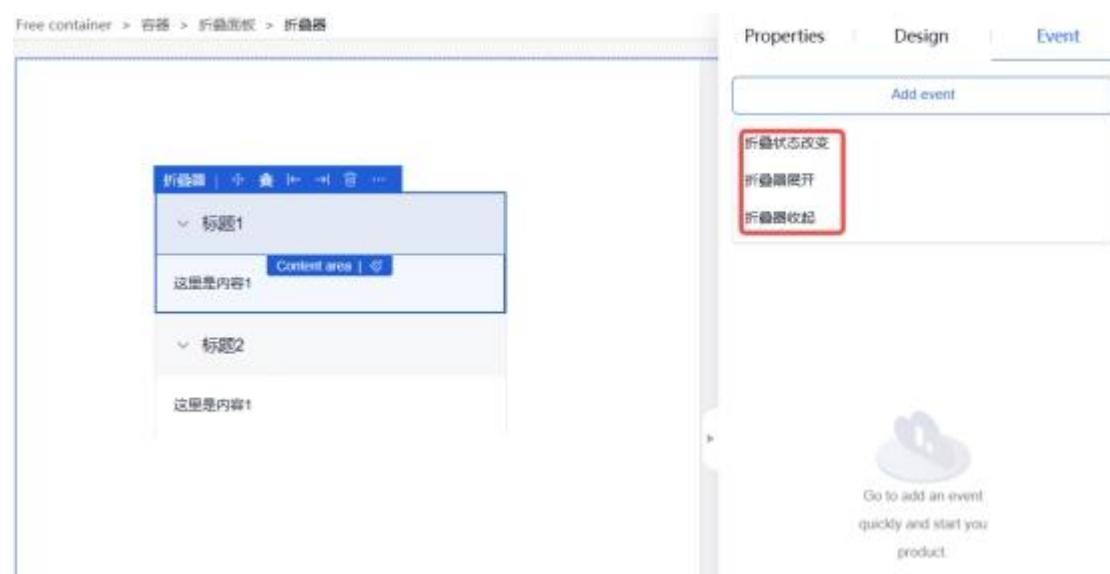
5.4.16.1 Content modification

Click the part to be modified in the "Folder" to modify it. The position can be adjusted in the property bar on the right.



5.4.16.2 Folder events

New folder events: folding state change, folder expansion, and folder collapse.



5.4.17 Tab

It can be used for content grouping. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Field name
		Tabs
		Title Hint
		Initial tab
		Active tab
	Status	Visibility
		Hiddenness
Events	Tab switch	

Operation instance

5.4.17.1 Tab editing

Click “Tab” to edit the contents of the tab in the “Properties” section of the property bar.



5.4.17.2 Tab switch

Click "Tab" to add the event "Tab Switch" in "Event" section of the property bar, which can be used for tab switch



5.4.18 Cascade selector

It supports cascaded selector with sub-items in options, and also supports multiple selections through source drop-down options. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Title
		Description
		Visible
		Disable
		Fixation
	Options	Custom options
		Channel data
		External interfaces
Events	Change in value	
	Getting focus	
	Loss of focus	
	Verification succeeded	
	Verification failed	

Operation instance

5.4.18.1 Custom options

1. Click “Cascade Selector”, and click custom options in the “Data” section in the “Properties” section of the property bar.



2. Add the required cascade options on the management selection page, and click “OK” to save.

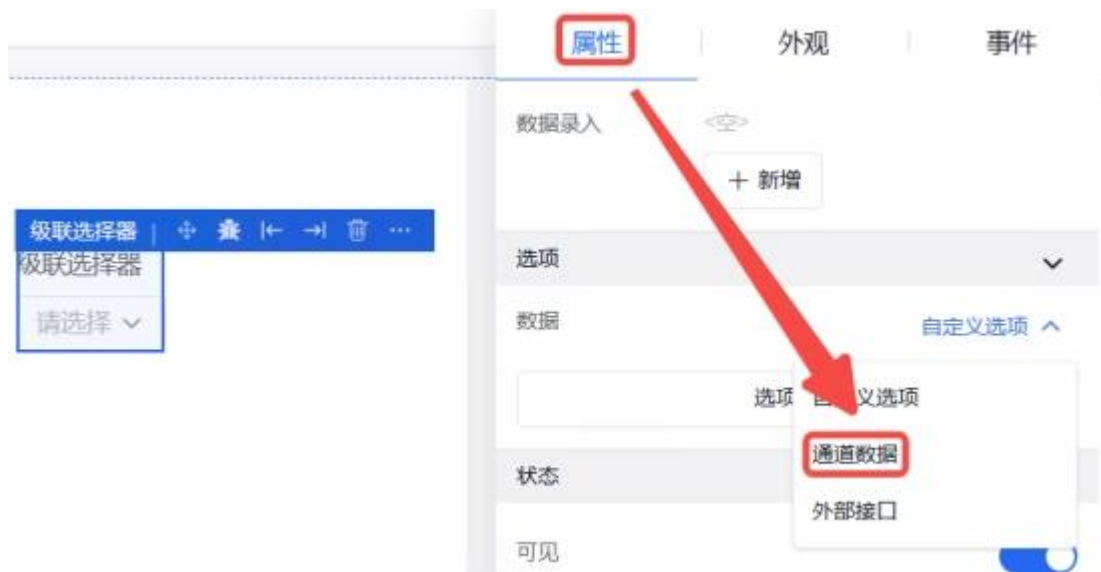
选项管理

选项A	A	+	✕
选项B	B	+	✕
选项b1	b1	+	✕
选项b2	b2	+	✕
选项C	C	+	✕
选项c1	c1	+	✕
选项c2	c2	+	✕

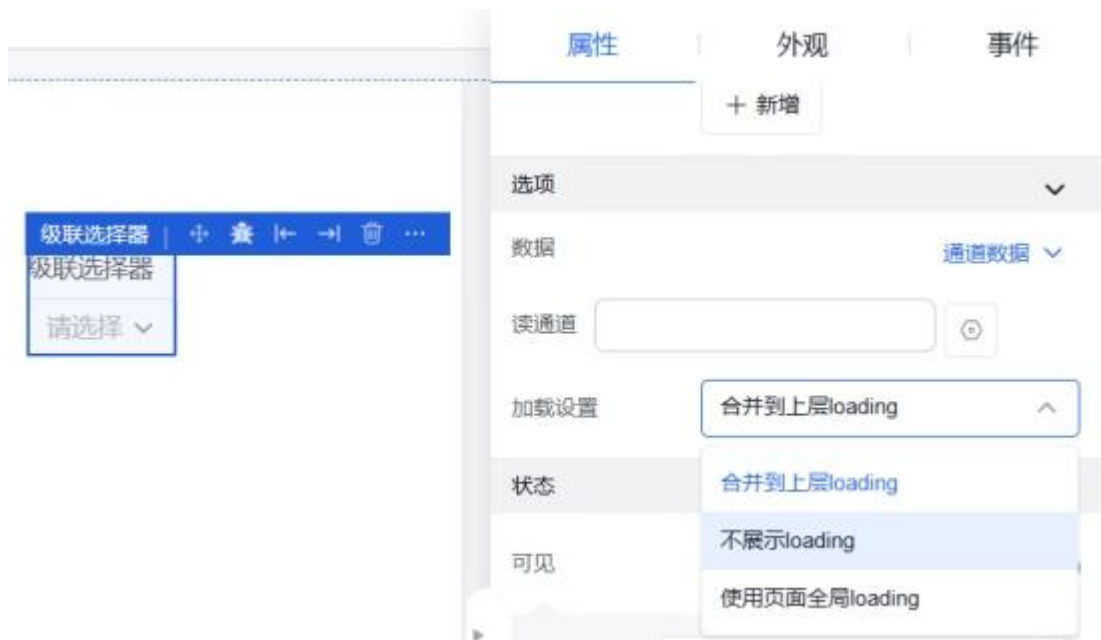
取消 确认

5.4.18.2 Channel data

1. Click “Cascade Selector” and click on the channel data in the “Data” section in the “Properties” section of the property bar.



- For the use of read channel, refer to the read channel binding operation of the text input box in 5.4.1. The loading status is divided into "Merge to the upper layer loading", "Do not show loading" and "Use page global loading".



5.4.19 Check box

It is a check box component. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position

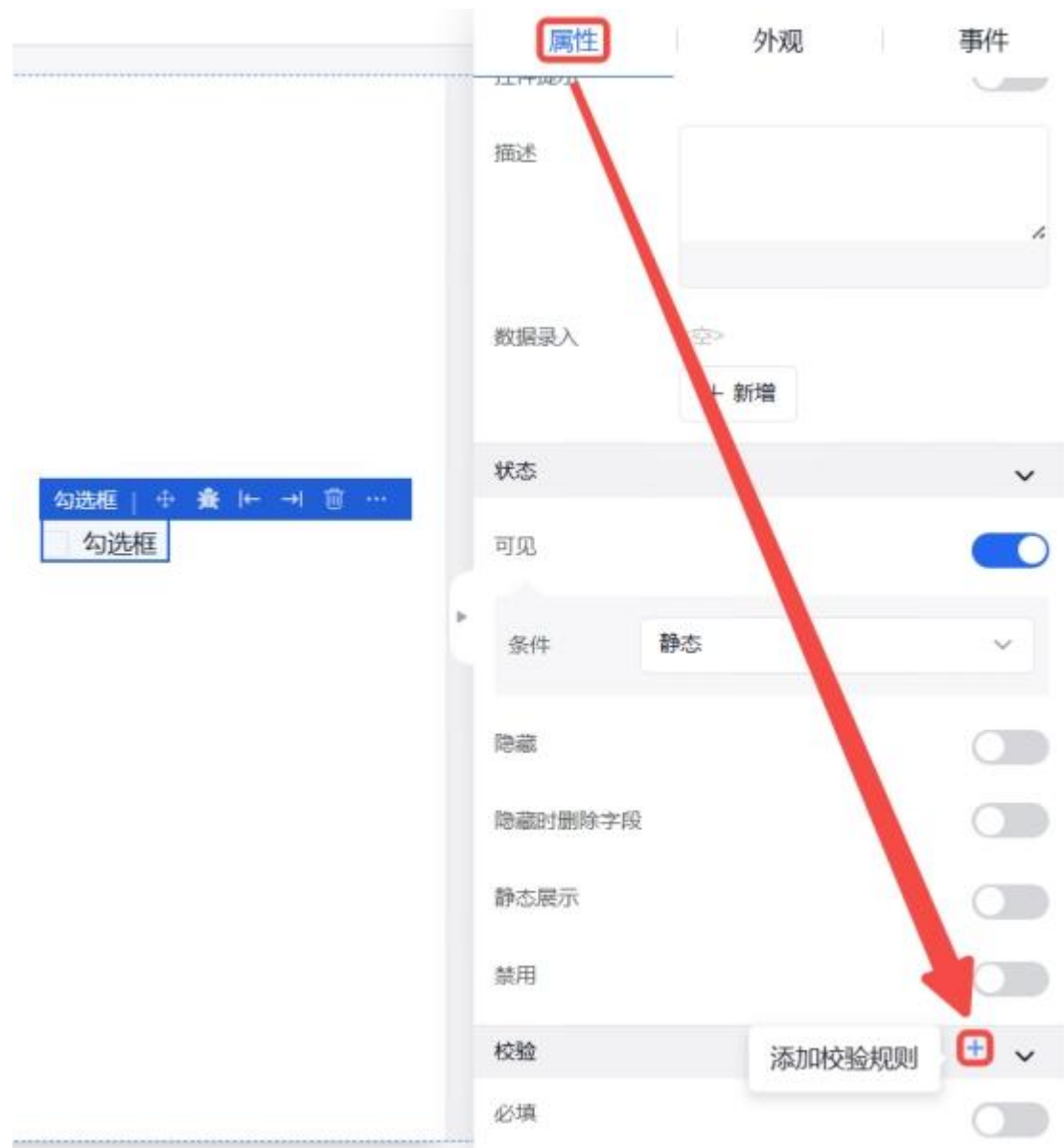
		Field name
		Title
		Description
		Value format
		Default selection
		Title Hint
		Control Hint
		Description
		Data Entry
	Status	Visibility
		Hiddenness
		Prohibitibility
Events	Change in value	
	Verification succeeded	
	Verification failed	

5.4.19.1 Value format

Refer to the setting operation of switch value formats in 5.4.6.

5.4.19.2 Verification rules

1. Click the “Check Box” and click “+” in the “Properties” section of the property bar.



2. Multiple validation rules can be selected.

CSS style of the content area

Add sub-form item

Sub-form presentation mode

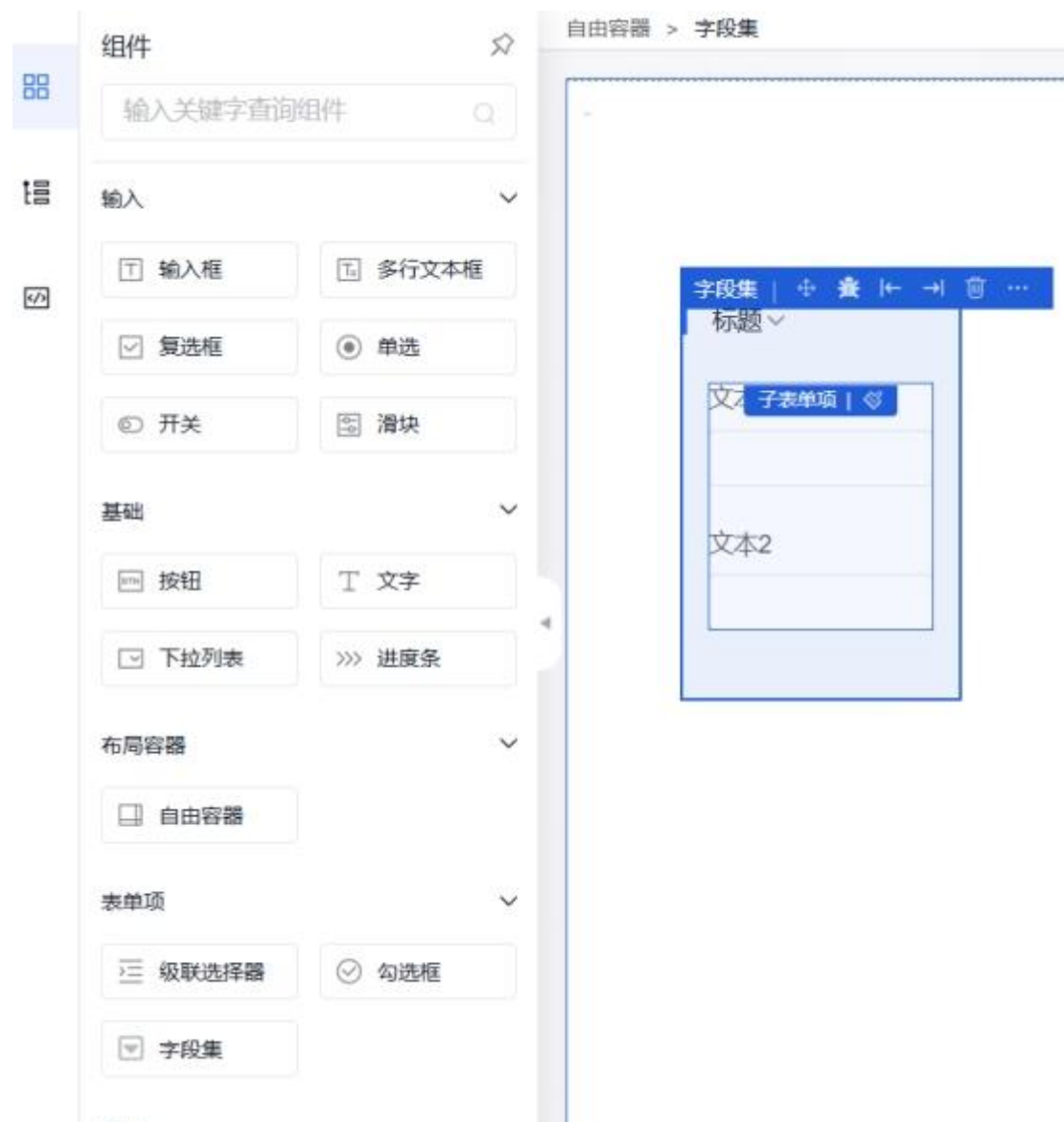
5.4.20.1 Adding and modifying subcomponents

1. Click “Field Set” and select the form item to be added from the form items of the component on the left after clicking “Add Sub-form Item” in the “Properties” section.

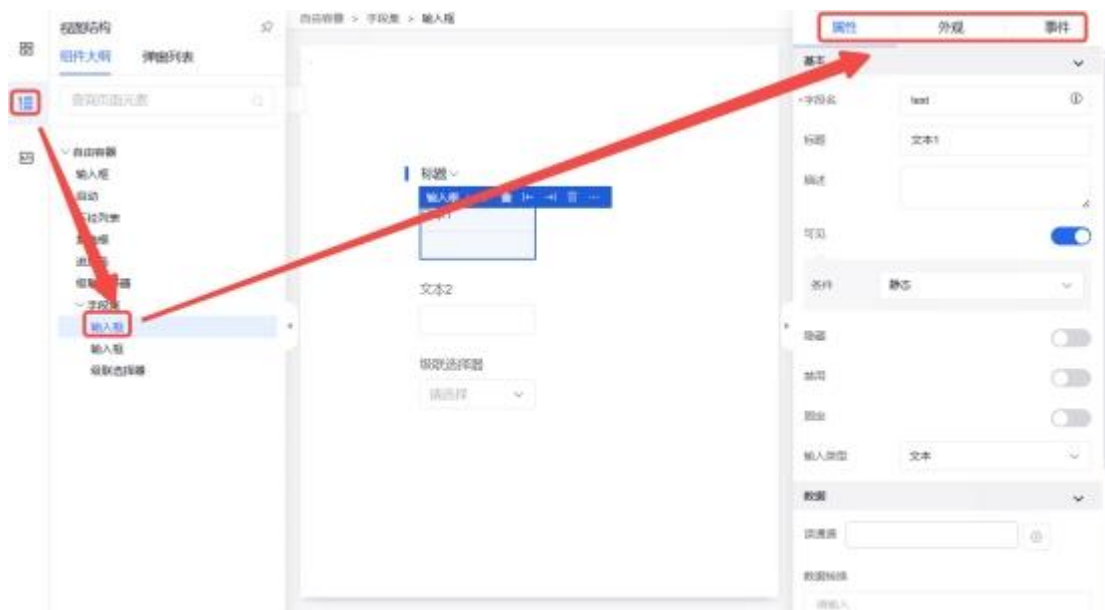


2. After the "Show" entry is deleted from the search box of the component bar, it will show all the components that can be added to the card.

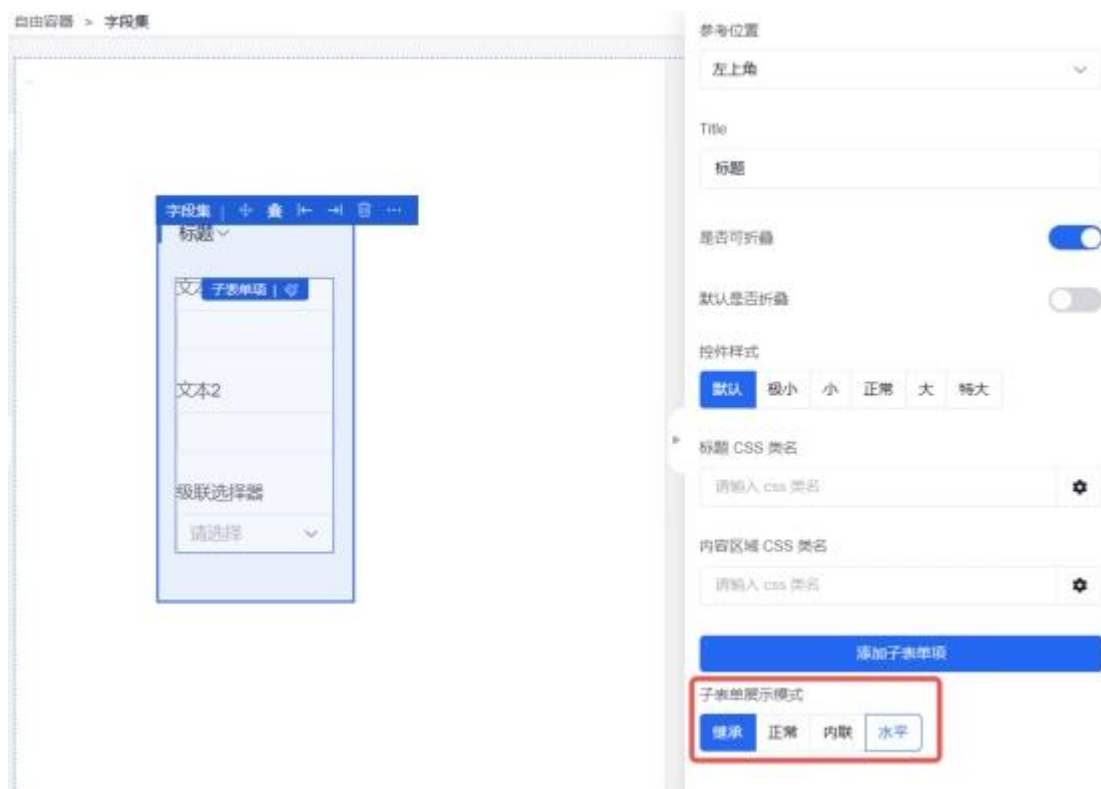




3. Click the component outline of the left view structure and click the component title to be edited. Then the property bar of the individual component will appear on the right. The operation mode of each component is the same as that of the original component.



- Click “Field Set” to select the presentation relationship of each component in the sub-form presentation mode.



5.4.21 Text prompt

It is a text display component. Like a container, it can place multiple renderers together. When the user hovers or clicks on the container, a floating layer of text prompt will be displayed. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Prompt title
		Prompt content
		Trigger mode
		Prompt location
		Theme color
		Container embedding

5.4.22 iframe

It can be used to embed an existing page. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Prompt title
		Page address
	Status	Visibility
		Hiddenness

5.4.23 Card

It supports the embedding of card components of other components. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Title
		Subtitle
		Picture address
		Open external links
		Description
		Highlight expression or not
	Status	Visibility
		Hiddenness

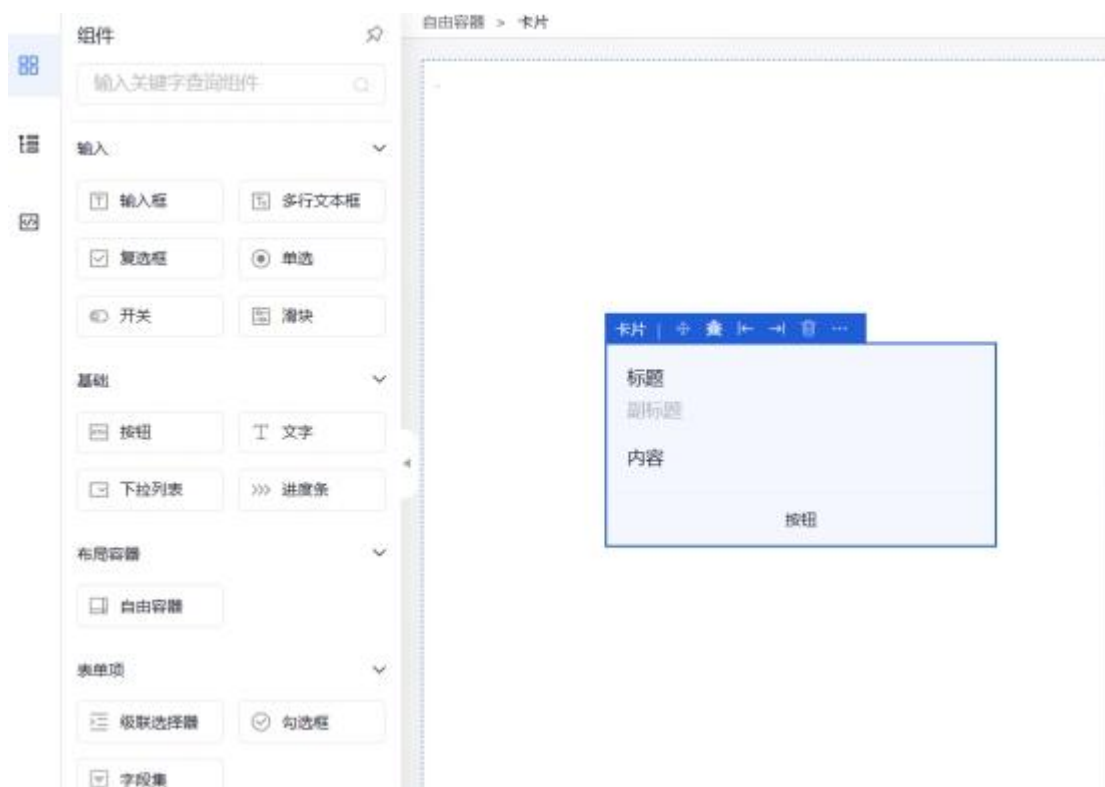
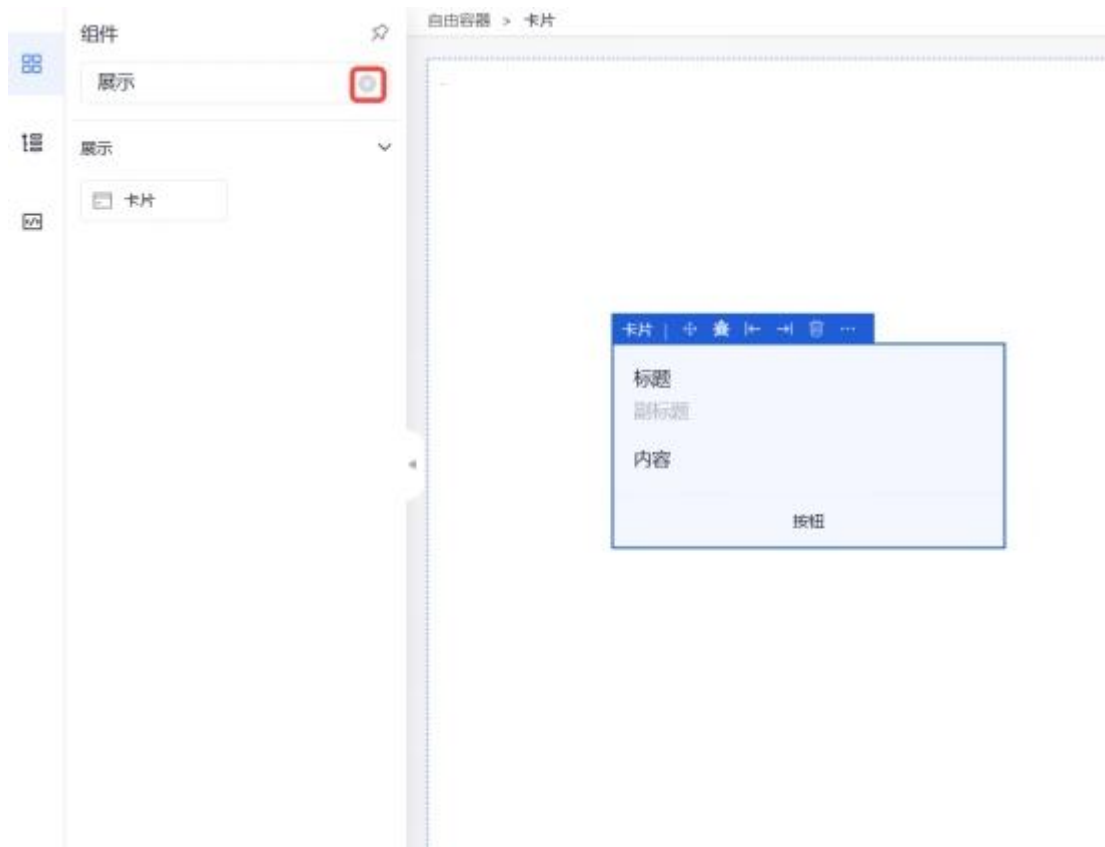
Operation instance

5.4.23.1 Adding and modifying subcomponents

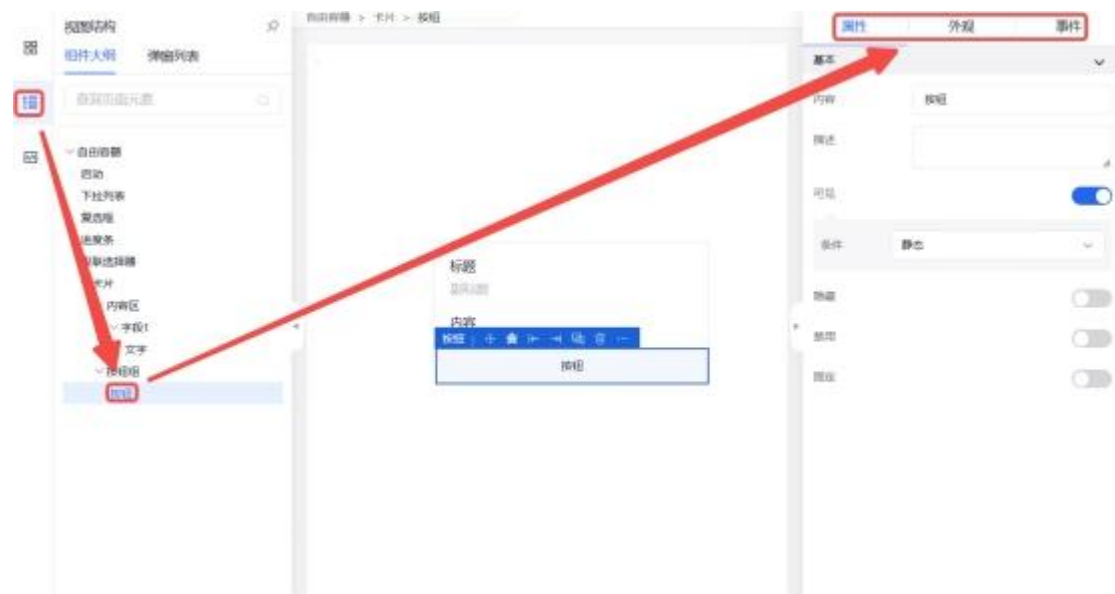
1. Click “Card” and select the form item to be added from the form items of the component on the left after clicking “Add” in the “Properties” section of the property bar.



2. After the "Show" entry is deleted from the search box of the component bar, it will show all the components that can be added to the card.



3. Click the component outline of the left view structure and click the component title to be edited. Then the property bar of the individual component will appear on the right. The operation mode of each component is the same as that of the original component.



5.4.24 Status display

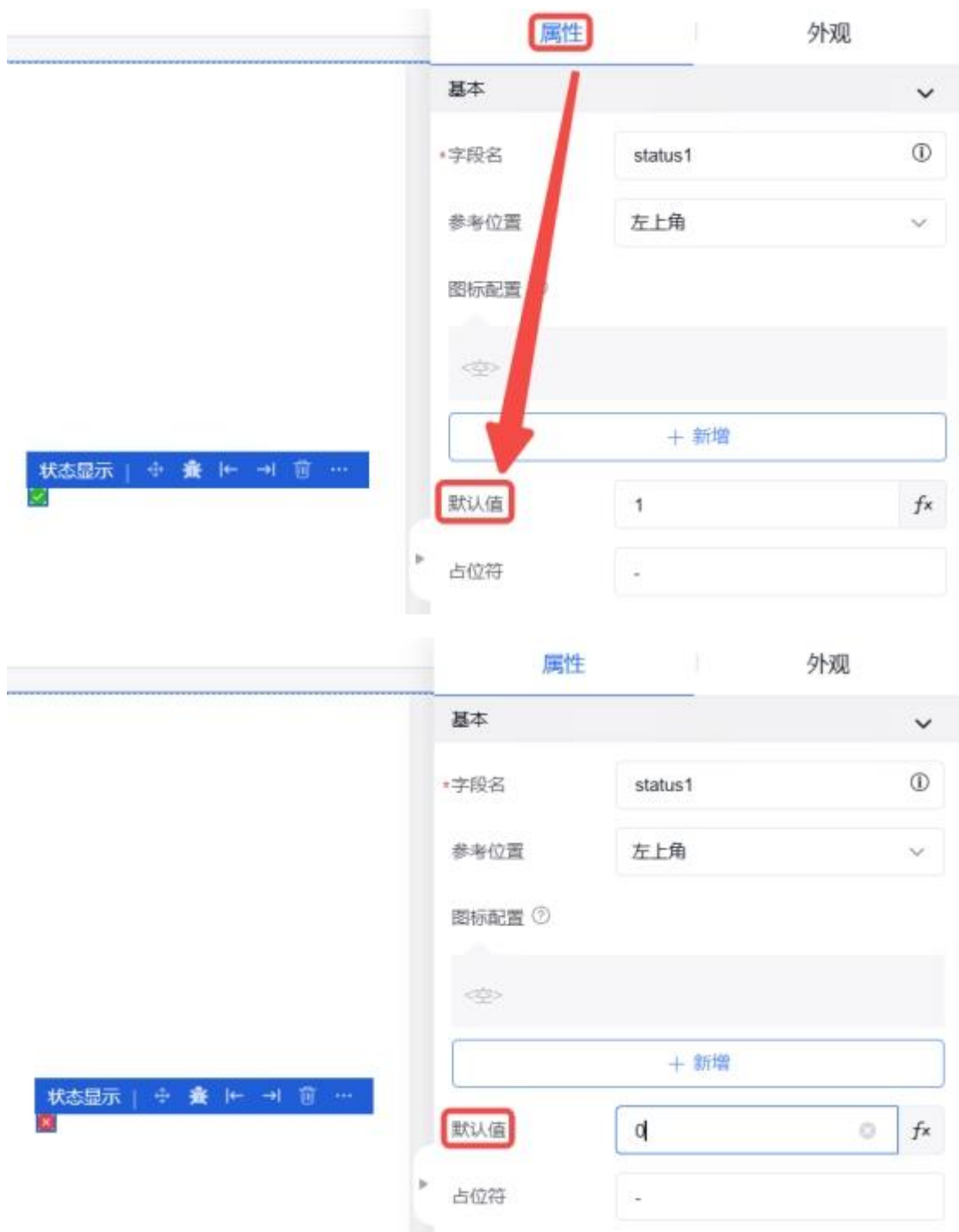
It is an icon component that is used to display the status based on associated fields, such as 1 displayed $\sqrt{}$, 0 displayed x. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Reference position
		Title
		Icon configuration
		Default value
		Placeholder
	Status	Visibility
		Hiddenness

Operation instance

5.4.24.1 Status change

Click "Status Display", and modify the default value item in the "Properties" section of the property bar. When the default value is 1, it will be displayed as √, and when the default value is 0, it will be displayed as x.



5.4.25 Step bar

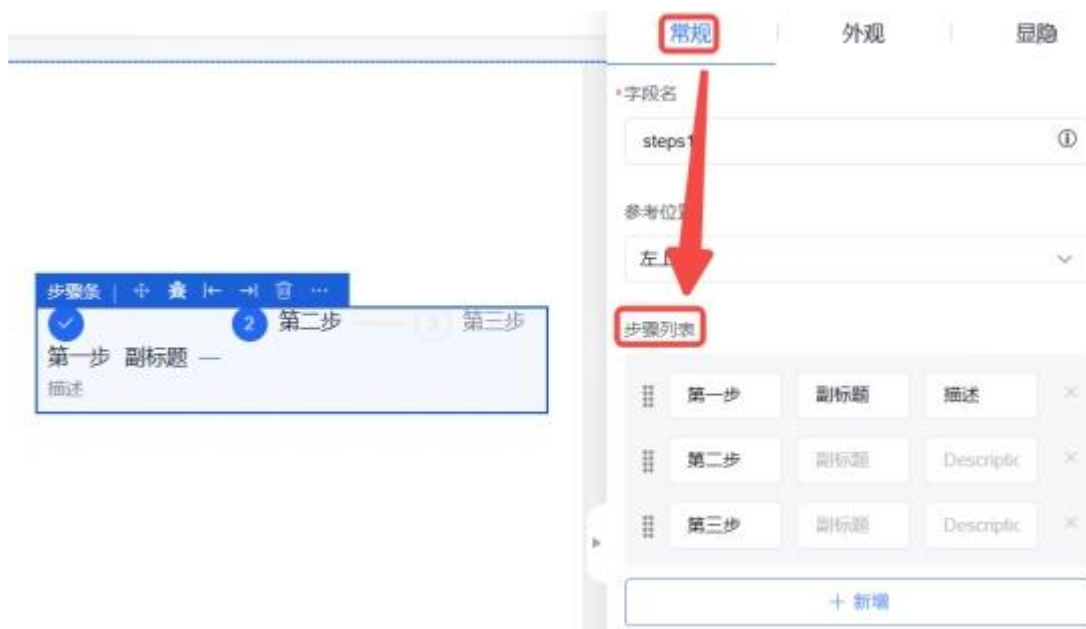
It is a step bar component. The properties and events are shown in the following table:

Type	Content	Name
Properties	Basic	Field name
		Reference position
		Step list
		Current step
		Current status
	Status	Visibility
		Hiddenness

Operation instance

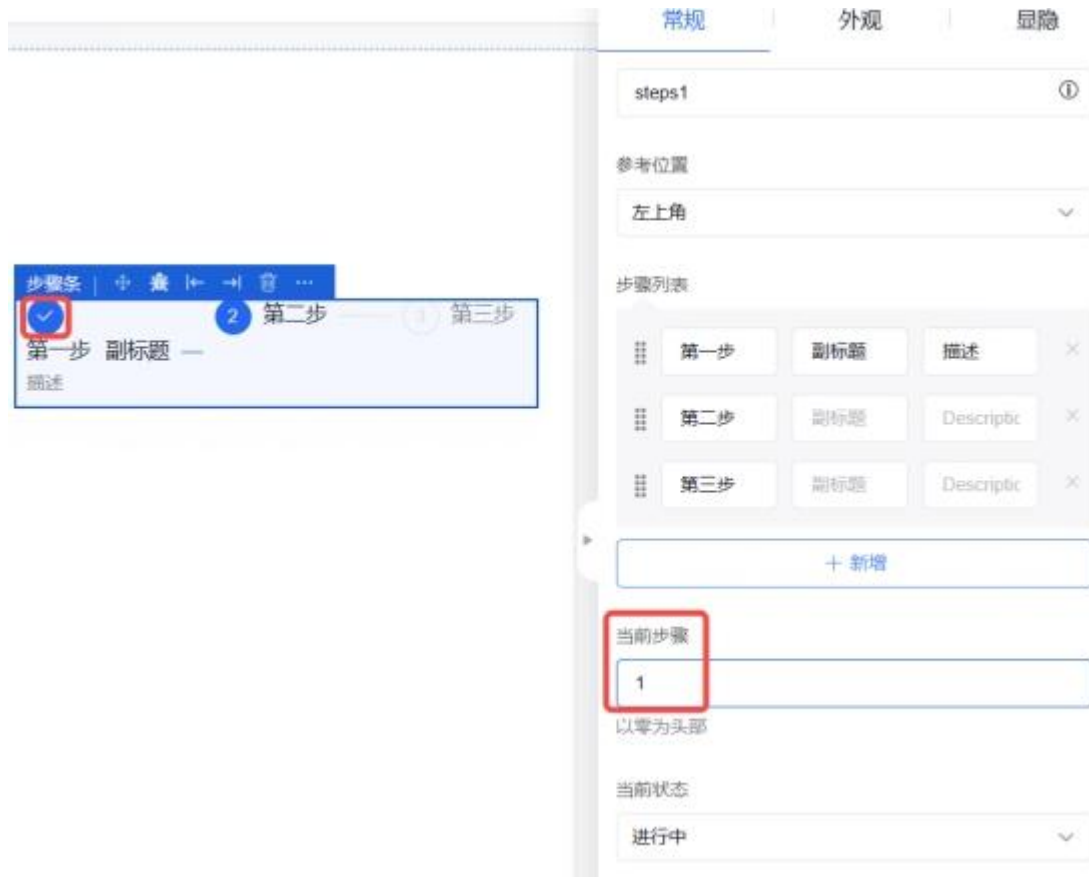
5.4.25.1 Modify the list of steps

Click the “Step Bar” and modify the list of steps in the “General” section of the property bar. You can not only modify the subtitle and description, but also add a new step.



5.4.25.2 Modify the step status

1. Click the “Step Bar” and edit the serial number at the current step in the “General” section of the property bar.



2. Edit the selected serial number step in the “Current Step” section. There are four statuses: “In Progress”, “Waiting”, “Done”, and “Error”.

步骤条

第一步 副标题 — 描述

2 第二步

第三步

常规

外观

显隐

steps1

参考位置

左上角

步骤列表

第一步

副标题

描述

第二步

副标题

Descriptic

第三步

副标题

Descriptic

+ 新增

当前步骤

1

以零为头部

当前状态

进行中

步骤条

第一步 副标题 — 描述

2 第二步

第三步

常规

外观

显隐

steps1

参考位置

左上角

步骤列表

第一步

副标题

描述

第二步

副标题

Descriptic

第三步

副标题

Descriptic

+ 新增选项

进行中

等待

完成

出错

进行中

6. Instance management

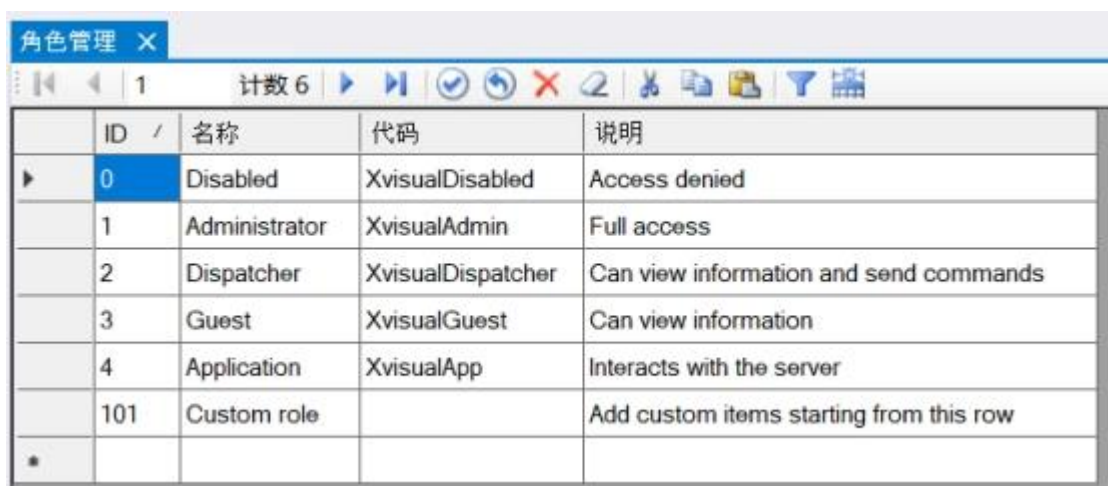
Instance management can only be viewed but not modified.



角色管理		实例管理			
计数 1					
	编号	名称	代码	父对象	说明
▶	1	Enterprise	Ent		

7. Role management

A role is a named set of permissions. It should be noted that the permissions are assigned to custom roles only. For built-in roles, permission checking has been already implemented in the application.



角色管理				
计数 6				
	ID /	名称	代码	说明
▶	0	Disabled	XvisualDisabled	Access denied
	1	Administrator	XvisualAdmin	Full access
	2	Dispatcher	XvisualDispatcher	Can view information and send commands
	3	Guest	XvisualGuest	Can view information
	4	Application	XvisualApp	Interacts with the server
	101	Custom role		Add custom items starting from this row
*				

7.1 Built-in roles

A built-in role is a system-built role, and its name and permissions cannot be changed. It applies to all objects.

ID /	名称	代码	说明
0	Disabled	XvisualDisabled	Access denied
1	Administrator	XvisualAdmin	Full access
2	Dispatcher	XvisualDispatcher	Can view information and send commands
3	Guest	XvisualGuest	Can view information
4	Application	XvisualApp	Interacts with the server
101	Custom role		Add custom items starting from this row

The following table lists the built-in roles whose permissions are hardcoded into the application.

ID	Name	Permission
0	Disabled	Access denied
1	Administrator	Full permissions
2	dispatcher	View information, and send commands
3	guest	Information that can be viewed
4	Application	Interacting with the server

7.2 Custom roles

Custom roles are created by project developers. Role creation is performed in the Administrator application in the user table. Click the blank line directly to create a custom role. It is recommended to assign id to custom roles starting at 101.

角色管理 X				
7 计数 7				
	ID /	名称	代码	说明
	0	Disabled	XvisualDisabled	Access denied
	1	Administrator	XvisualAdmin	Full access
	2	Dispatcher	XvisualDispatcher	Can view information and send commands
	3	Guest	XvisualGuest	Can view information
	4	Application	XvisualApp	Interacts with the server
	101	Custom role		Add custom items starting from this row
▶	102			

8. User management

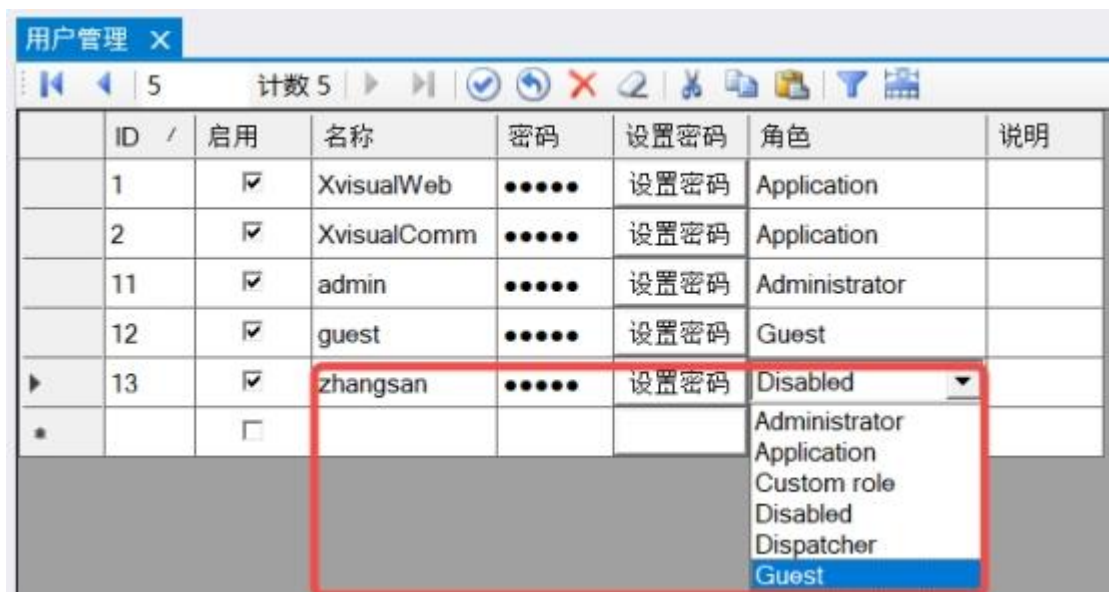
8.1 Create a user

Creating a user is performed in the Administrator application in the user table. Click the blank line directly to create a user. By default, the users with ids of 1 to 12 are created from the project template. XvisualWeb and XvisualComm users are used to connect Webstation and Communicator client applications to the Server application. The admin and guest users using the built-in roles are used to log in to the system.

用户管理 X							
1 计数 5							
	ID /	启用	名称	密码	设置密码	角色	说明
▶	1	☒	XvisualWeb	•••••	设置密码	Application	
	2	☒	XvisualComm	•••••	设置密码	Application	
	11	☒	admin	•••••	设置密码	Administrator	
	12	☒	guest	•••••	设置密码	Guest	
	13	☒			设置密码	Disabled	
•		☐					

8.2 Edit a user

Editing a user is performed in the Administrator application in the users table. Click the corresponding line directly to modify it. The ID is automatically generated by the system, and the name, password and role can be modified by you. After modification, you need to click “Submit”. Otherwise, you will be reminded that it has not been saved.



8.3 Delete a user

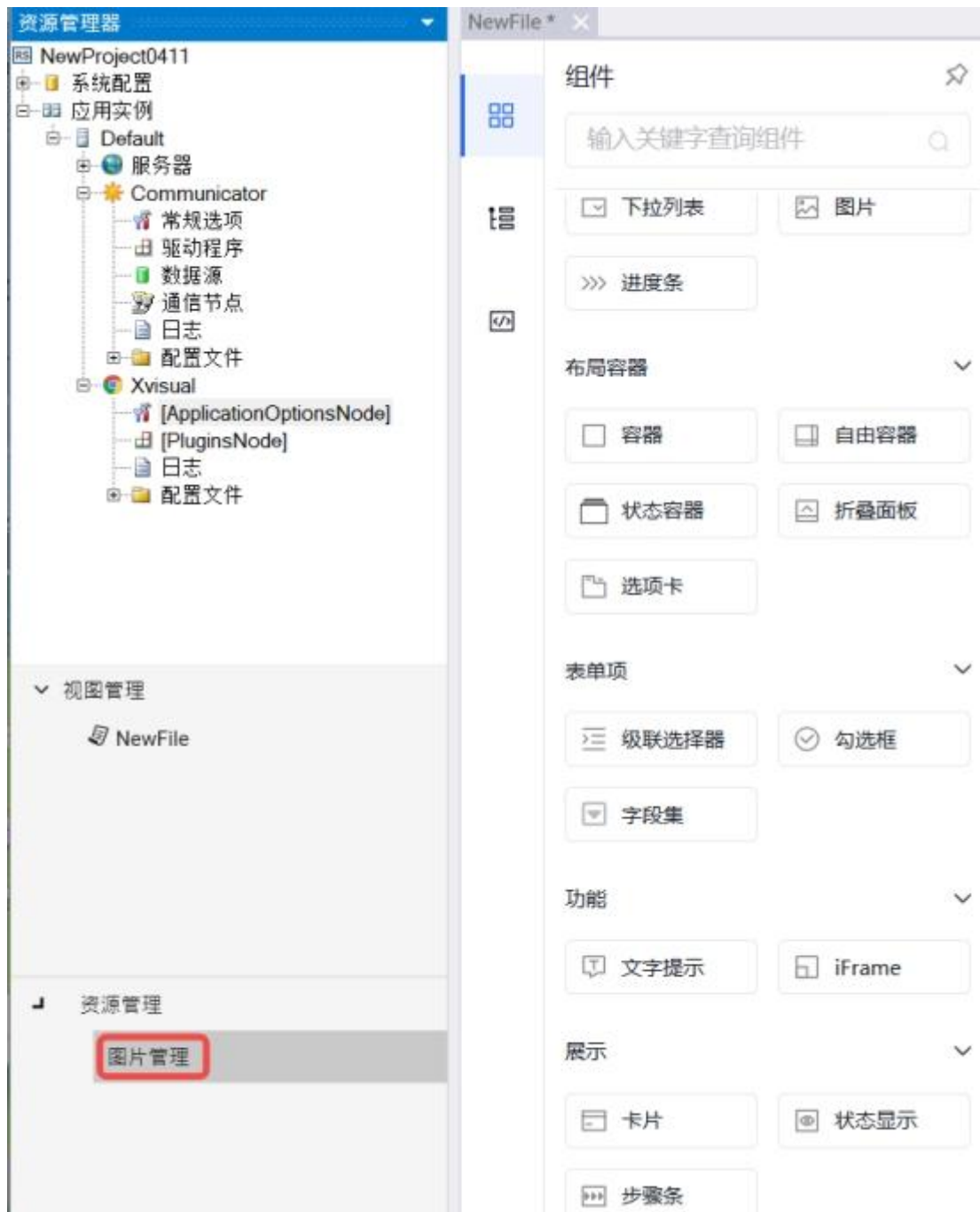
Deleting a user is performed in the Administrator application in the users table. Select the user data to be deleted and click the “x” button above to complete the deletion.



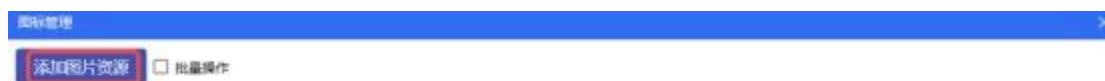
9. Resource management - picture management

9.1 Add a picture

1. Click to enter the picture management in the resource management.



2. Click “Add Picture Resource” to add pictures, or add pictures in batches.

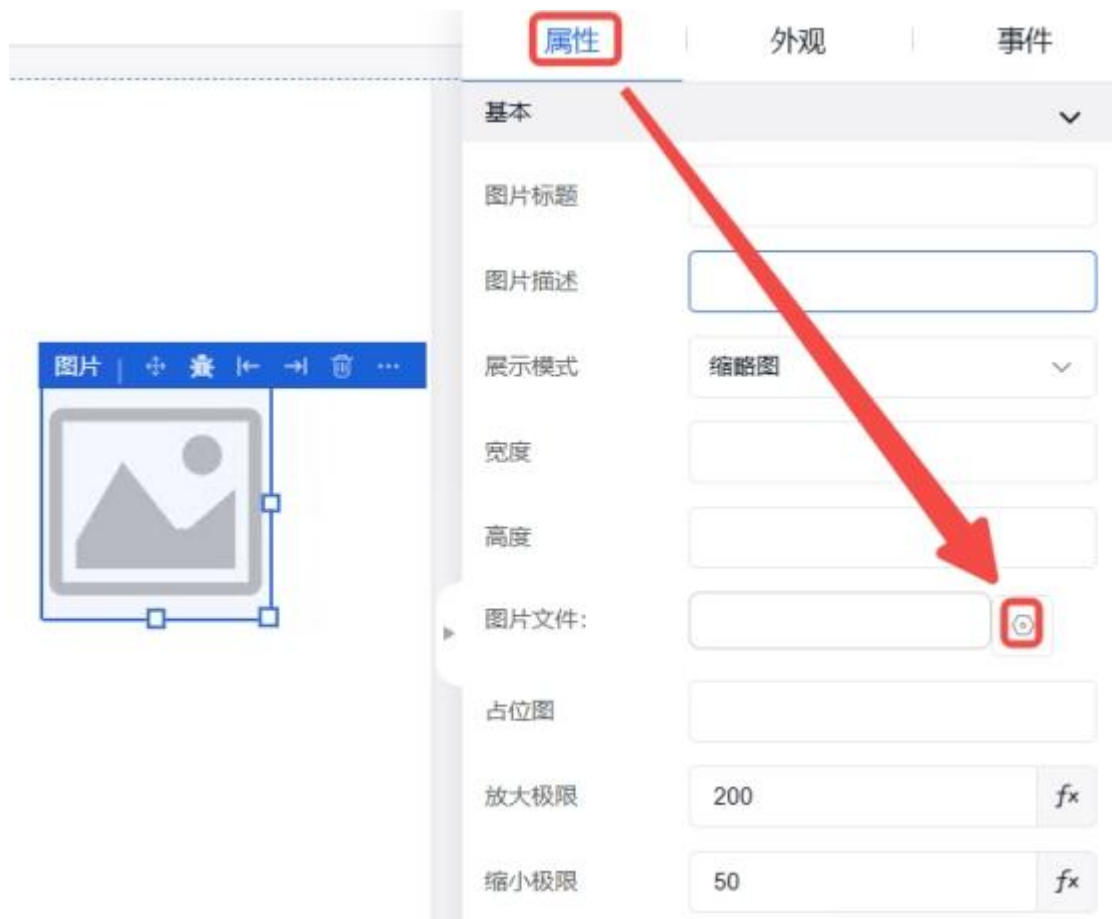


3. After the picture is added successfully, it will be displayed in the icon management.



9.2 Use a picture

1. Click the “Picture” component and click the “Edit” button on the picture file in the “Properties” section of the property bar.



2. Enter the picture selection interface, select the picture you want to upload, and click “Save” to complete the operation.



3. Upon completion of the upload, the picture component content in the view is updated to the picture of the user's choice.



Appendices

I. Usage Properties and Annotations

The following table provides an overview of all compilations and annotations that can be configured in the server. These can be defined in various real-time environments in <Product Name> to implement different functions. Refer to the following table for a detailed description of the properties and annotations.

Symbol description/legend	Meaning

II. Error Diagnosis

Extended diagnostics may be required if undesirable operating behavior occurs. According to specific circumstance, this may include the following actions: <Action 1>, <Action 2>, <Action n>. Details can be found in the table below:

Action	Use cases	Link

The following table provides an overview of possible unexpected operation behaviors and solutions.

Unintended operation behavior	Solution

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III. Support and Service

SINSEGYE provides comprehensive support and services for the company's products and solutions, ensuring a fast and professional response to relevant issues.

Data download

Our data download section covers a wealth of file resources, including application cases, technical documentation, and product descriptions, to meet your diverse needs.

Data download address: <https://help.sinsegye.com.cn/>

Get support

Please contact us for local support and service of SINSEGYE products. You can visit our official website www.sinsegye.com.cn to find out the branch address of SINSEGYE and get more information about SINSEGYE.

In addition, you can contact us by:

- Hotline: 400-013-2158
- Email Address: support@sinsegye.com.cn